Introduction to Machine Learning

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Acknowledgement
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Outlines

- Introduction to Machine Learning
- Recap data preparation phase
- Classification and Regression
- Unsupervised Learning & Dimensionality Reduction
- Deploy Machine Learning model

Introduction

Nowadays we have:

Machine Learning (ML)

- Various data types
  - Structured data (Relational Data Base)
  - Semi-structured data (XML, JSON)
  - Unstructured data (Video, Image, Text)
- Massive data (Big data) generated everyday from deferent sources (IOT).
- High computation process (Time and resources).

Tasks that are too complex to program

- Human/Animal tasks: speech recognition, image understanding, driving
- Tasks beyond human: very large and complex data, e.g., astronomical data, weather prediction…etc

Adaptivity

- **Tasks changing over time**: spam detection, product recommendation

Introduction

What is learning?

- "Learning denotes changes in a system that ... enable a system to do the same task more efficiently the next time."

  Herbert Simon

- "Learning is any process by which a system improves performance from experience." **Herbert Simon**
- "Learning is constructing or modifying representations of what is being experienced." **Ryszard Michalski**
- "Learning is making useful changes in our minds." **Marvin Minsky**

The process of turning **experience** into **expertise**

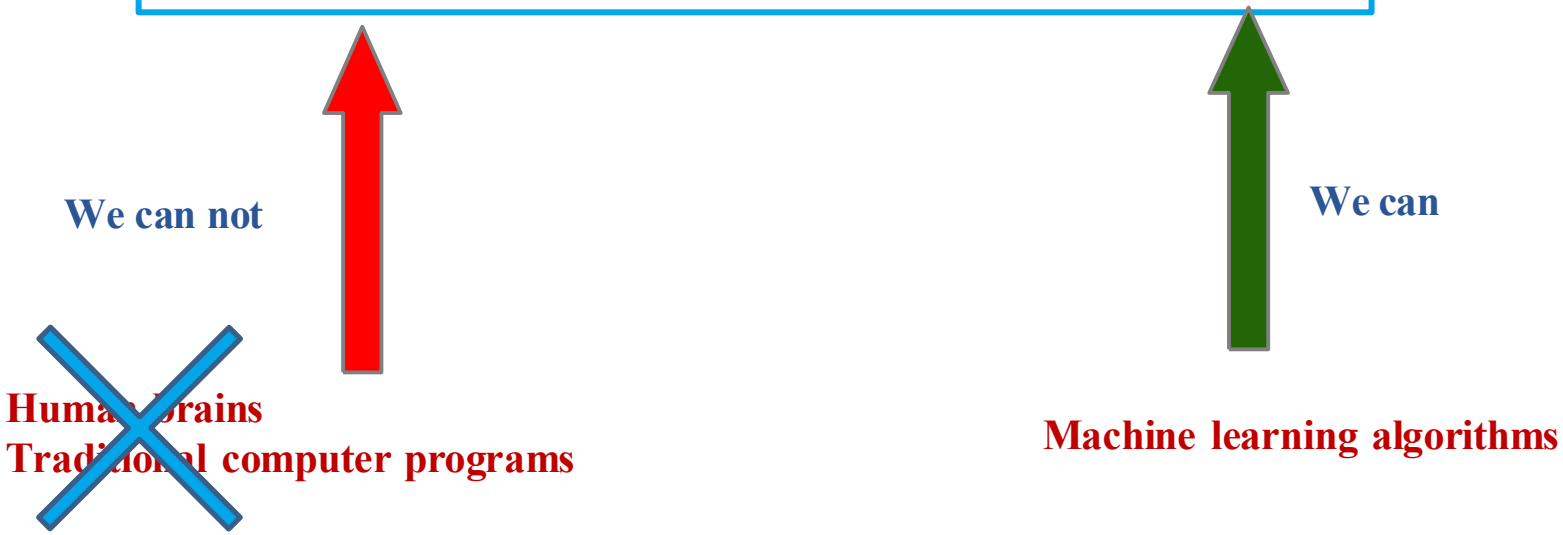
- Experience: training data, measurements, observations
- Expertise: the ability to perform some task

https://courses.cs.washington.edu/courses/cse446/15sp/slides/intro.ppt https://www.cmpe.boun.edu.tr/~ethem/i2ml2e/

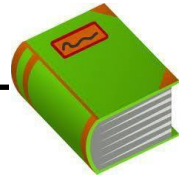
Introduction

Traditional learning limitations:

- **Too much diversity** - learning by memorization fails
- We need to use some **prior knowledge** about the problem
- Problem **complexity**



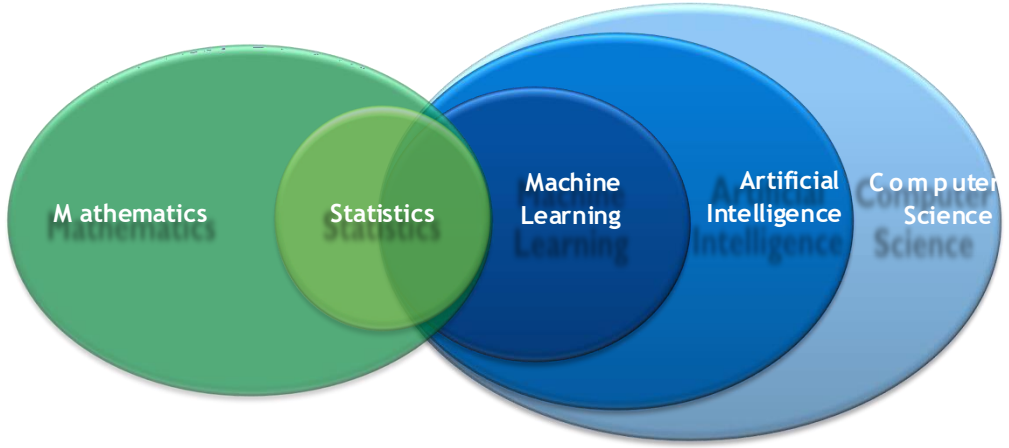
What is Machine learning?



Definition: ML

is a field of study that gives machines the ability to learn using examples without being explicitly programmed
ML is a subfield of AI
Embedding knowledge to machines without hard-coding it

Role of Computer science: Efficient algorithms to
- Solve the optimization problem
- Representing and evaluating the model for inference



Optimize a performance criterion using example data or past experience

Role of Mathematics: Linear algebra and calculus to
- Solve regression problem
- Optimization functions

Role of Statistics

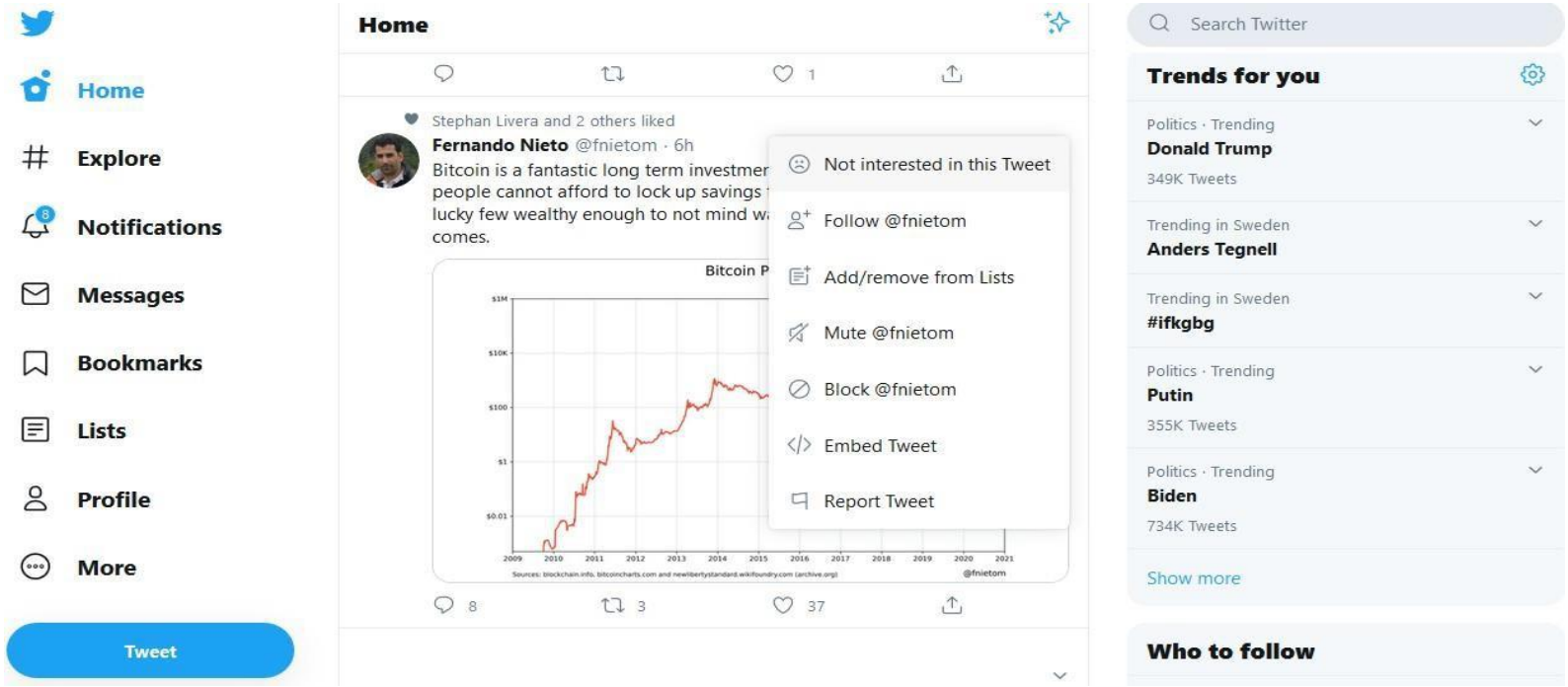
Email spam detection, Face recognition, Weather forecasting, Activities recognition

What is Machine learning?

- A program is said to learn from **experience *E*** with respect to some **tasks *T*** and **performance measure *P*** if its performance on *T*, as measured by *P*, improves with experience *E*.

Suppose that Twitter watches which Tweets you do or do not marked as "uninteresting", And based on that, learns how To better filter Tweets for you.

What is the experience E, task T, and performance measure P?



1. Classifying Tweets as interesting or uninteresting.
2. Watching you label Tweets as interesting or uninteresting.
3. The fraction of Tweets correctly classified as interesting/uninteresting.

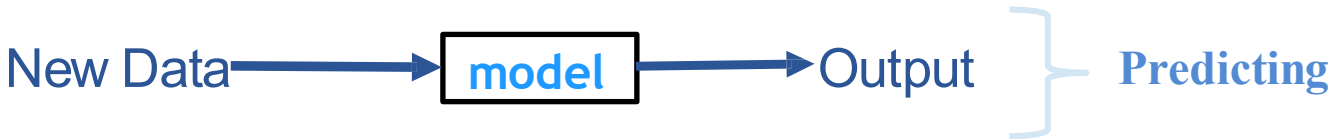
What is Machine learning?

Traditional Programming



Machine Learning (Supervised)





Machine learning algorithms build a **model** from the **training data**, then uses this model to make **predictions** or take decisions.

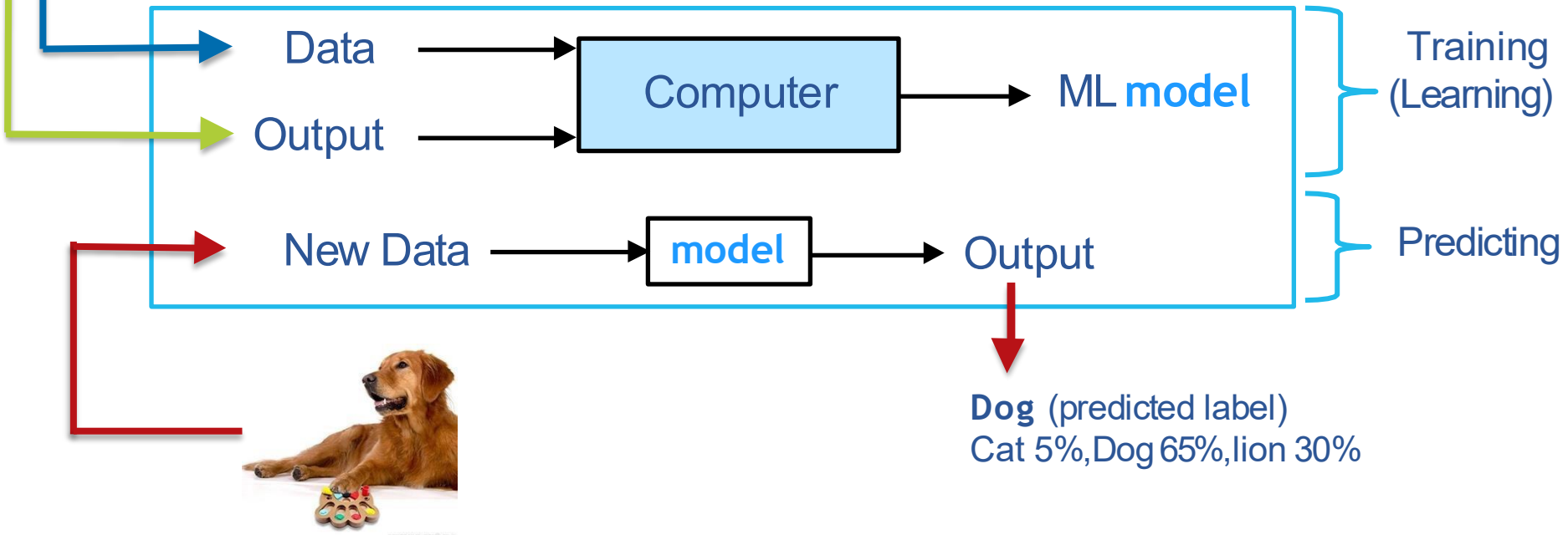
What is Machine learning?



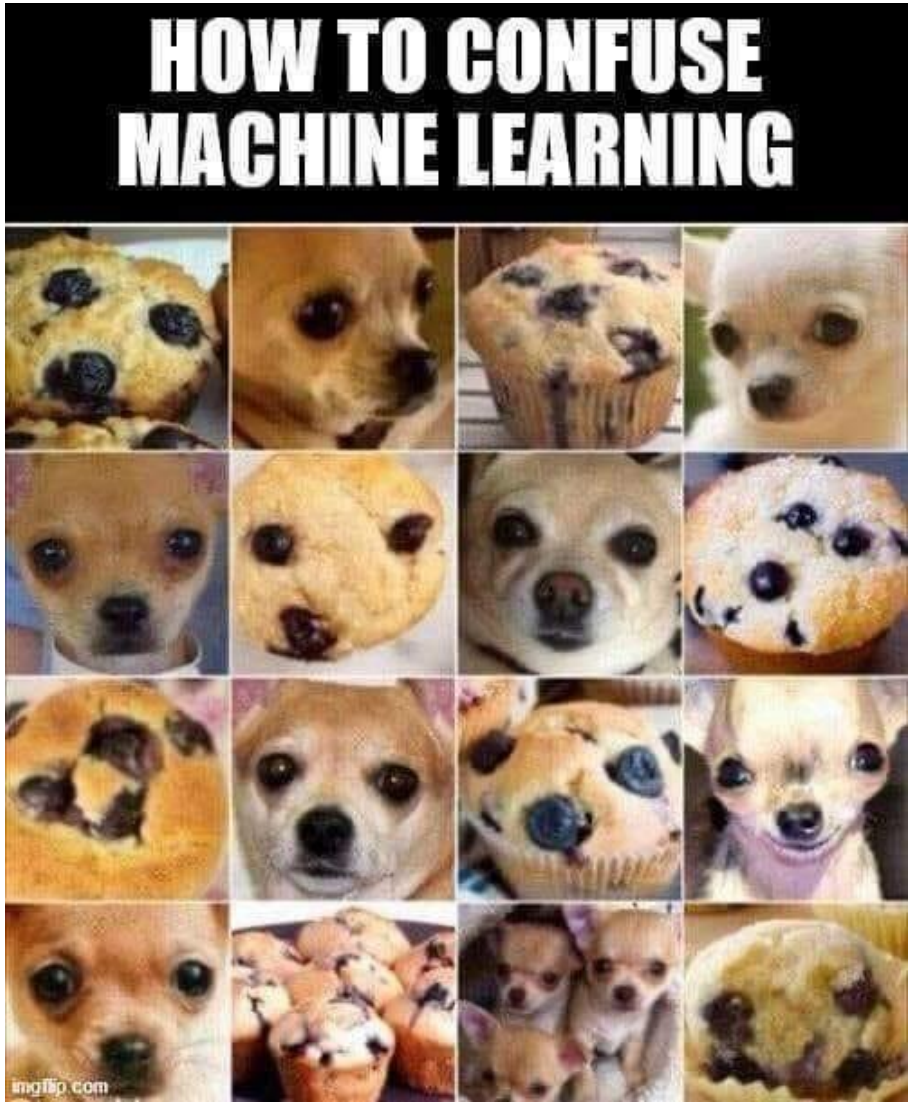
Example:
The data consist of images …

The output consists of labels (Cat, Dog or Lion)

- (Supervised) Machine learning



What is Machine learning?







It is about the Data!!

Example (self-driving car)

Consider the following problem:

- You have a camera on your car that periodically captures images of the road and send them to your app.
- You want your app to recognise what each image contains (pedestrians, bikes, other cars, etc.)



Why do we need machine learning:

- It is extremely hard to solve this without machine learning.
- We can not manually define/code general rules that allow us to recognize what an image contains.



When Do We Use Machine Learning?

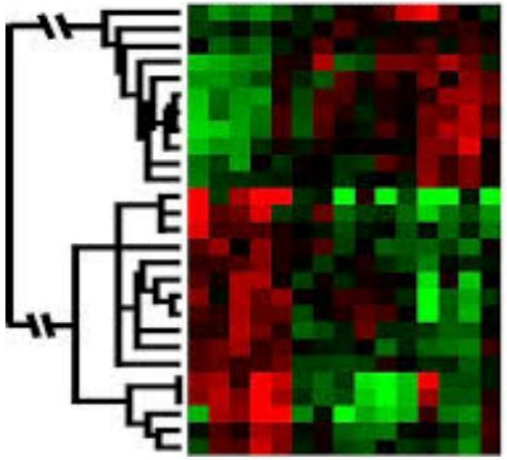
ML is used when:

- Human expertise does not exist (**navigating on Mars**)
- Humans can't explain their expertise (**speech recognition**)
- Models must be customized (**personalized medicine**)
- Models are based on huge amounts of data (**genomics**)

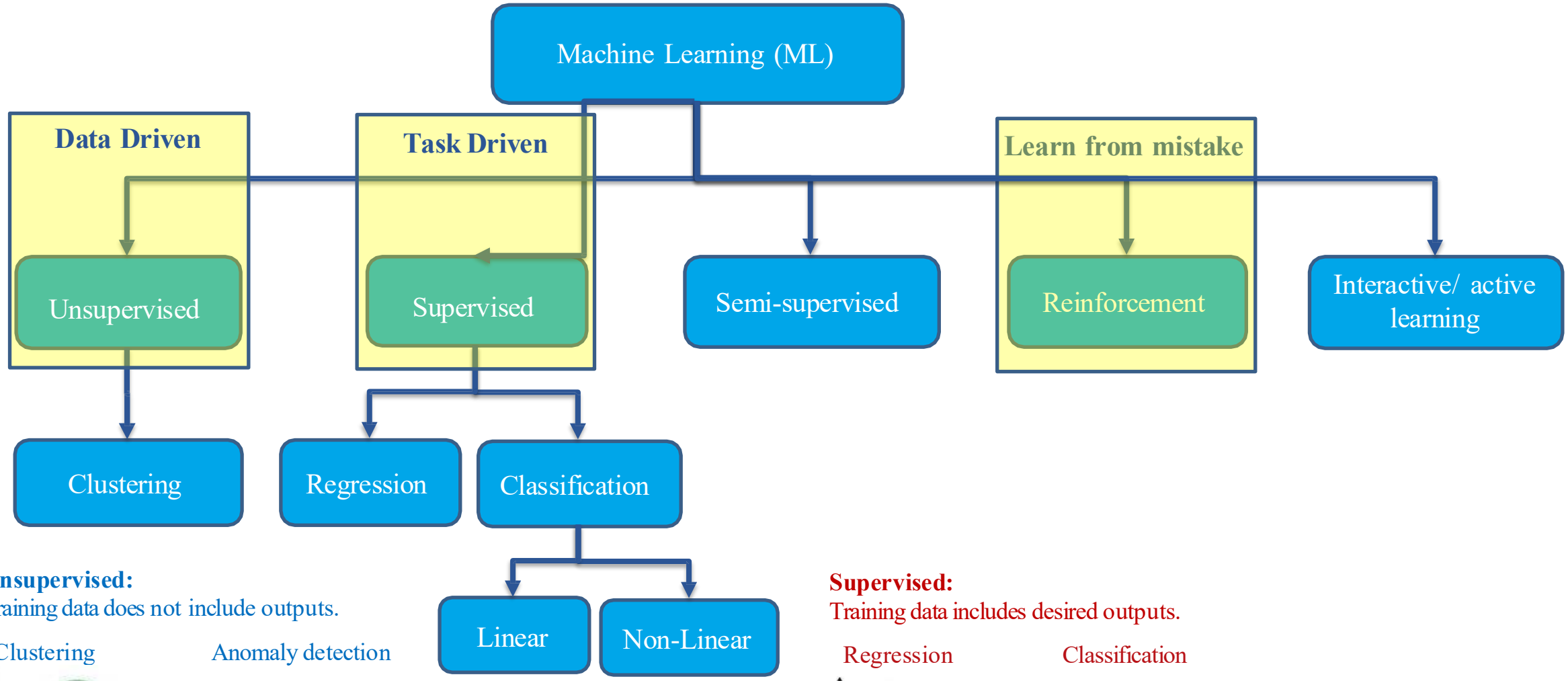






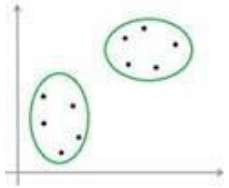


Machine learning types (based on learning)

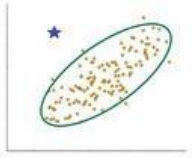


Unsupervised:
Training data does not include outputs.

Clustering

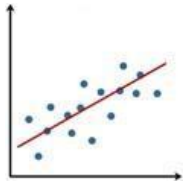


Anomaly detection

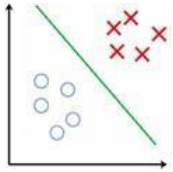


Supervised:
Training data includes desired outputs.

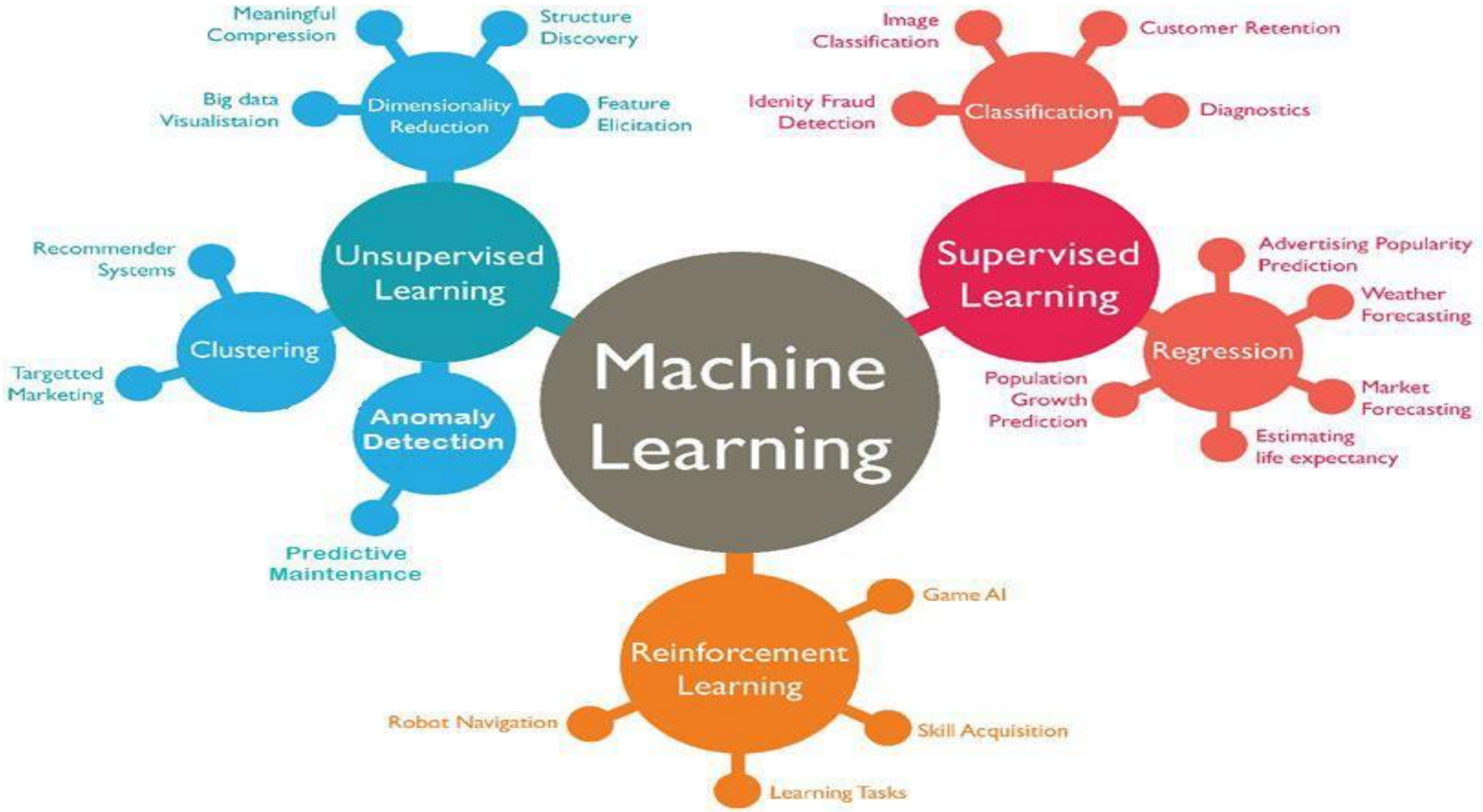
Regression



Classification



Machine learning types (based on learning)



Introduction to Supervised ML

AI system

AI system = Code + Data



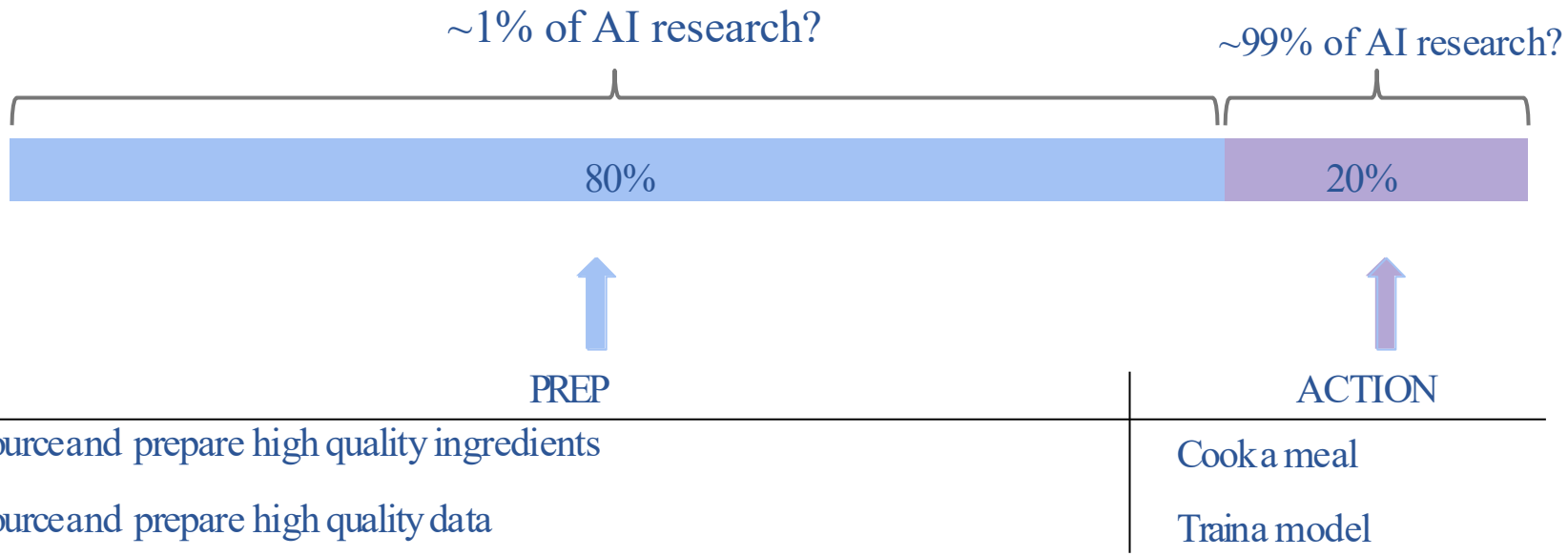




(Model/algorithm)

Machine Learning

Data is the food for AI



https://www.deeplearning.ai/wp-content/uploads/2021/06/MLOps-From-Model-centric-to-Data-centric-AI.pdf

Supervised learning



- Linear and nonlinear **regression** and **classification** methods, including:
  - Logistic Regression
  - SVMs
  - Random Forest
  - Neural Networks Deep NN (CNN, RNN, LSTM …) … and many more.
- **Evaluate** their performance.
- Achieve a good **generalization**.
- **Use** them in real-world use cases.

Supervised learning

- The dataset is labelled (output).
- In the training phase the input and the output are known.
- Comparison of the right output with the output produced by the ML algorithm.
- It guides the learning process to modify the model in order to reduce the error (difference between right and produced output).
- Supervised learning algorithms are for classification and regression.
- Classification problems can be linearly separable or not.

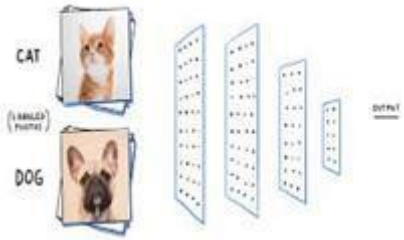
Text	Category
Congratulation you are selected	ham
Congrats you won lottery	spam
travel for free	spam
selected for credit cards	spam
very Good	ham
Good night	ham
lottery	spam

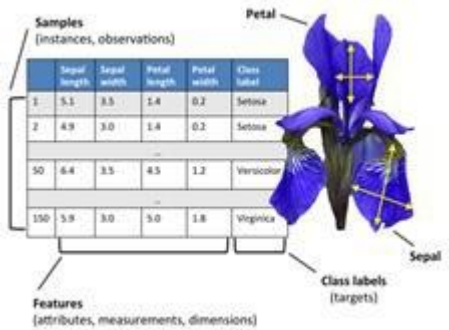


Supervised learning

What I should do before applying the supervised learning?

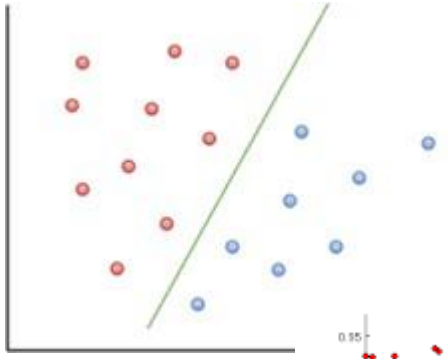
1. Identify the problem type
   - Classification (discrete class label output)
     - Linear separability
     - Non-linear separability
   - Regression (continuous quantity output)

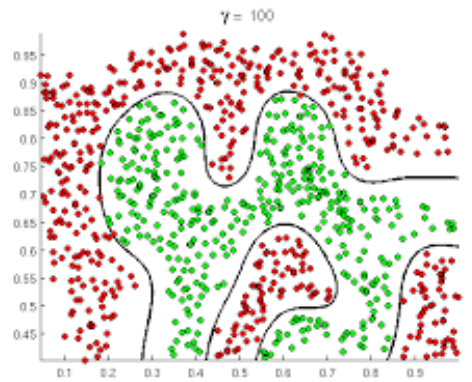




1. Data pre-preparation
   - Select data
   - Preprocess Data
     - Formatting
     - Cleaning
     - Sampling
   - Transform Data
     - Scaling
     - Decomposition
     - Aggregation

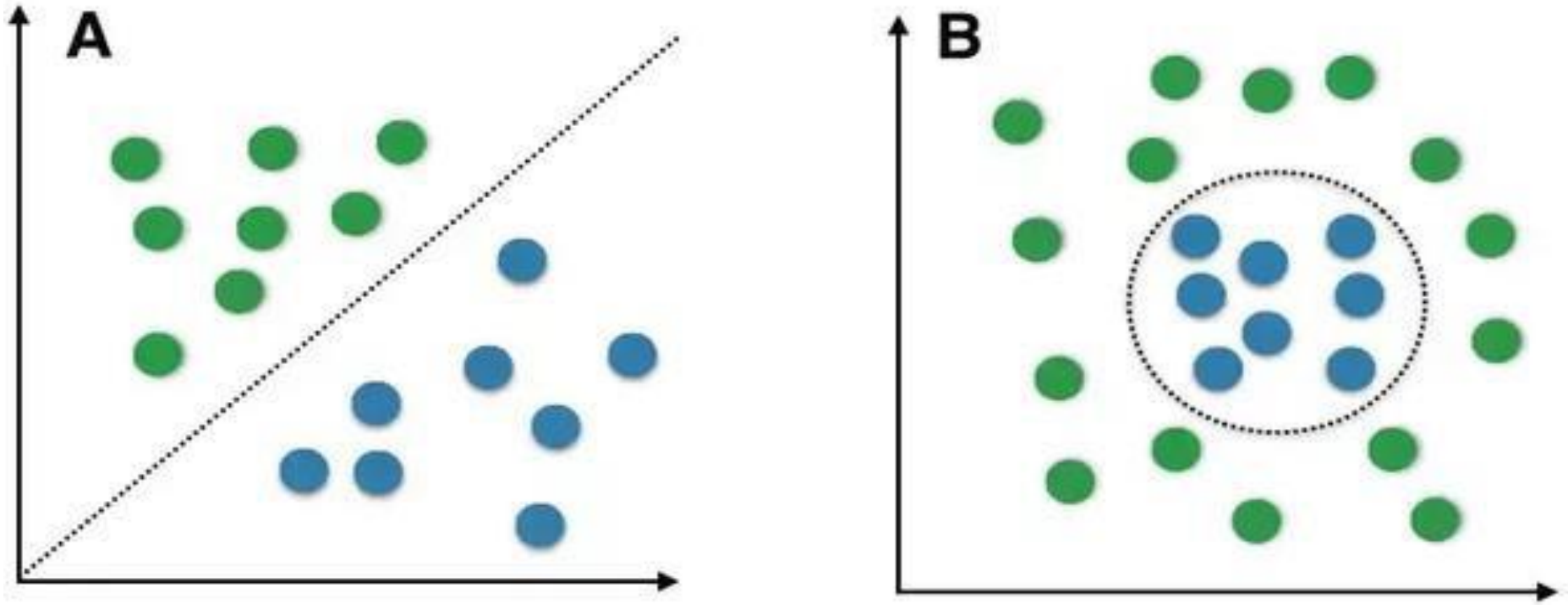
date	BombSta	ActuatorSt	OfBombSt	WaterT	AirCon	AirTep	HumiAirCOn	ExternalHum	SolarRad	OutSideTem	Output (Y)
1443658200	1	1	1	22.93	40	20.38	65.11	94.5	0	18.37	23.51
1443658800	1	1	1	22.92	40	20.38	65145	94.6	0	18.37	23.51
1443659400	1	1	1	22.92	40	20.38	65215	95.2	0	18.37	23.51
1443660000	1	1	1	22.93	40	20.38	65.3	94.9	0	17.87	23.51
1443660600	1	1	1	22.94	40	20.38	65.16	94.7	0	17.87	23.51
1443661200	1	1	1	22.93	40	20.38	65185	94.4	0	17.87	23.51
1443661800	1	1	1	22.91	40	20.38	65186	94.3	0	17.36	23.51
1443662400	1	1	1	22.91	40	20.38	64.91	94.7	0	17.36	23.51
1443663000	1	1	1	22.92	40	20.38	64.88	94.8	0	17.36	23.51
1443663600	1	1	1	22.93	40	20.38	64875	95.1	0	17.36	23.51
1443664200	1	1	1	22.93	40	20.38	64825	94.8	0	17.36	23.51
1443664800	1	1	1	22.91	40	20.38	64635	94.8	0	17.36	23.51
1443665400	1	1	1	22.9	40	20.38	64595	94	0	17.36	23.51
1443666000	1	1	1	22.9	40	20.38	64.52	92.8	0	16.84	23.51
1443666600	1	1	1	22.91	40	20.38	64425	92.3	0	16.84	23.51
1443667200	1	1	1	22.92	40	20.38	64.41	91.9	0	16.84	23.51
1443667800	1	1	1	22.9	40	20.38	64395	91.4	0	16.84	23.51
1443668400	1	1	1	22.88	40	20.38	64375	91	0	16.84	23.51
1443669000	1	1	1	22.87	40	20.38	64325	90.7	0	16.84	23.51
1443669600	1	1	1	22.88	40	20.38	64195	90.8	0	16.84	23.51
1443670200	1	1	1	22.88	40	20.38	64.04	91.3	0	16.34	23.51





https://machinelearningmastery.com/how-to-prepare-data-for-machine-learning/
https://medium.com/@jebaseelanravi96/machine-learning-iris-classification-33aa18a4a983

Supervised Learning: problem identification



Based on rule of Thumb by using simple methods first such as decision trees, SVM with linear kernel,.. etc. **If the results are not good,** we should try non-linear classifiers, such as SVM with Gaussian kernel, Random Forest, MLP, etc.

http://sebastianraschka.com/Articles/2014_naive_bayes_1.html
https://www.quora.com/In-machine-learning-how-can-we-determine-whether-a-problem-is-linear-nonlinear

Supervised Learning: data preparation

1 Select data:

- Understand the problem that we have.
- Identify the correct feature to use from your data (select all data or subset of all available featurs).
  - Feature extraction
  - Data correlation
- Sometimes we need to
  - Derive or simulate some data (Problems with small dataset) using algorithms such as:
    - Generate Synthetic Data: appending new artificial patterns based on data behaviour
  - Exclude data







Supervised Learning: data preparation
Dataset example (ZOO)

```
termite,0,0,1,0,0,0,0,0,0,1,0,0,6,0,0,0,6
toad,0,0,1,0,0,1,0,1,1,1,0,0,4,0,0,0,5
tortoise,0,0,1,0,0,0,0,0,1,1,0,0,4,1,0,1,3
tuatara,0,0,1,0,0,0,1,1,1,1,0,0,4,1,0,0,3
tuna,0,0,1,0,0,1,1,1,1,0,0,1,0,1,0,1,4
vampire,1,0,0,1,1,0,0,1,1,1,0,0,2,1,0,0,1
vole,1,0,0,1,0,0,0,1,1,1,0,0,4,1,0,0,1
vulture,0,1,1,0,1,0,1,0,1,1,0,0,2,1,0,1,2
wallaby,1,0,0,1,0,0,0,1,1,1,0,0,2,1,0,1,1
wasp,1,0,1,0,1,0,0,0,0,1,1,0,6,0,0,0,6
wolf,1,0,0,1,0,0,1,1,1,1,0,0,4,1,0,1,1
worm,0,0,1,0,0,0,0,0,0,1,0,0,0,0,0,0,7
wren,0,1,1,0,1,0,0,0,1,1,0,0,2,1,0,0,2
```

Original data before the pre-processing (101 patterns, 17 features and 6 classes)

https://archive.ics.uci.edu/ml/datasets/zoo

Supervised Learning: data preparation

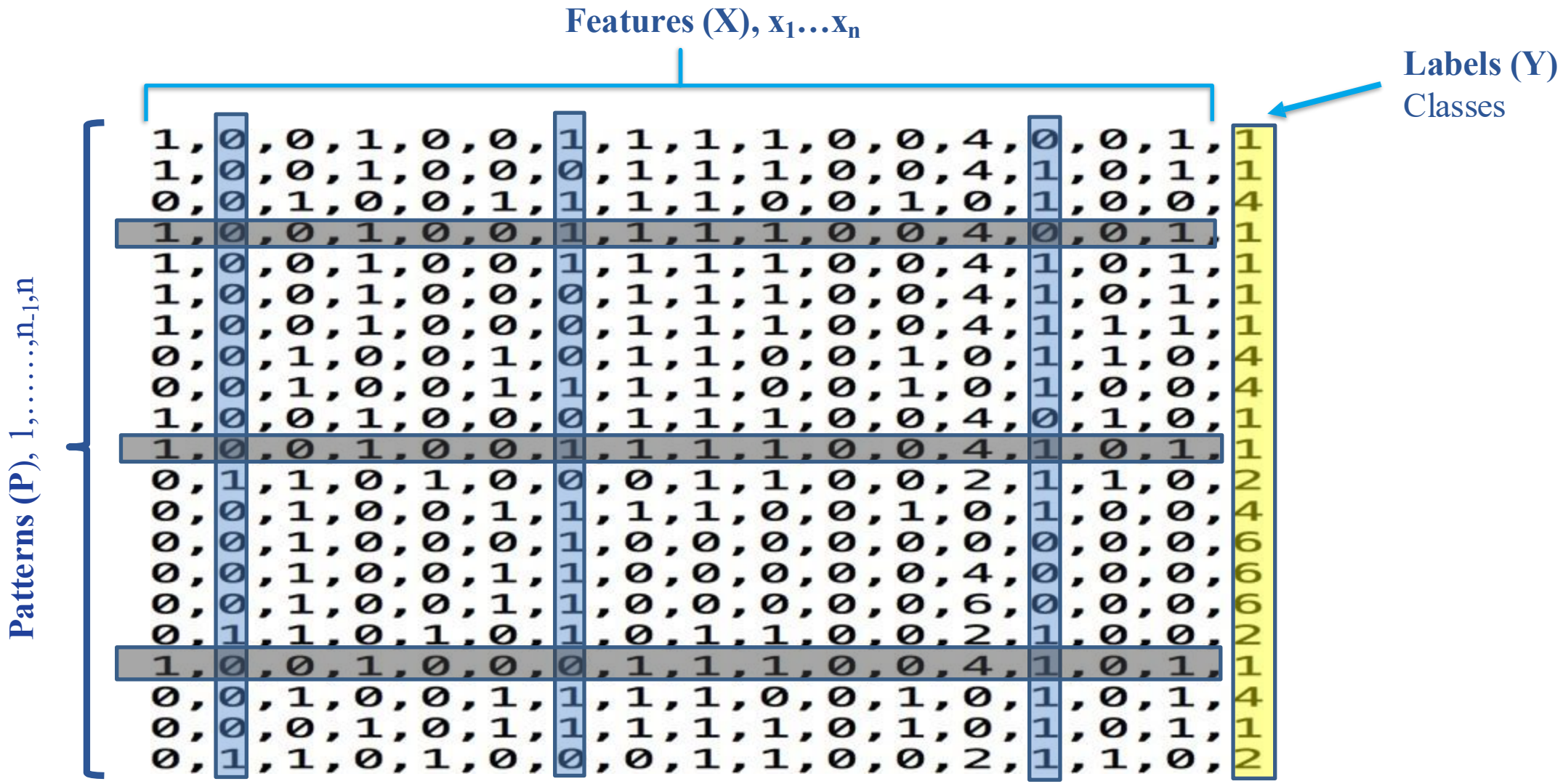
2. Preprocess Data

- Formatting :
  - Relational database data ⟹ Flat file
  - Proprietary file format ⟹ Text file, Relational database…etc
- Cleaning :
  - Remove outliers values, repeated patterns (Based on the algorithm we used and the problem we have), noise or fixing the missing data
- Sampling
  - Take a small representative sample of the selected data.

Reduce the running time and the required resources (computational and memory)

Exploring and prototyping the solutions much faster before considering the whole dataset

Supervised Learning: data preparation
Dataset example (ZOO)



Zoo dataset after the pre-processing phase (55 patterns, 16 features and 6 classes)

https://archive.ics.uci.edu/ml/datasets/zoo

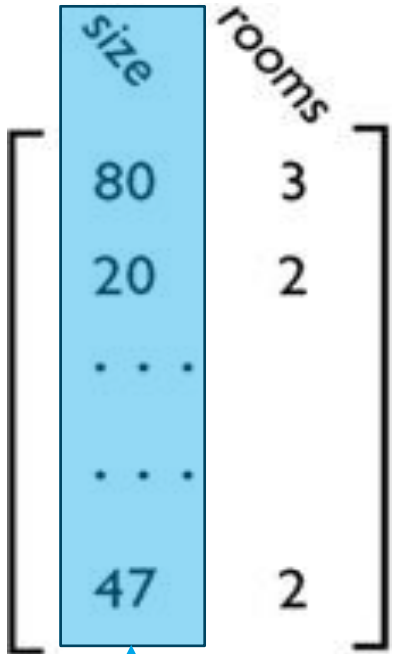
Supervised Learning: data preparation

3. Features Normalization

- Feature normalization is a pre-processing step used to normalize the range of the features.
- It is important when the features have very different scales.
  - ✓ For example, if the values of ***feature 1*** are ∈ [0, 1] but the values of ***feature 2*** are ∈ [120, 190], then normalizing the features is important.
- **Motivation**:
  - ✓ Suppose that some ML algorithm computes the Euclidean distance between two points. If one of the features has a **broad range of values**, the distance will be **governed by** this particular feature. Therefore, the range of all features **should be normalized** so that each **feature contributes approximately** proportionately to the final distance.

Supervised Learning: data preparation

min-max Features scaling



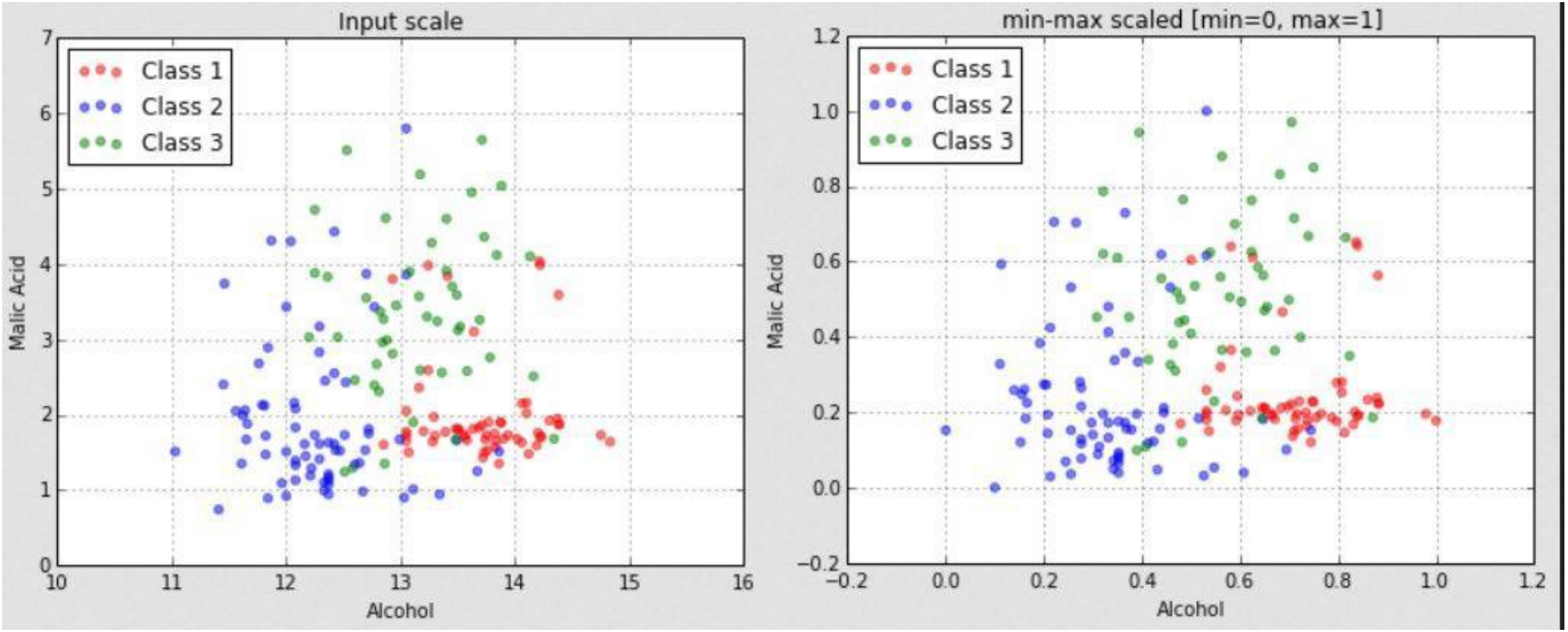
$$v'_j = \frac{v_j - \min(v_j)}{\max(v_j) - \min(v_j)}$$

- v_j is a column (corresponding to feature$j$) from the data matrix X.
- v'_j are the normalized values of feature j. These values will be $\in[0,1]$

Supervised Learning: data preparation

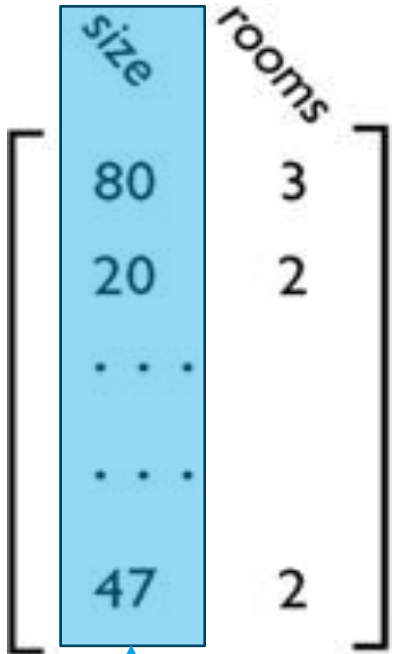
Before scaling

After scaling



Supervised Learning: data preparation

Features Standardization



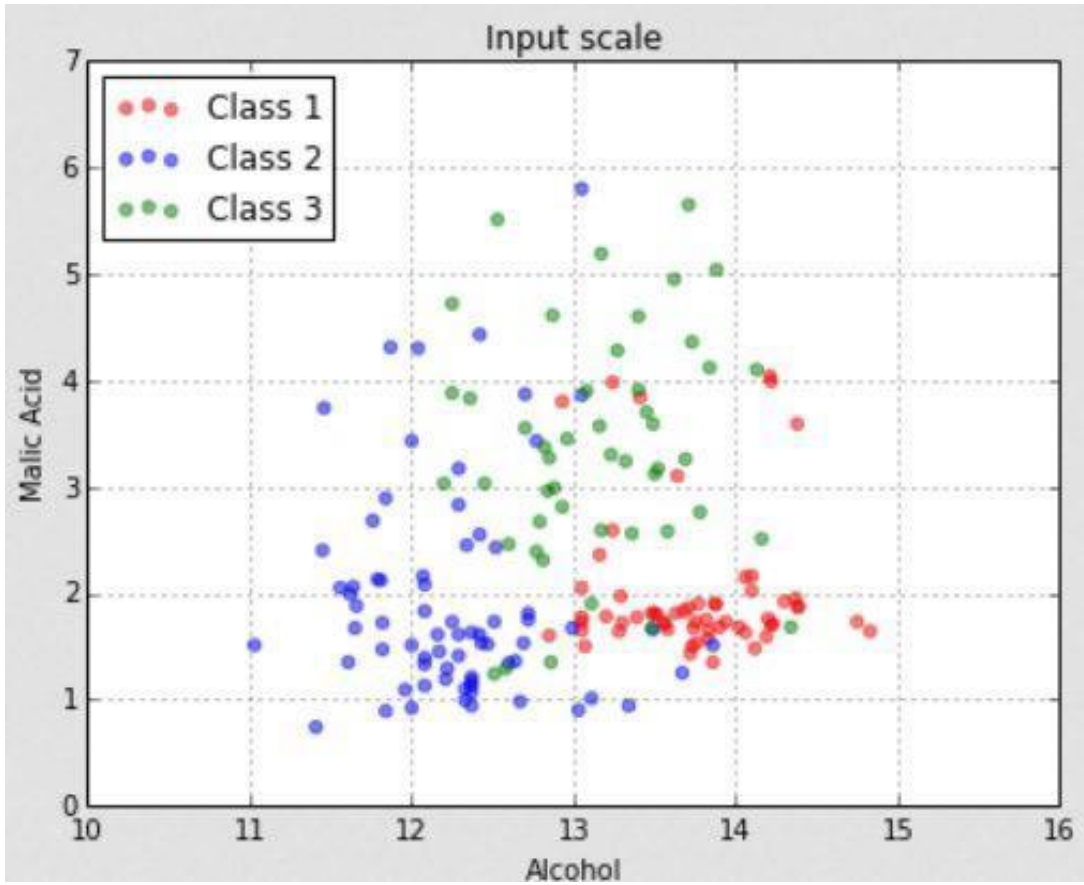
$$v'_j = \frac{v_j - \text{mean}(v_j)}{\text{stdev}(v_j)}$$

- v_j is a column (corresponding to feature j) from the data matrix X.
- v'_j are the normalized values of feature j. These values will be $\in[0,1]$
- To normalize, we just subtract the mean and divide by the standard deviation.

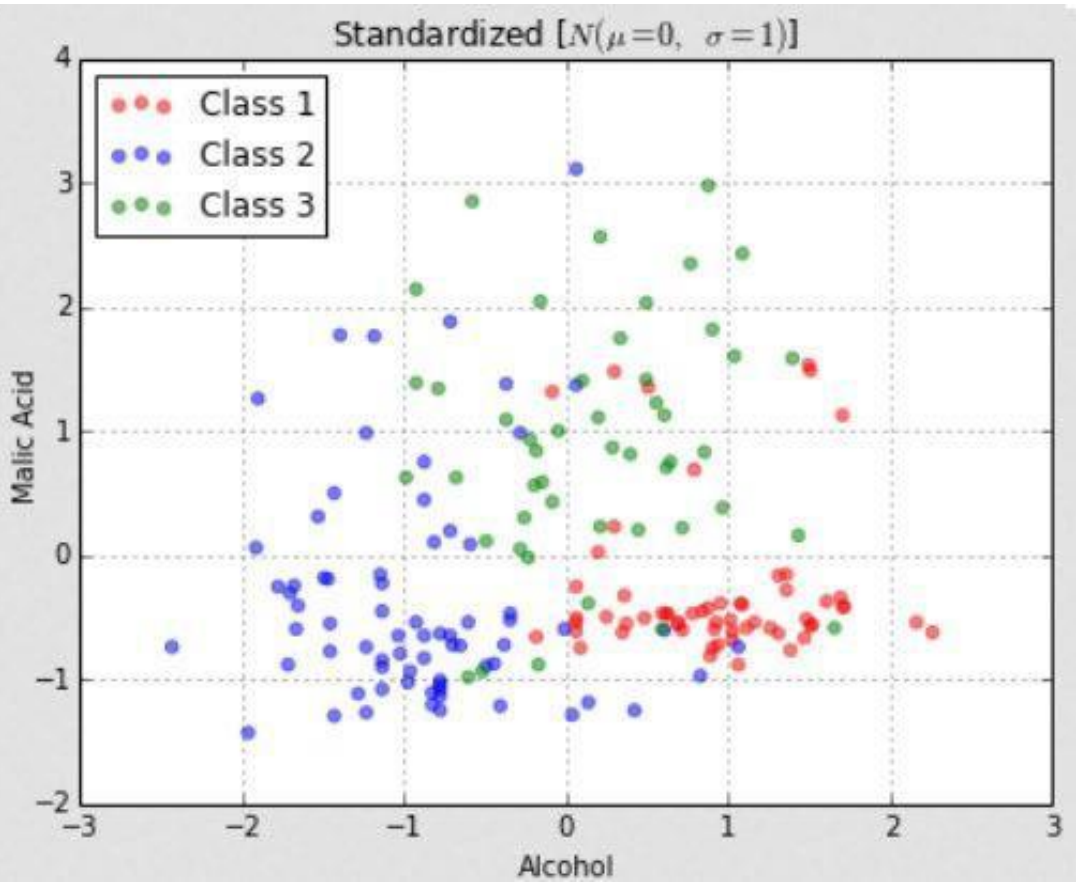
Supervised Learning: data preparation

Features Standardization

Before scaling



After scaling



Supervised Learning: data preparation

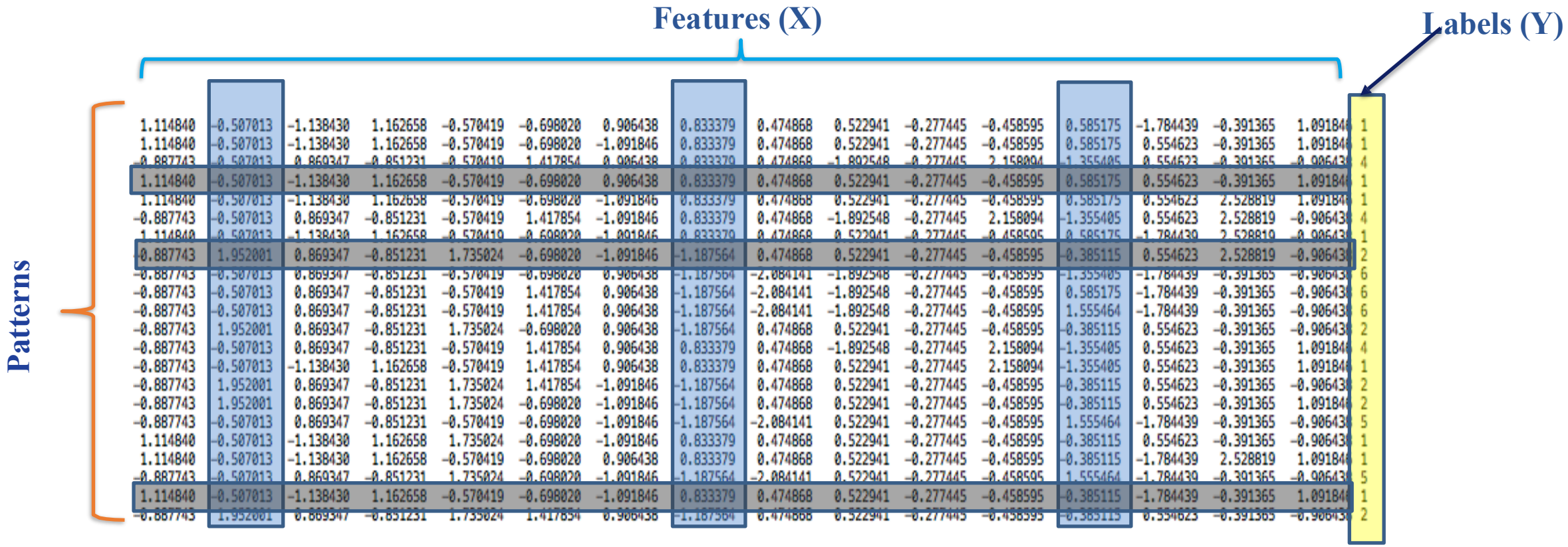
"NOTE: do not rescale or standardize the output (target variable)."

Transform Data

- Decomposition: complex feature → Split → constituent parts (date that may have day and time)
- Aggregation: merge two or more features to gather (air condition status on/off and its temperature)

1. Famili, A., et al. "Data preprocessing and intelligent data analysis." *Intelligent data analysis* 1.1 (1997): 3-23.
2. https://www.quora.com/What-is-the-meaning-of-min-max-normalization
3. https://stats.stackexchange.com/questions/59502/why-normalization-standardization-values-should-be-smaller-than-1
4. Mohamad, Ismail Bin, and Dauda Usman. "Standardization and its effects on K-means clustering algorithm." *Research Journal of Applied Sciences, Engineering and Technology*6.17 (2013): 3299-3303.

Supervised Learning: data preparation
Dataset example (ZOO)



Each row is an example (Patterns) Learning based on examples.

Dataset ZOO with:

- 55 Patterns
- 16 Features
- 6 Classes

https://archive.ics.uci.edu/ml/datasets/zoo

Alawadi, Sadi, Manuel Fernández Delgado, and David Mera Pérez. "Machine learning algorithms for pattern visualization in classification tasks and for automatic indoor temperature prediction." PhD diss., Universidade de Santiago de Compostela, 2018.

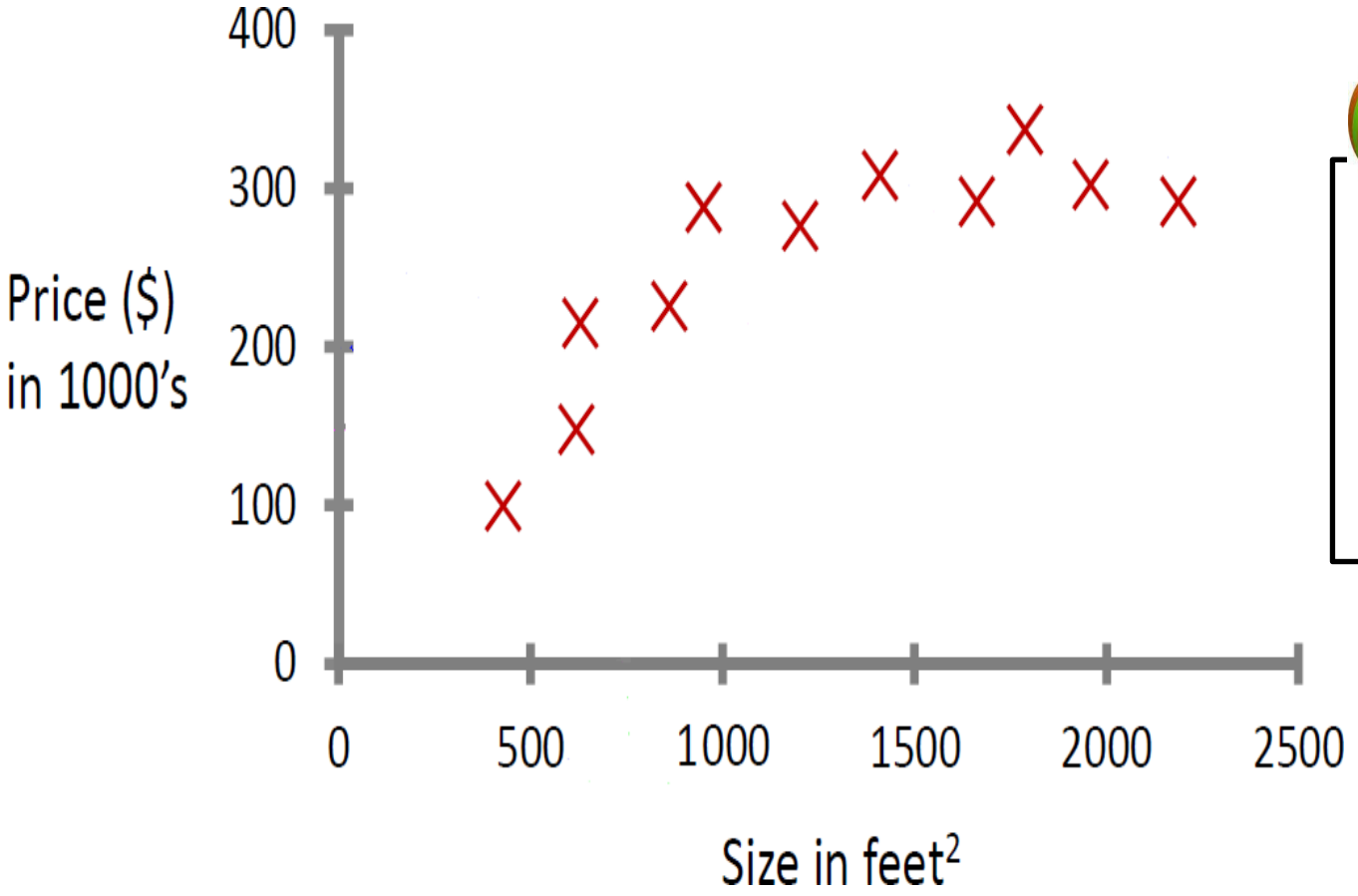
Supervised Learning: data preparation

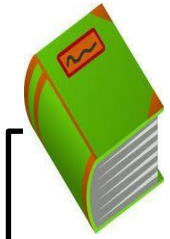
- Generate data partitions
    - Training (for algorithm learning)
    - Validation (for hyperparameter tuning)
    - Testing (for quality evaluation)
- Model tuning parameters and training.
- Export the trained model for deployment of ML algorithm in real life.

Introduction to Supervised ML Regression

What is Regression?

Housing price prediction (regression)





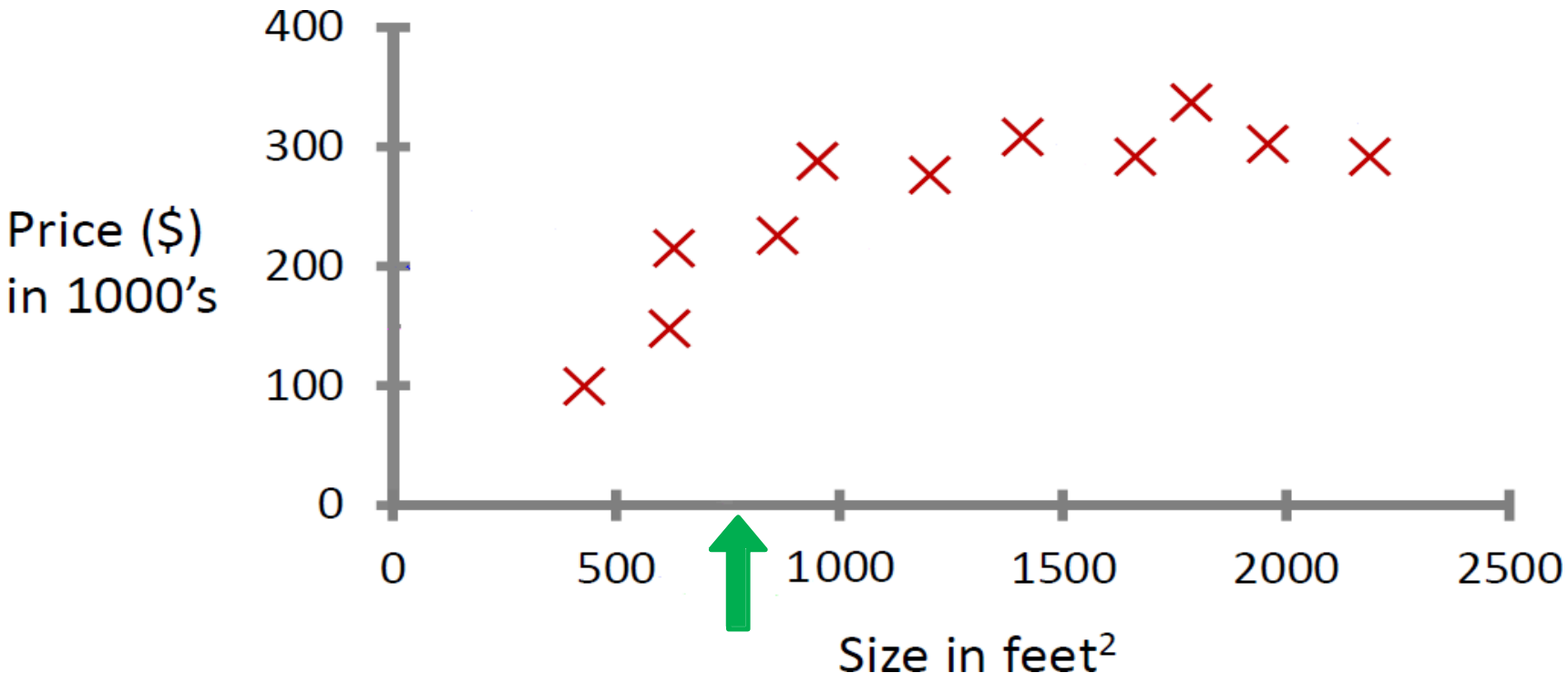
Definition: Regression

is a form of predictive modelling technique which investigates the relationship between a dependent variable[Output] and several independent variables[Inputs]

What is Regression?

Housing price prediction (regression)

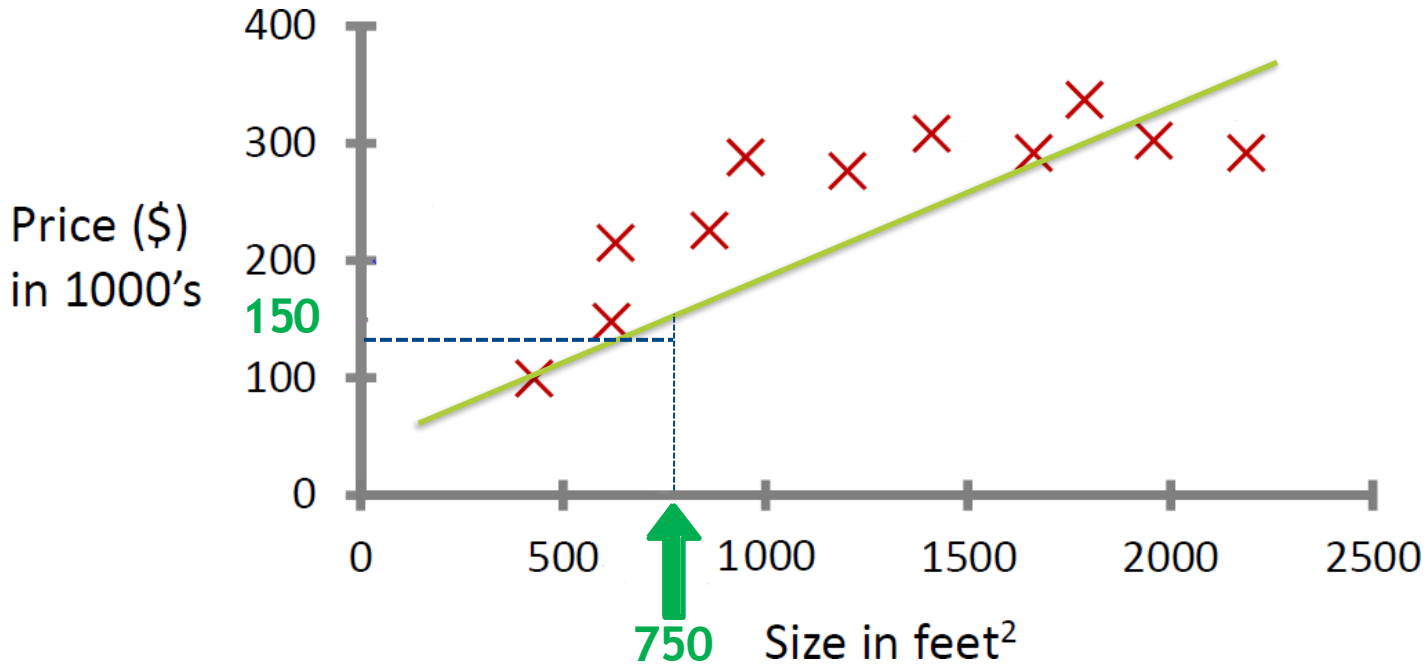
- Suppose that you want to sell a house of **size 750 feet²** and want to know how much you can get for this house, i.e. **predict its price**.
- How can a learning algorithm help you?



What is Regression?

Housing price prediction (regression)

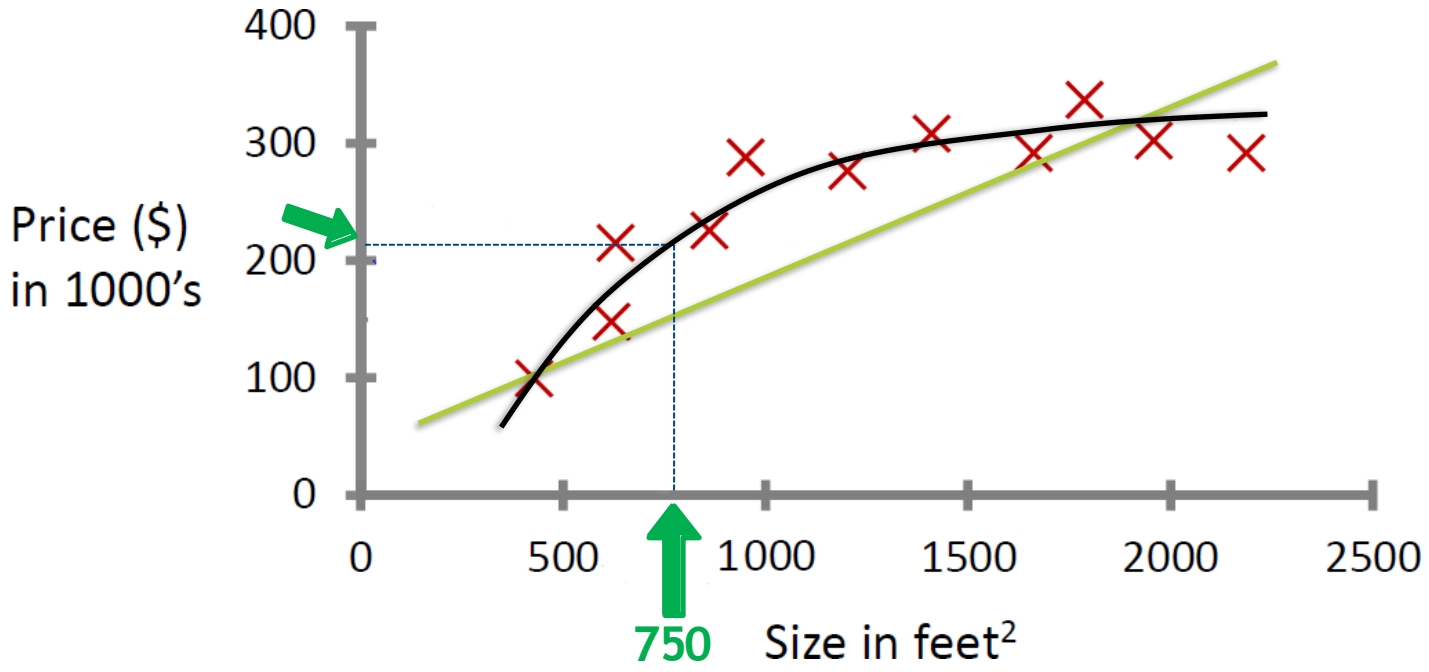
- you can fit a straight line to the data and predict the price of the house.



What is Regression?

Housing price prediction (regression)

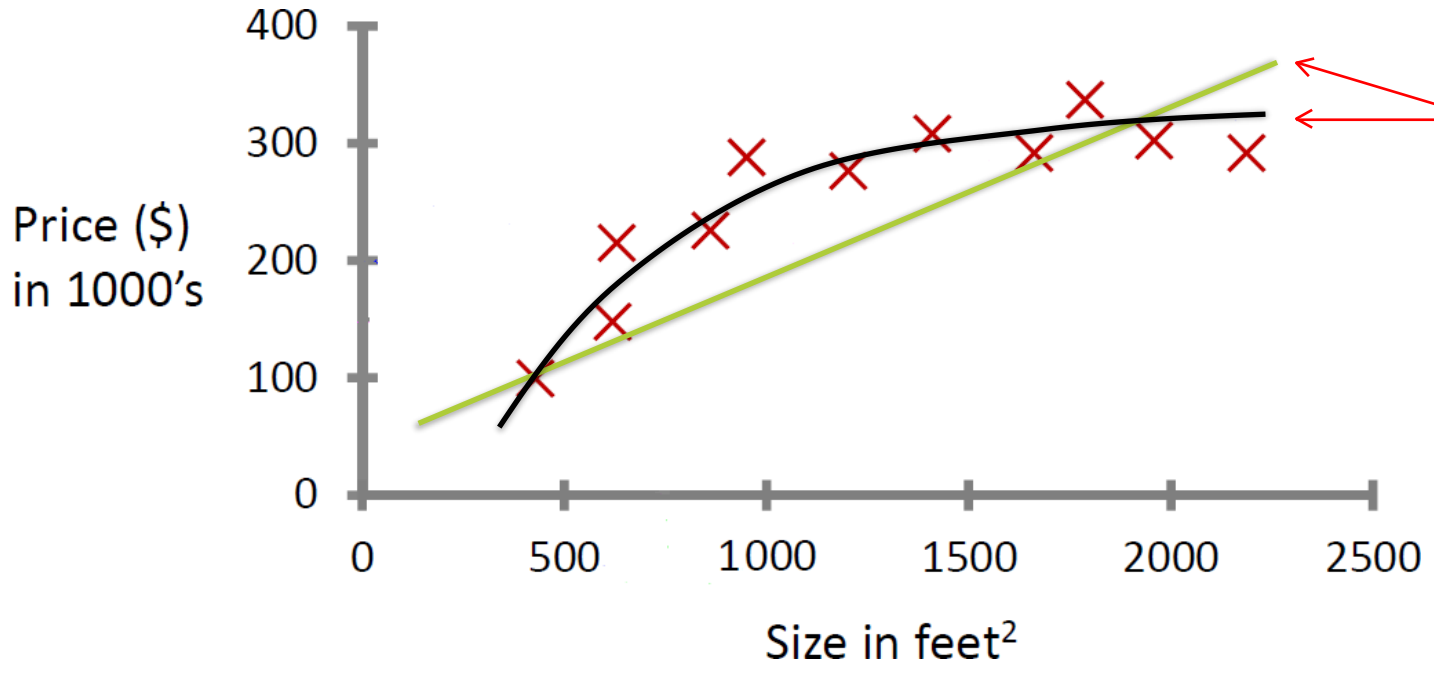
- you can fit a straight line to the data and predict the price of the house.
- or maybe it's better to fit a quadratic function (2nd-order polynomial).



What is Regression?

Housing price prediction (regression)

- you can fit a straight line to the data and predict the price of the house.
- or maybe it's better to fit a quadratic function (2nd-order polynomial).

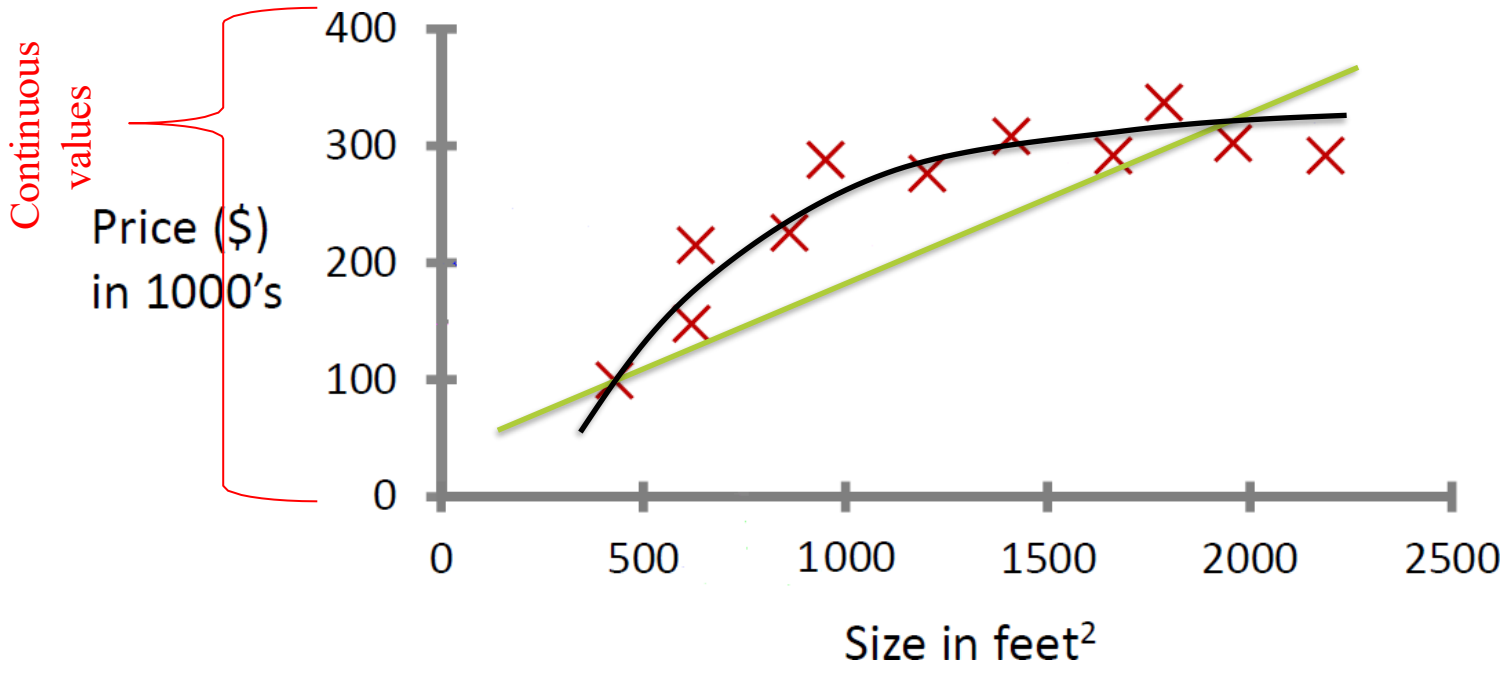


How to decide which model to choose for this dataset ?

What is Regression?

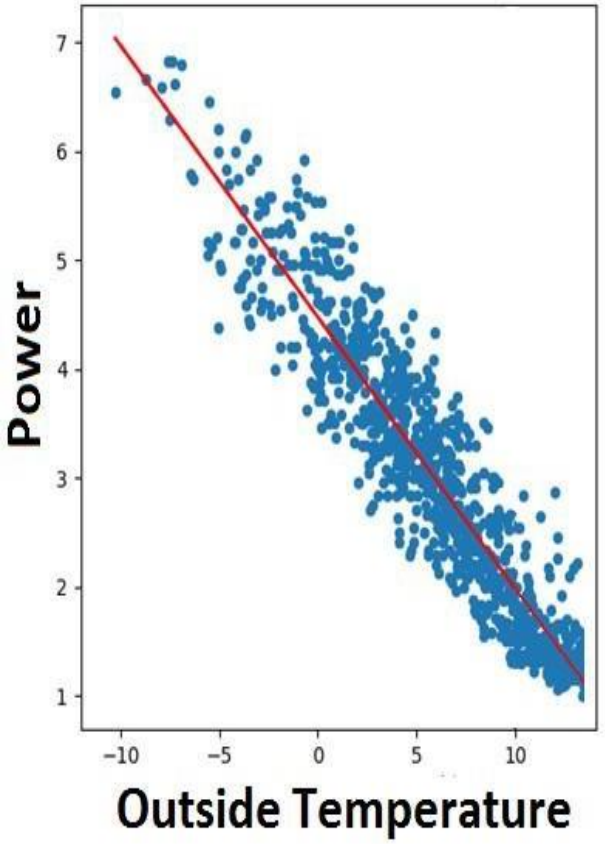
Housing price prediction (regression)

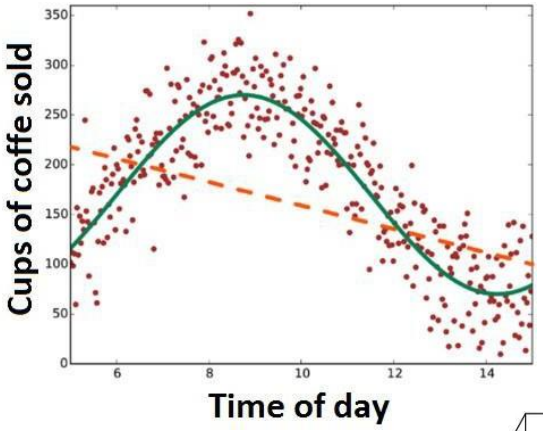
- This is an example of a **supervised learning** algorithm:
  - The right answers (here, the prices) are given in the training dataset.
- More specifically, this example was a **regression problem**:
  - Predicting a continuous valued output (price).

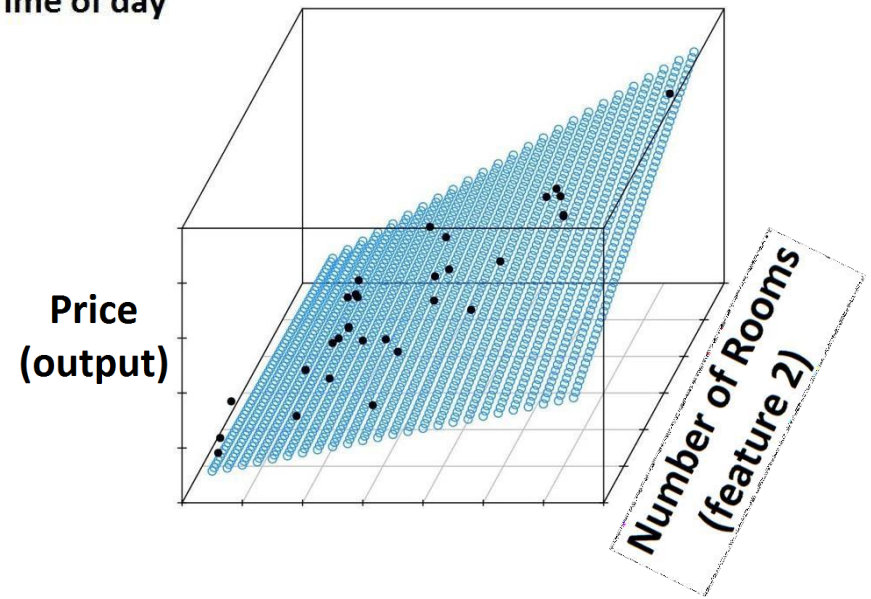


What is Regression?

More regression examples







Regression Algorithms

- **Linear regression:** linear model (lm).
- **Least squares:** non-negative least squares regression (nnls).
- **Least absolute shrinkage and selection operator** (lasso).
- **Least absolute shrinkage and selection operator** (lasso).
- **Partial least squares:**
  - Sparse partial least squares regression (spls).
  - Simpls method (simpls).
- **Generalized linear regression:**
  - Generalized linear model (glm).
  - Penalized linear model (penalized).
  - Lasso and elastic-net regularized GLM (glmnet).
- **Ridge regression with forward, backward and sparse input selection (foba).**
- **Neural networks**
  - Multi-layer perceptron with one hidden layer and weight decay (mlpWeightDecay).
  - Model averaged neural network (avNNET).
  - Generalized regression neural network (grnn).
  - Extreme learning machine (elm).
  - ELM neural network with Gaussian kernel (elm-kernel).
  - Deep belief network with one hidden layer (dnn).
- **Support vector regression with Gaussian kernel (svmRadial).**
- **Boosting ensemble of generalised linear models (randomGLM).**

Fernández-Delgado, M., M. S. Sirsat, E. Cernadas, S. Alawadi, S. Barro, and M. Febrero-Bande. "An extensive experimental survey of regression methods." *Neural Networks*(2018).

Regression Algorithms

- **Regression trees:**
  - Recursive partitioning and regression tree (rpart).
  - Model tree (m5).
- **Bagging ensembles:**
  - Bagging of conditional inference regression trees (bag).
  - Bagged MARS (bagEarth).
  - Bagged CART (treebag).
- **Random forests:**
  - Forest of random regression trees (rf).
  - Quantile regression forest (qrf).
  - Ensemble of extremely randomized regression trees (extraTrees) .
- **Gradient boosting machine:**
  - GBM with linear base regressors (BstLM).
  - GBM with smoothing splines (bstSm).
  - GBM with regression trees (bstTree).
  - Generalized boosting regression model (gbm).
- **Gaussian processes**
  - Gaussian process regression with linear kernel (GausprLinear).
  - Gaussian process regression with polynomial kernel (GausprPoly).
- **Quantile regression with LASSO penalty (rqlasso).**
- **Other methods:**
  - multivariate adaptive regression spline (earth).
  - projection pursuit regression (ppr).
- **Bayesian models:**
  - Bayesian GLM (bayesglm).
  - Bayesian regularized neural network (brnn).
  - Bayesian additive regression tree (bartMachine).
- **Independent component regression (icr).**
- **Generalized additive model with splines (gam).**
- **Prototype models: M5 rule-based model with corrections based on nearest neighbors (cubist).**

Fernández-Delgado, M., M. S. Sirsat, E. Cernadas, S. Alawadi, S. Barro, and M. Febrero-Bande. "An extensive experimental survey of regression methods." *Neural Networks*(2018).

Regressors tunable parameters and packages

Regressor	Hyperp. (values)	Packages	Regressor	Hyperp. (values)	Packages
lm	–	MASS	randomGLM	maxInterationOrder(3)	randomGLM
glm	–	gbm, plyr	bag	–	caret
penalized	$\lambda_1(5)$, $\lambda_2(4)$	penalized	BstLm	mstop(10)	bst, plyr
glmnet	$\alpha(7)$, $\lambda(3)$	glmnet	bstSm	mstop(10)	bst, plyr
nnls	–	nnls	bstTree	mstop(4), maxdepth(5)	bst, plyr
spls	$K(3)$,η, $\kappa(7)$	spls	gbm	n.trees(5) interaction.depth(5)	gbm, plyr
simpls	ncomp(10)	pls	rf	mtry(10)	randomForest
lasso	–	elasticnet	qrf	mtry(2)	quantregForest
foba	$k(2)$,$\lambda(10)$	foba	extraTrees	mtry(10) numRandomCuts(2)	extraTrees
MLP	n.hidden(20)	nnet	cubist	committees(3) neighbors(3)	Cubist
avNNet	size(7) decay(3)	nnet	bayesglm	–	arm
grnn	spread(14)	Matlab	brnn	neurons(15)	brnn
elm	nhid(20) actfun(4)	elmNN	bartMachine	$K(3)$,$\alpha(3)$	bartMachine
dnn	layer1(10) layer2(10) layer3(10)	deepnet	icr	ncomp(10)	fastICA
elm-kernel	$\sigma(25)$,$C(25)$	Matlab	gam	select(2)	gam
svmRadial	$\sigma(5)$,$C(4)$	kernlab	gausssprLinear	–	kernlab
rpart	complexity(10)	rpart	gausssprPoly	degree(5),scale(3)	kernlab
M5	pruned(2) smoothed(2) rules(2)	RWeka	rqlasso	$\lambda(10)$	rqPen
bagEarth	nprune(10)	caret	earth	nprune(15)	earth
treebag	–	ipred, plyr e1071	ppr	nterms(10)	stats

Alawadi, Sadi, Manuel Fernández Delgado, and David Mera Pérez. "Machine learning algorithms for pattern visualization in classification tasks and for automatic indoor temperature prediction." PhD diss., Universidade de Santiago de Compostela, 2018.

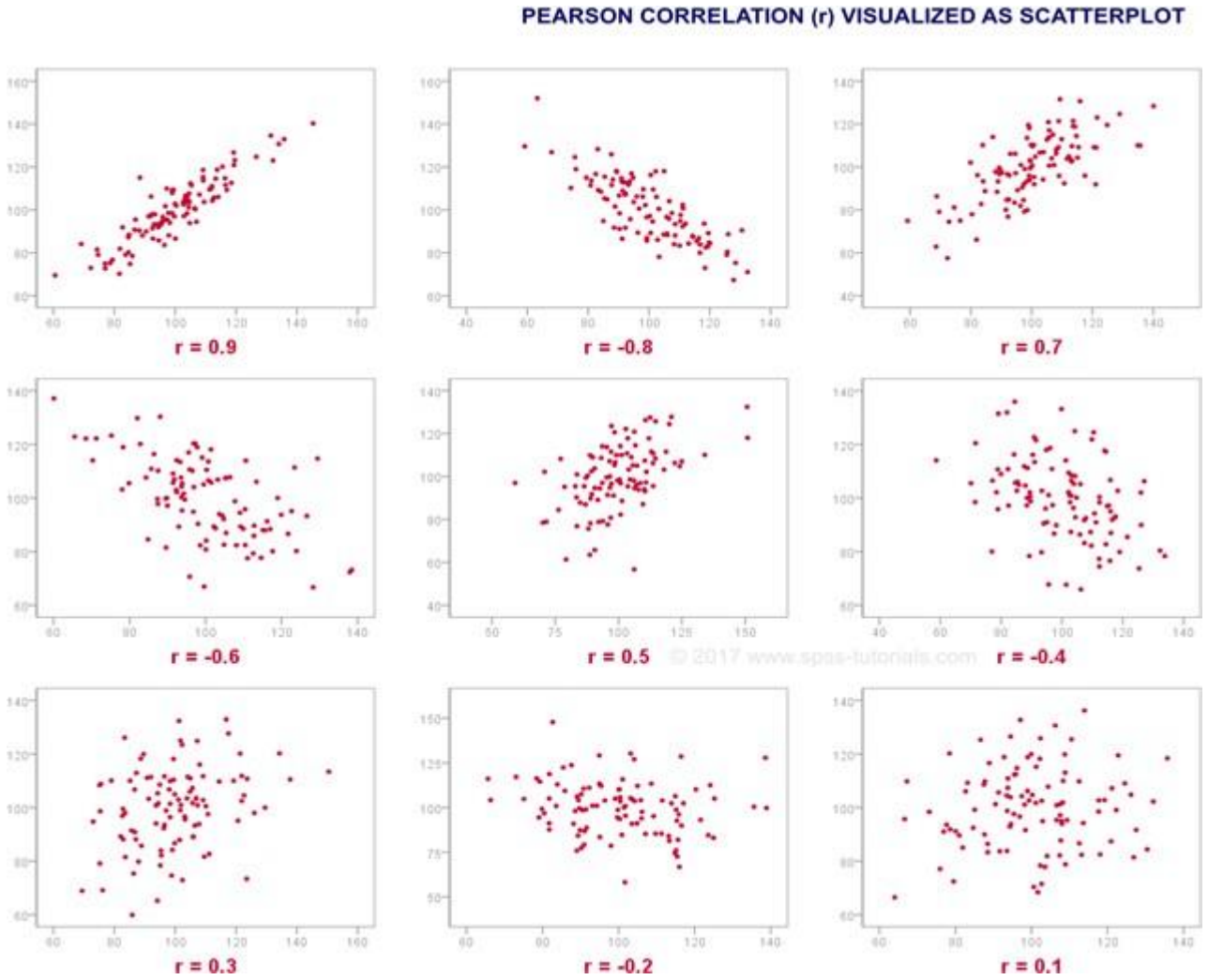
Regressor performance evaluation metrics

Pearson correlation coefficient (R-coefficient)

$$\rho(\hat{Y}, Y) = \frac{1}{N-1}\sum_{i=1}^{N}\left(\frac{\hat{Y} - \mu_{\hat{Y}}}{\sigma_{\hat{Y}}}\right)\left(\frac{Y_i - \mu_Y}{\sigma_Y}\right)$$

Mean Squared Error

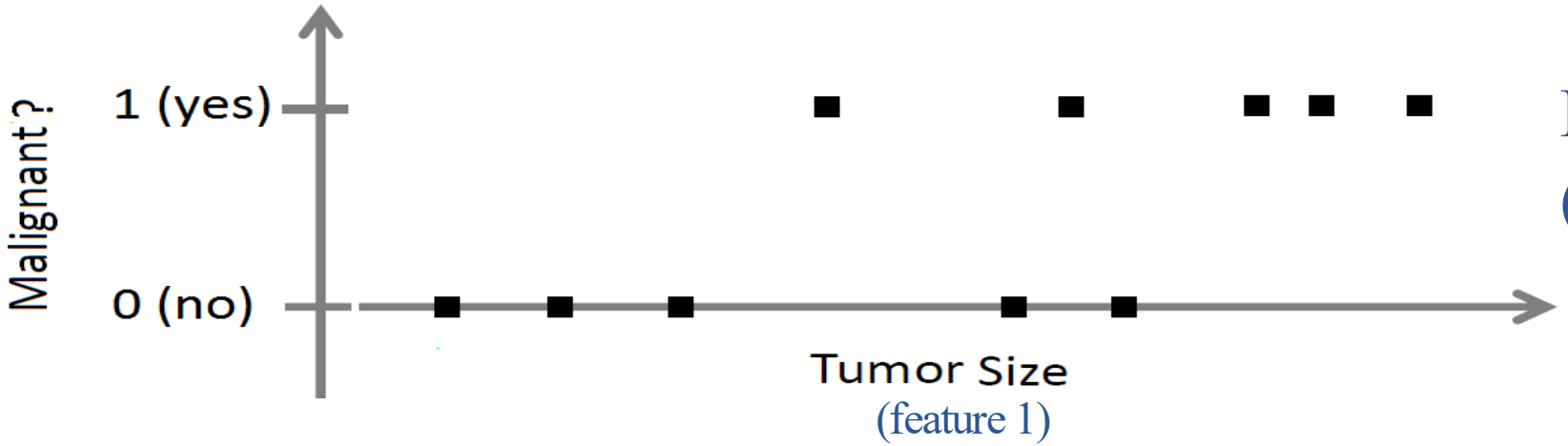
$$MSE = \frac{1}{N-1}\sum_{i=1}^{N}\left(\hat{Y}_i - Y\right)^2$$



https://statistics.laerd.com/statistical-guides/pearson-correlation-coefficient-statistical-guide.php

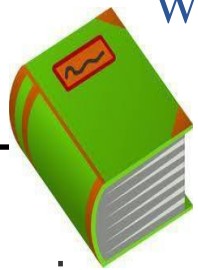
Introduction to Supervised ML Classification

What is Classification?



Breast cancer malignant/benign (classification)

We only have discrete output values (in this example: 1 or 0)



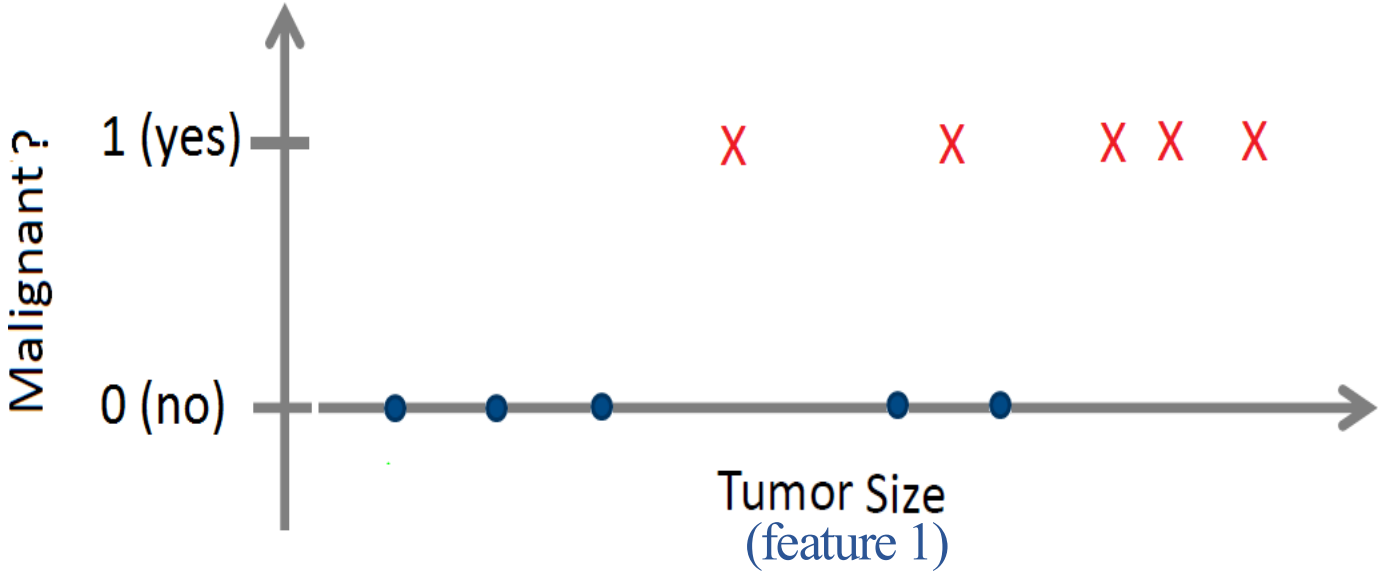
Definition:Classification

is a predictive modeling problem where the class label is anticipated for a specific example of input data. For example, in determining handwriting characters, identifying spam, and so on, the classification requires training data with a large number of datasets of input and output.

What is Classification?

Breast cancer malignant/benign (classification)

We don't need the second axis if we simply visualize the classes with different colors or shapes.

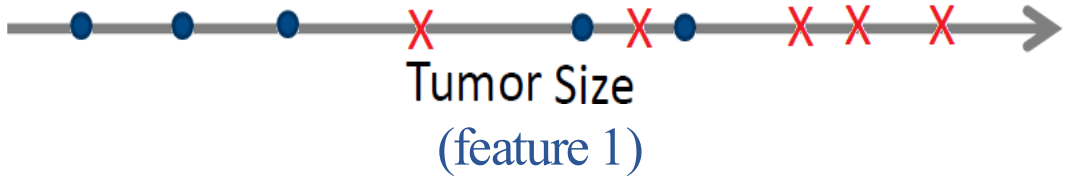


Two axis

Malignant ?

X = 1 (yes)

● = 0 (no)

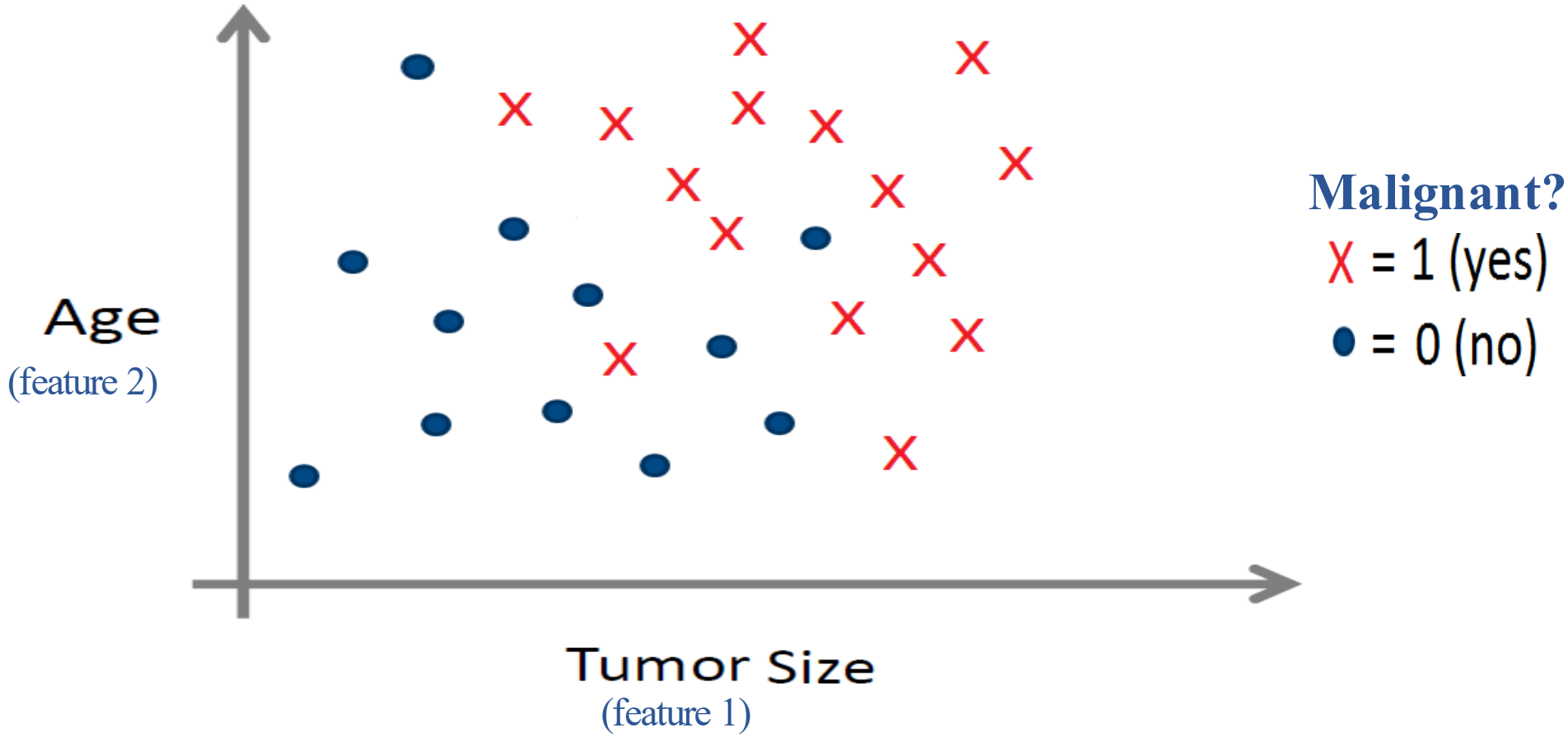


Only one axis

What is Classification?

Breast cancer malignant/benign (classification)

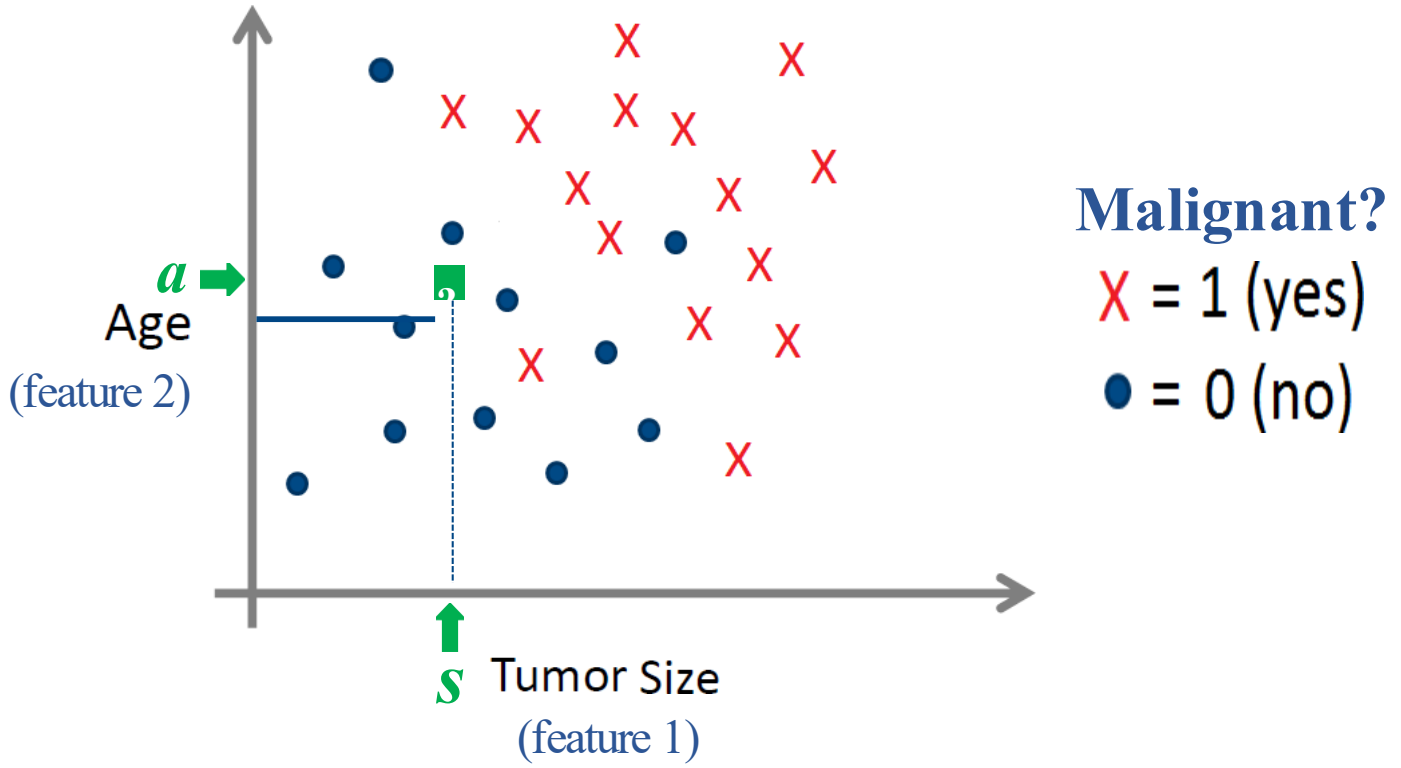
- The patient's data can be characterised by more than one feature
  - e.g. Tumor size and Age …



What is Classification?

Breast cancer malignant/benign (classification)

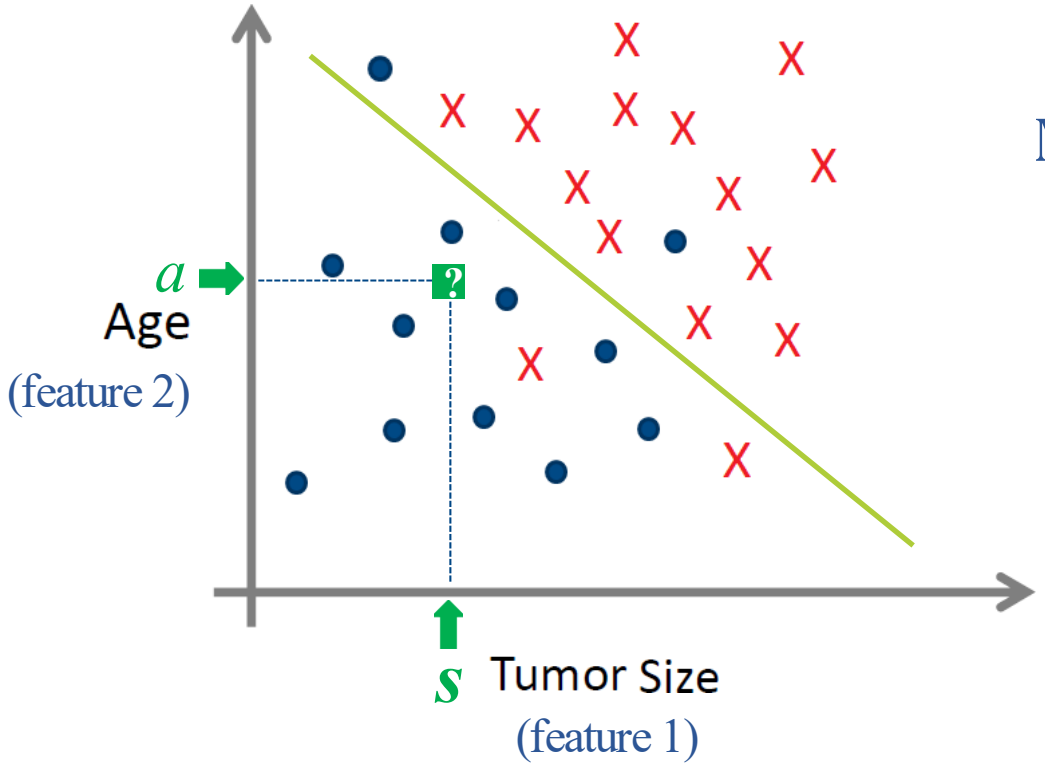
– Suppose that you get a new patient who has some tumor size ***s*** and age *a*, and you want to predict if it is malignant or benign. How can a learning algorithm help you?



What is Classification?

Breast cancer malignant/benign (classification)

– You can fit a linear model to the training data, and then predict the class of the new patient.



Malignant?

X = 1 (yes)

● = 0 (no)

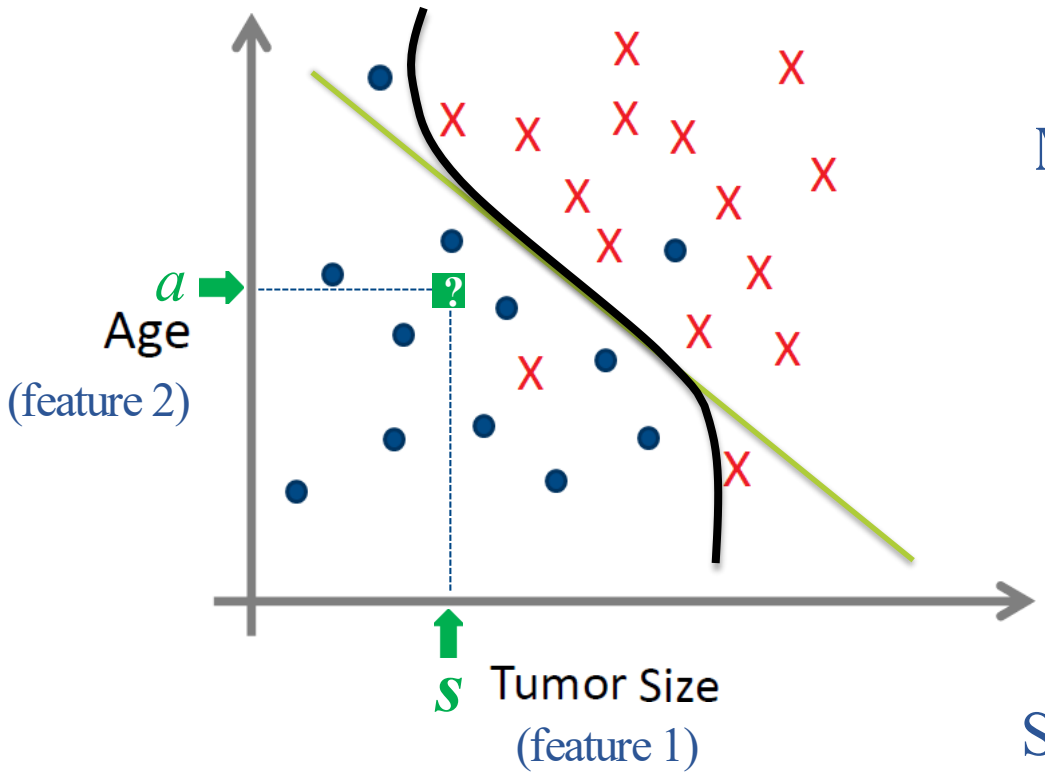
So for patient ***(s, a),*** we would predict the class "***benign***".

What is Classification?

Breast cancer malignant/benign (classification)

- you can fit a linear model to the training data, then predict the class of the new patient
- or you can fit a non-linear model to the training data

…

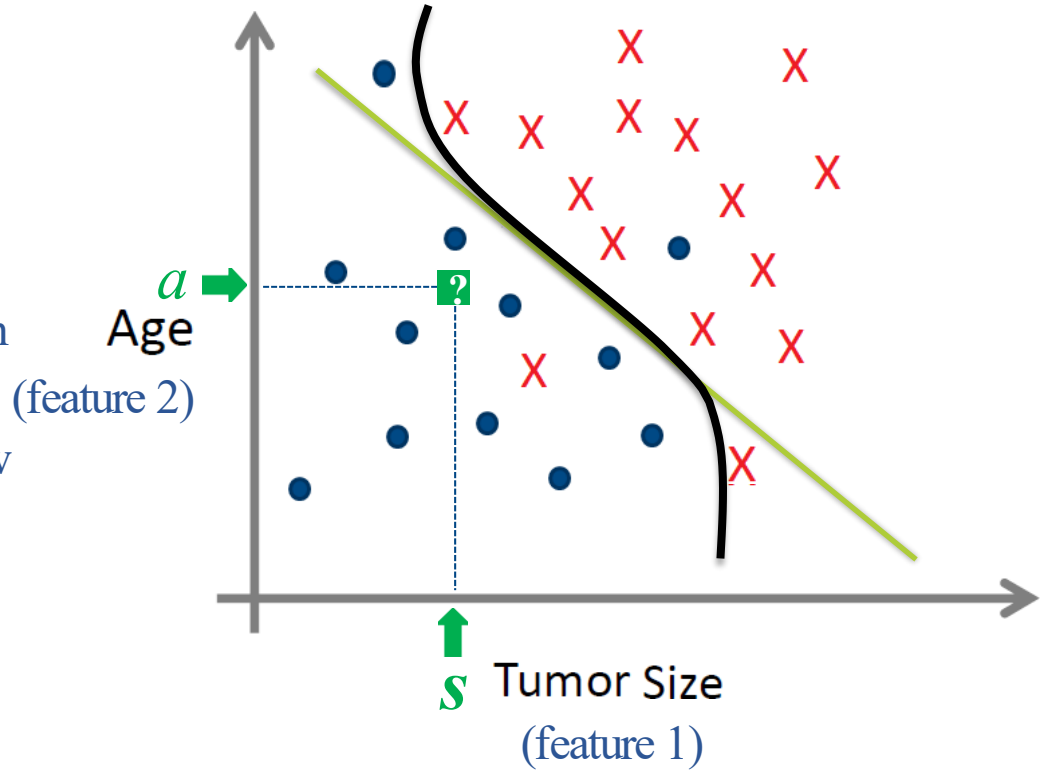


So for patient ***(s, a)***, we would predict the class "***benign***".

What is Classification?

Breast cancer malignant/benign (classification)

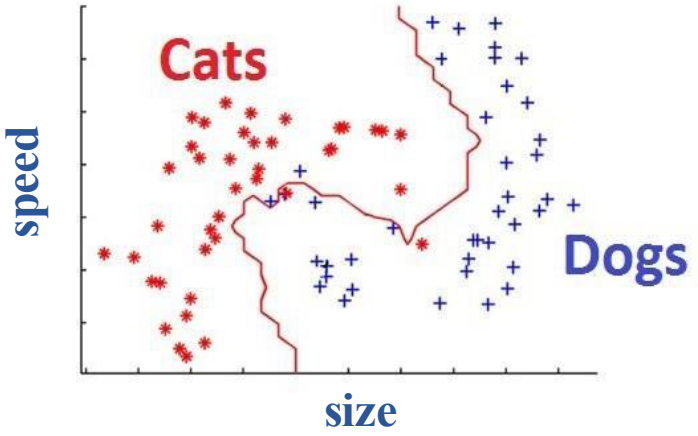
- ✓ Again, this is an example of a supervised learning algorithm:
  - ➢ The right answers (here, the classes malignant/benign) are given with the training dataset.
  - ➢ i.e. for each patient (data-point) in the training dataset, we know if he is has a malignant or benign cancer.
- ✓ However, this example was a classification problem:
  - ➢ Predicting a discrete-valued output (malignant/benign).

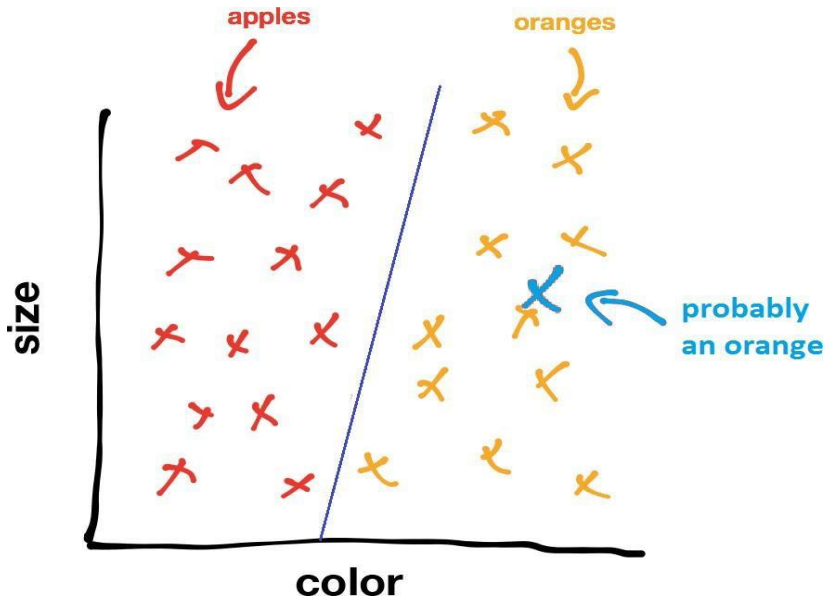


Note: In this example, we had two features (age and size), but we will see ML algorithms that can easily deal with many features …

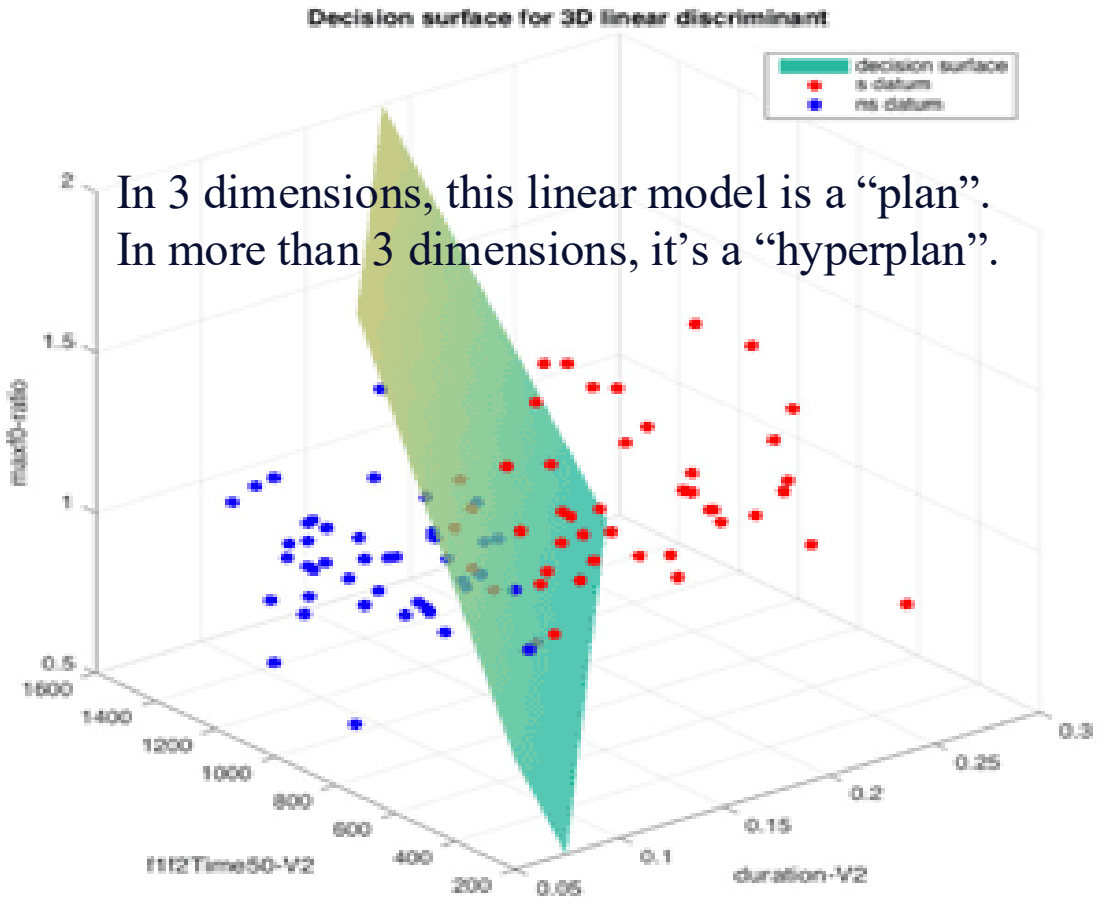
What is Classifiaction?

More classification examples





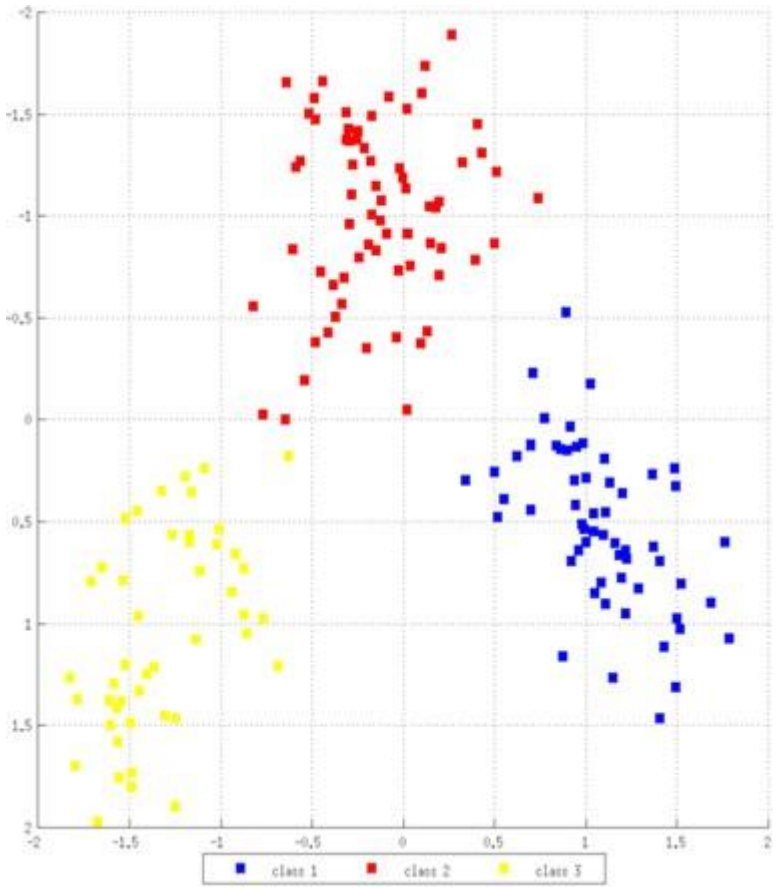
Classification is about learning decision boundaries and predicting the "class" of new data-point.

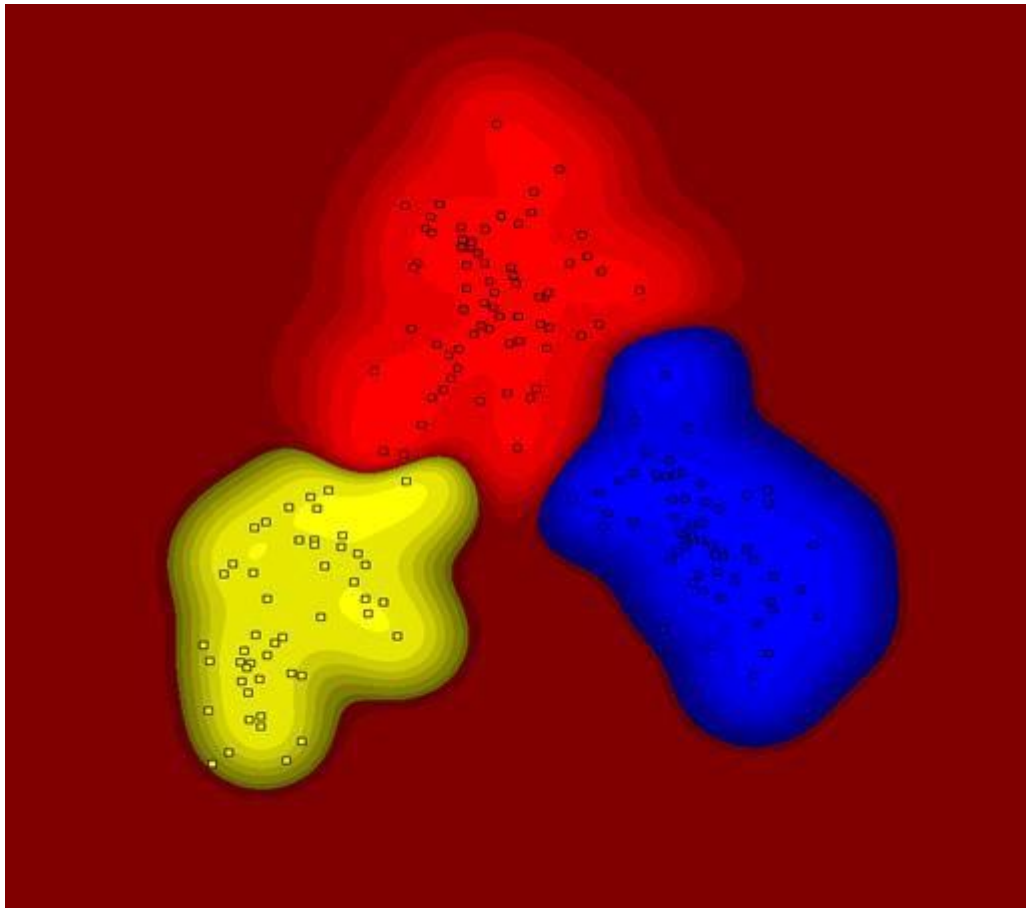


In 3 dimensions, this linear model is a "plan".
In more than 3 dimensions, it's a "hyperplan".

What is Classifiaction?

More classification examples



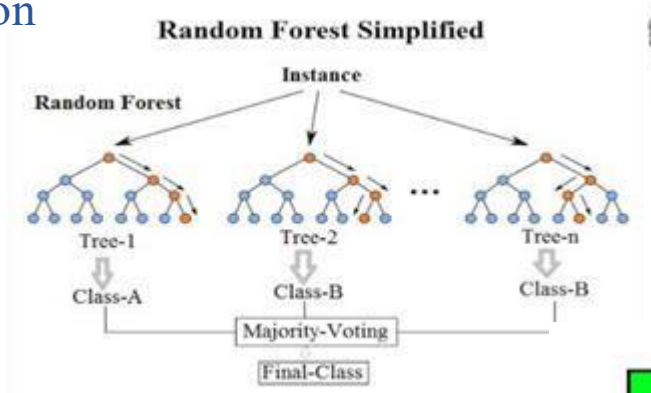


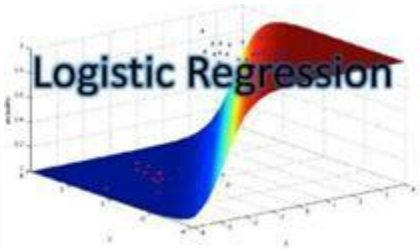
Classification maps for dataset **wine** (3 classes), small squares are examples (patterns)

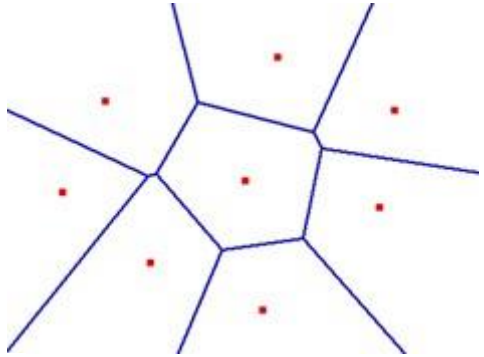
Classification algorithms

Popular classifiers:

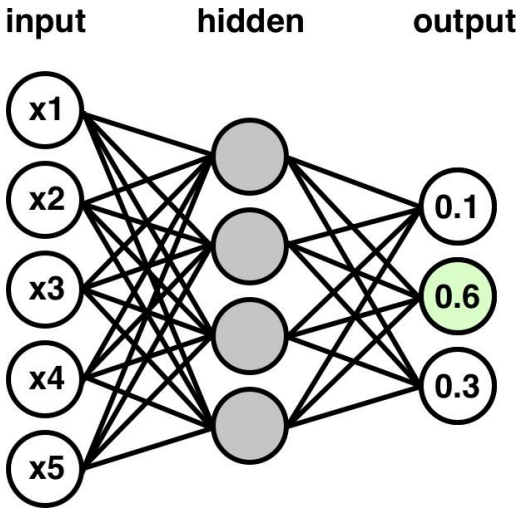
- Logistic Regression/Classification
- K-Nearest Neighbours
- Support Vector Machine
- Naive Bayes
- Decision Tree
- Random Forest
- Neural Networks (ANN)
- ….etc

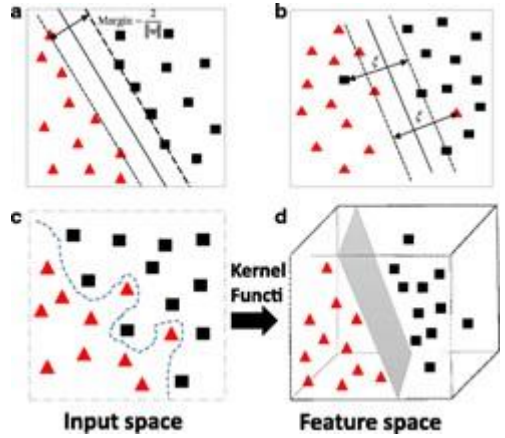












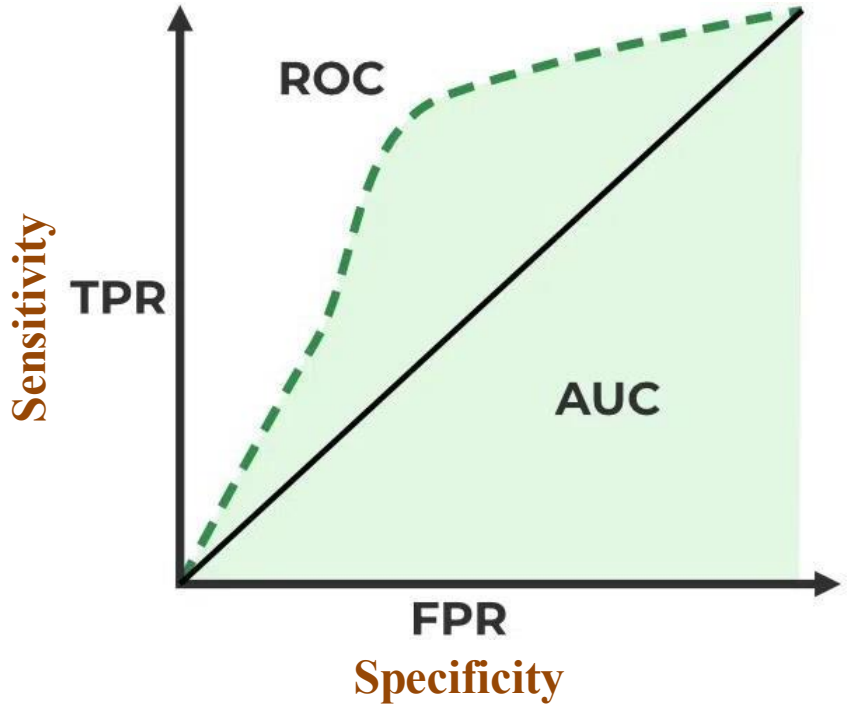
Fernández-Delgado, Manuel, Eva Cernadas, Senén Barro, and Dinani Amorim. "Do we need hundreds of classifiers to solve real world classification problems?." The Journal of Machine Learning Research 15, no. 1 (2014): 3133-3181.

Classifiers performance evaluation metrics

Confusion Matrix

Predicted Values \ Actual Values	Positive (1)	Negative (0)
Positive (1)	TP	FP
Negative (0)	FN	TN

$$F1\ score = 2 * \frac{Precision * Recall}{Precision + Recall}$$



- **Accuracy (all correct / all)** = (TP + TN)/(TP + TN + FP + FN)
- **Misclassification (all incorrect / all)** = (FP + FN)/(TP + TN + FP + FN)
- **Precision (true positives / predicted positives)** = TP /(TP + FP)
- **Sensitivity aka Recall (true positives / all actual positives)** = TP / (TP + FN)
- **Specificity (true negatives / all actual negatives)** =TN /(TN + FP)

Classifiers performance evaluation metrics

Performance measure for classification: Cohen kappa (n_{ij}=no. patterns of class ***i*** assigned to class ***j***).

$$\kappa = 100\frac{p_a - p_e}{s - p_e};\ p_a = \sum_{i=1}^{C} n_{ii};\ p_e = \frac{1}{s}\sum_{i=1}^{C}\left(\sum_{j=1}^{C} n_{ij}\right)\left(\sum_{k=1}^{C} n_{ki}\right);\ s = \sum_{i=1}^{C}\sum_{j=1}^{C} n_{ij}$$

During the training we can use ***K*-fold** cross validation method to evaluate the model

Just one measure.

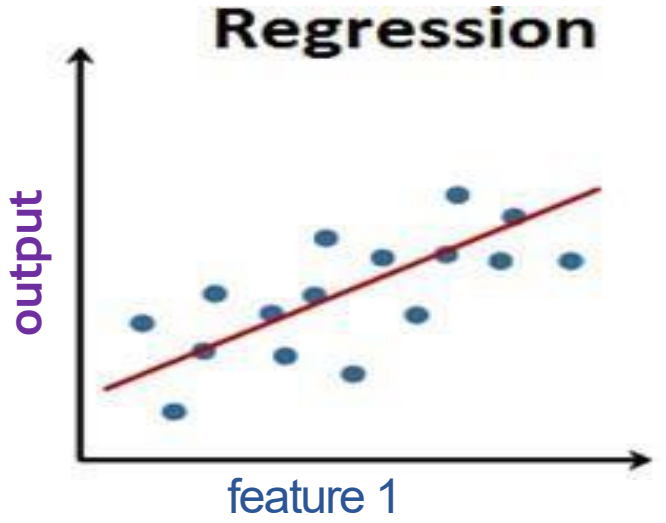
Discards patterns correctly classified by chance (e.g. due to class unbalance).

Kvålseth, Tarald O. "Note on Cohen's kappa." Psychological reports 65, no. 1 (1989): 223-226.

Difference between Regression and Classification

Regression:

- The output (i.e. target variable) is continuous. It consist of real values.
  - predicting the price of houses (in SEK)
  - predicting the power consumption (in kW)
  - predicting how much healthy the patient is (e.g. $\in [0, 1]$)
  - … etc.



Classification:

- The output (i.e. target variable) is discrete. It consists of classes (or categories).
  - predicting if an image contains a cat or a dog
  - predicting customer categories
  - good/bad, healthy/sick, red/green/blue, A/B/C/D, 0/1/2
  - … etc.



Difference between Regression and Classification

Would you treat the following as a classification or as a regression problem?

- Problem 1: Amazon wants to predict how many articles will be sold over the next 2 months.
- Problem 2:Volvo wants to detect whether the air pressure system of a vehicle (bus) is faulty or healthy, based on data collected from similar vehicles (know to be faulty or healthy).
- Problem 3:You want to predict the price of Bitcoin by tomorrow based on recent historical prices (as well as other features, e.g. news and user sentiments …).
- Problem 4:You have a large set of research articles downloaded from the Internet. You don't know the topic of each article, so you want to group them into groups of articles that talk about similar topics.

Difference between Regression and Classification

Would you treat the following as a classification or as a regression problem?

Regression

- Problem 1: Amazon wants to predict how many articles will be sold over the next 2 months.

Classification

- Problem 2:Volvo wants to detect whether the air pressure system of a vehicle (bus) is faulty or healthy, based on data collected from similar vehicles (know to be faulty or healthy).

Regression

- Problem 3:You want to predict the price of Bitcoin by tomorrow based on recent historical prices (as well as other features, e.g. news and user sentiments …).

Clustering

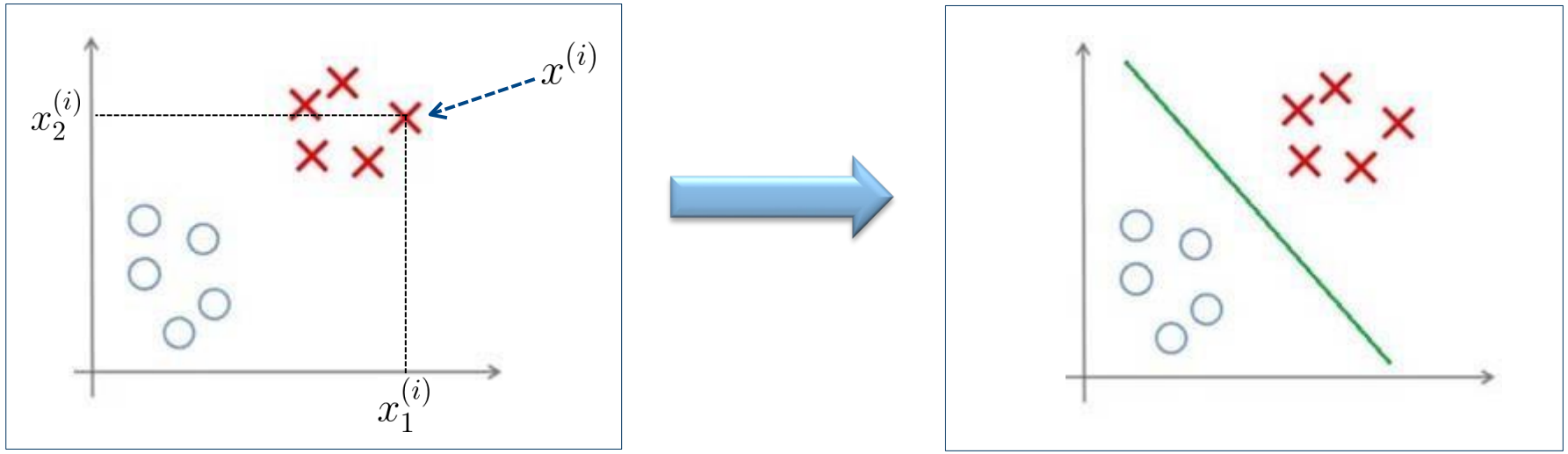
- Problem 4:You have a large set of research articles downloaded from the Internet. You don't know the topic of each article, so you want to group them into groups of articles that talk about similar topics.

Introduction to Unsupervised ML Clustering

Introduction to Unsupervised Learning

In supervised learning (e.g. classification), we have a labelled training dataset:

$$\{(x^{(1)}, y^{(1)}), (x^{(2)}, y^{(2)}), \ldots, (x^{(n)}, y^{(n)})\}$$

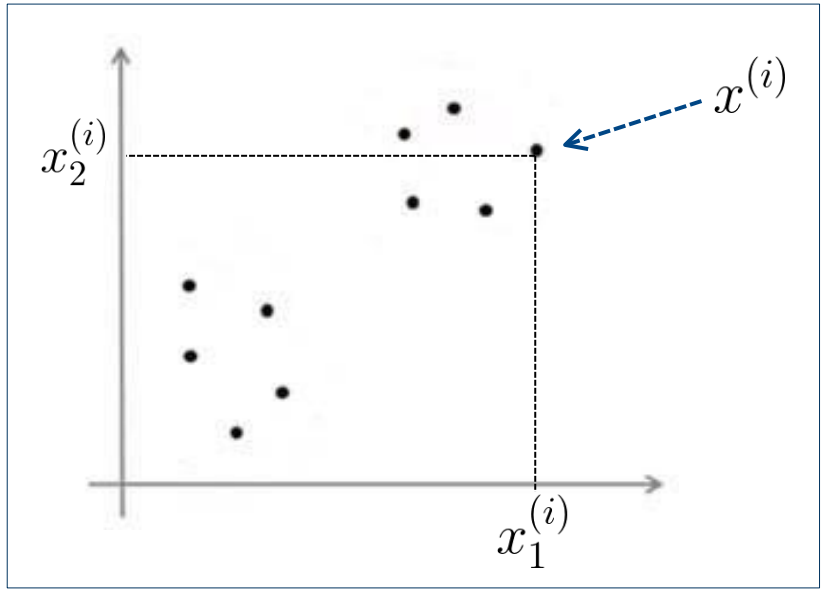


So, for each data-point $x^{(i)} \in R2$, we have the corresponding class-label $y^{(i)} \in \{\times, \circ\}$

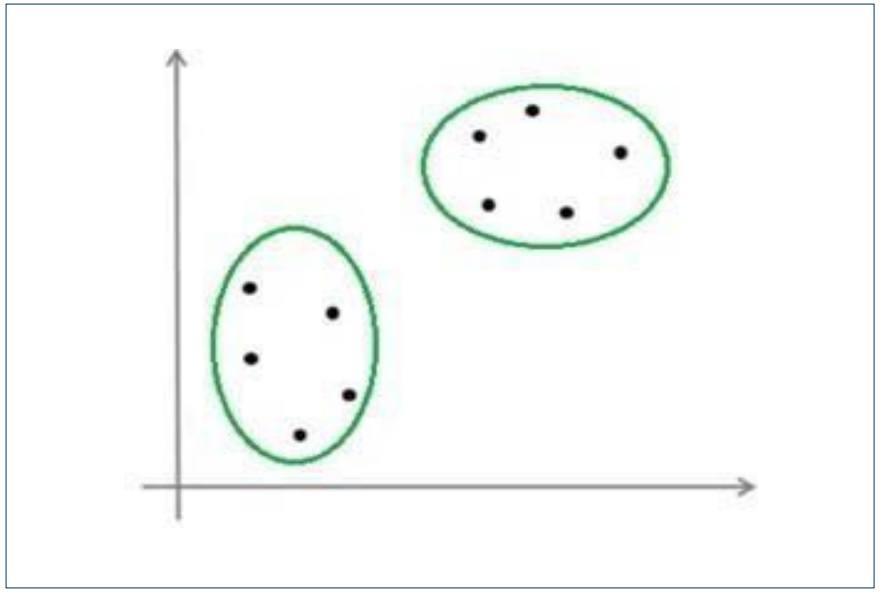
Introduction to Unsupervised Learning

In unsupervised learning (e.g. clustering) we have an unlabeled training dataset:

$$\{x^{(1)}, x^{(2)}, \ldots, x^{(n)}\}$$







We have only data-points $x^{(i)} \in R$

In clustering, we want to explore the data to find some intrinsic groups (clusters) in it. The clusters are not known beforehand.

Introduction to Unsupervised Learning



Data

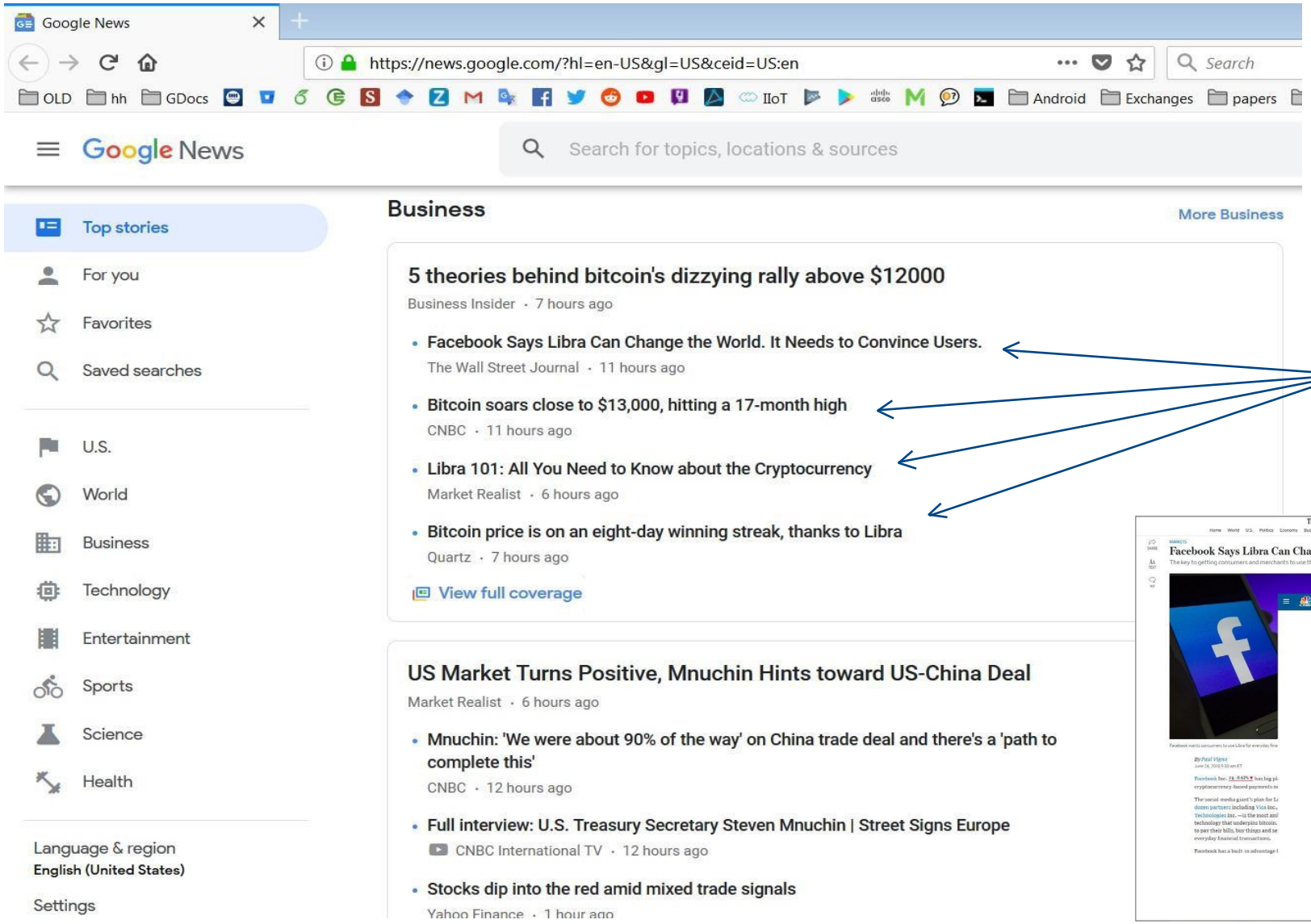




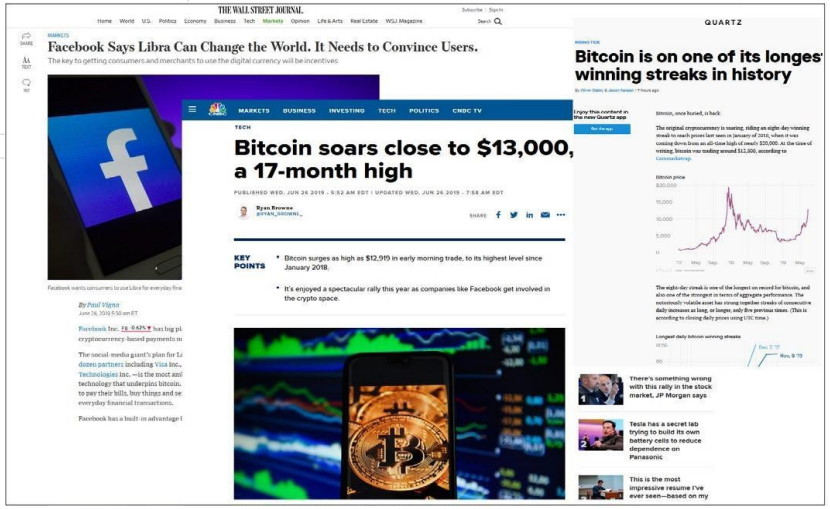
ML
(Learner)

- **Clustering**
  - k-means
  - GMM
  - DBSCAN
  - Hierarchical clustering, … etc.
- **Anomaly detection**
  - Density estimation, … etc.
- **Dimensionality reduction**
  - PCA
  - LDA
  - MDS
  - PKDA, … etc.
- **Representation learning**
  - Auto-encoders, … etc

Some applications of clustering



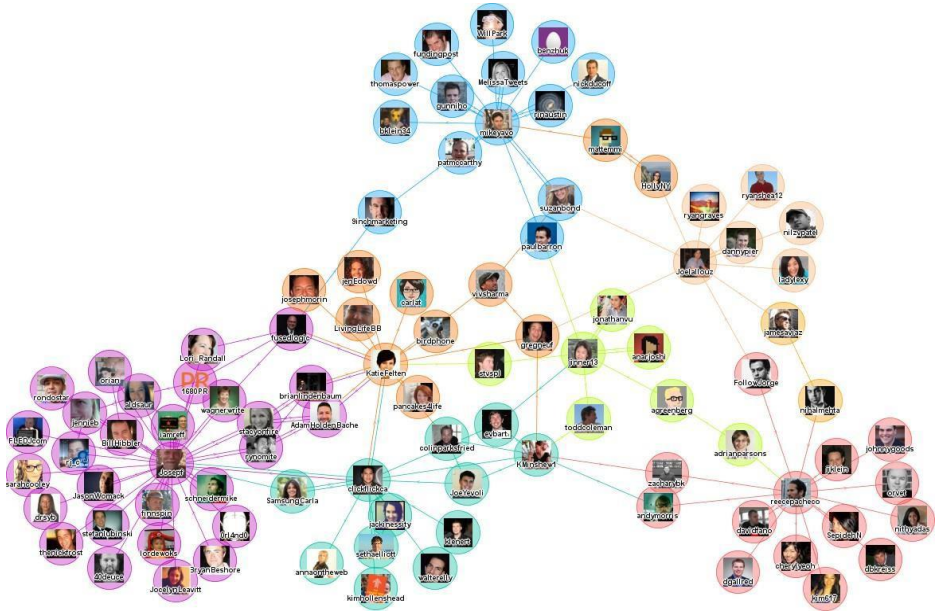
Automatically grouping together the stories (news articles on the Web) that talk about the same topic.



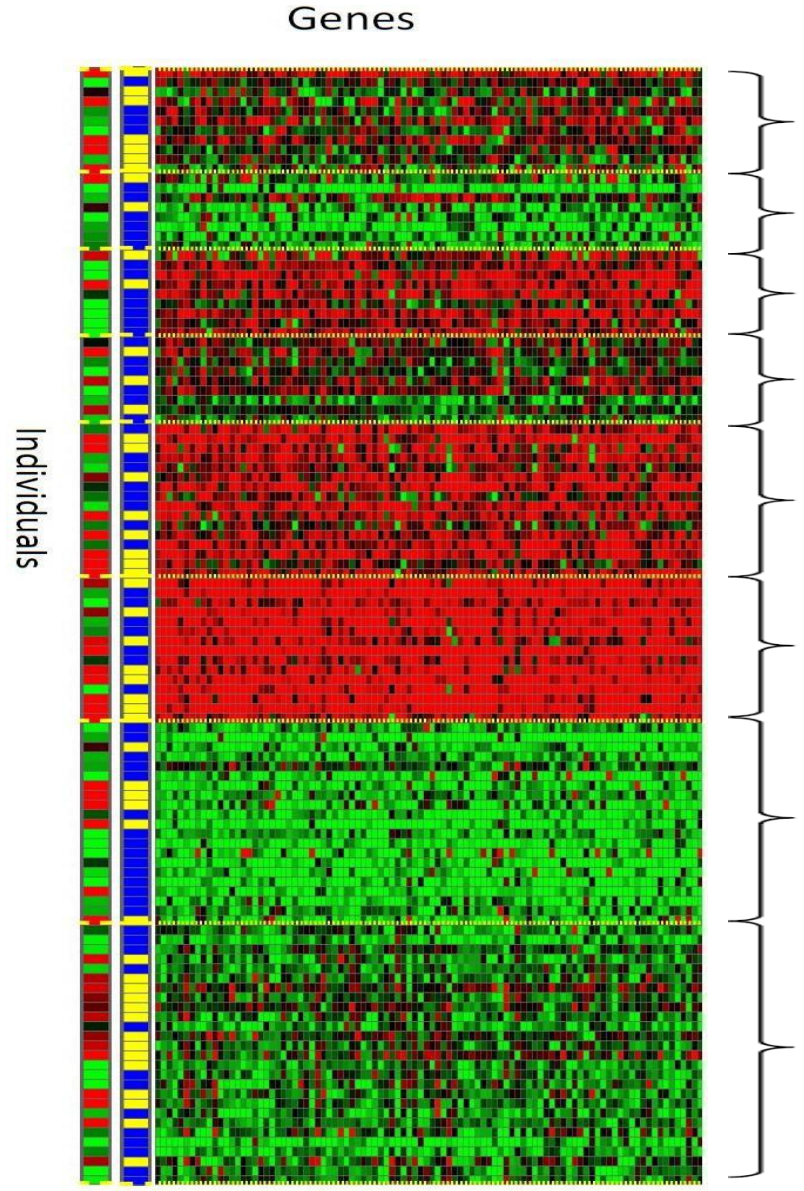
Some applications of clustering

Grouping people based on their genes:

- DNS Microarray data: colours here correspond to how much individuals do or do not have a certain gene.
- Run a clustering algorithm to group individuals into different groups/types of people.



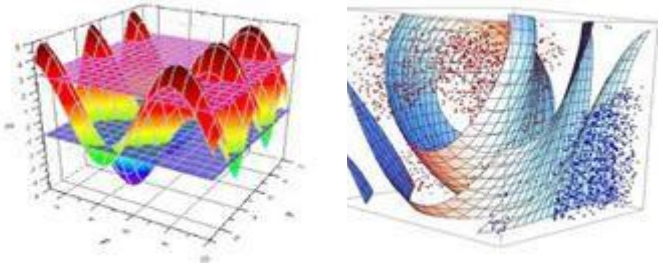
Social network analysis



Dimensionality Reduction

$\mathbf{x}_i \in \mathbb{R}^n$

Patterns in 3D



Patterns in nD



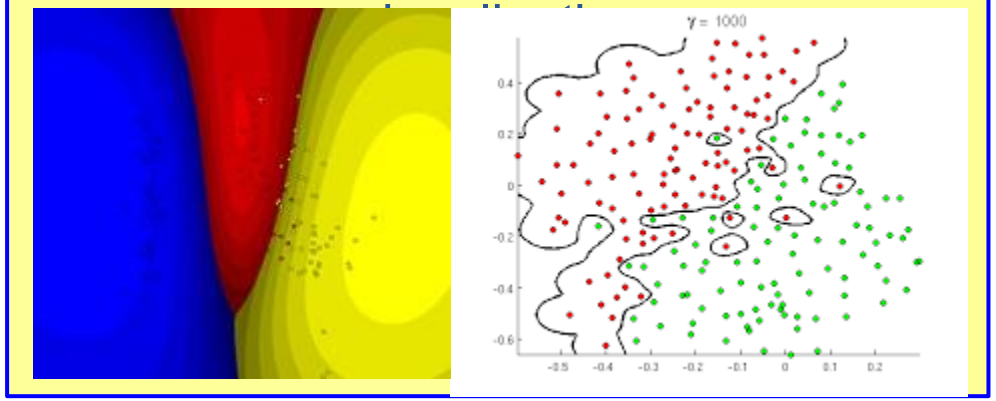
Related to dimensionality reduction

To propose methods of data visualization for classification or regression tasks preserving the performance

Mapping to 2D

Classifier
Regressor

Data



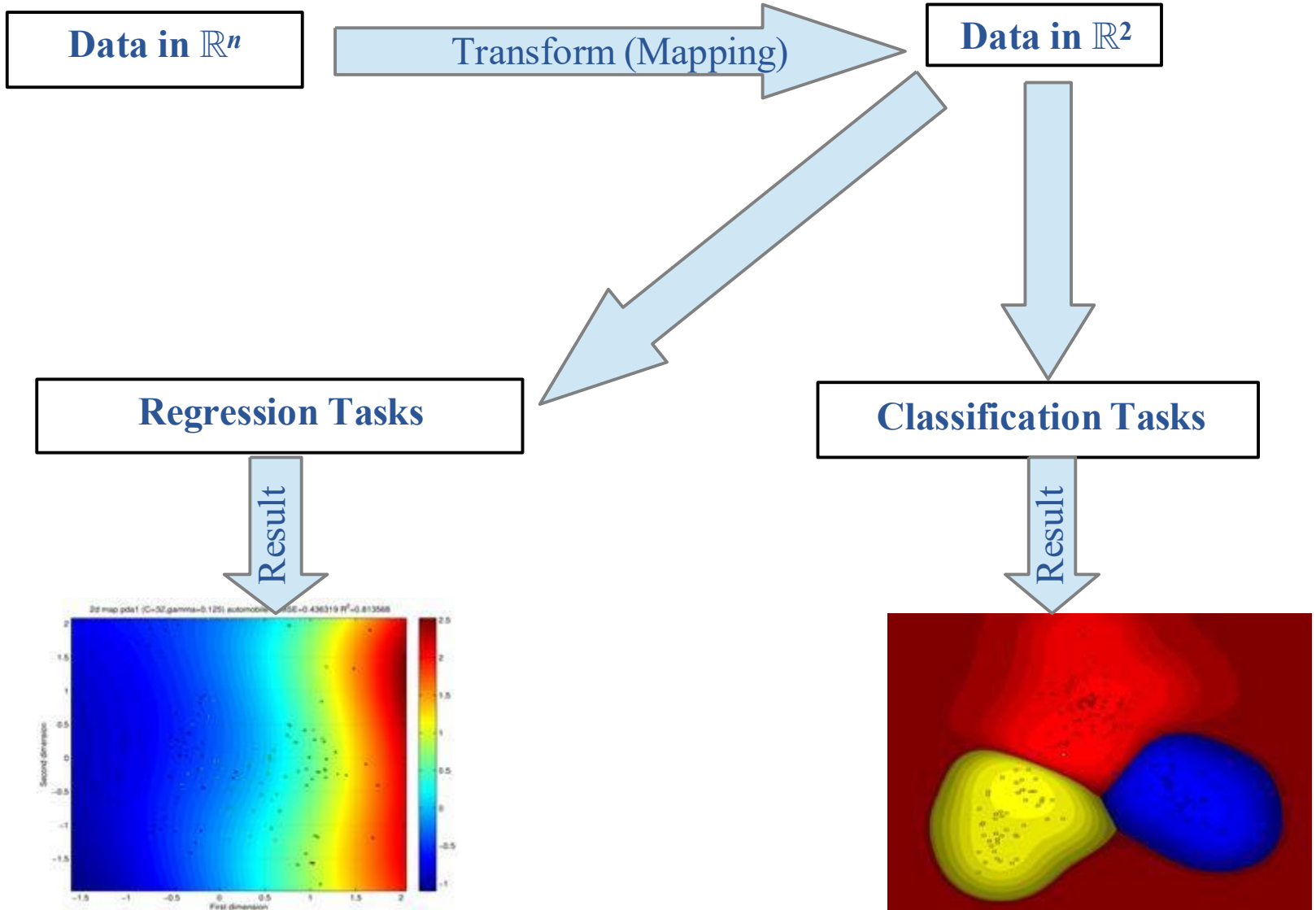
Dimensionality Reduction Methods

Method	Features
Classical methods	
LDA	Supervised
GDA	Supervised, kernel, not explicit
PCA	Linear, unsupervised
KernelPCA	Unsupervised, non-linear, kernel
ProbPCA	Unsupervised, Expectation-Maximization (EM), Gaussian processes
GPLVM	Unsupervised, Gaussian processes, not-explicit
FA	Latent variables, EM
Global non-linear methods	
Sammon, MDS, SPE, SNE, SSNE, t-SNE	Preserve pairwise distance, not explicit
Isomap, Landmark Isomap	Preserve pairwise geodesic distance, neighborhood graph
Diffusion map	Preserves distance, diffusion kernel, neighborhood graph, not explicit
Multilayer autoencoder	Neural network
Local non-linear methods	
LLE	Linear, Preserves neighbor weights, neighbor graph
Hessian LLE	Preserves neighbor weights, not explicit
NPE	Linear approach to LLE, preserves neighbor weights
LTSA	Local tangent space, nearest neighbors, not explicit
LLTSA	Linear approach to LTSA
Laplacian eigenmap	Preserves pairwise distances, neighbor graph
LPP	Linear approach to Laplacian eigenmap, neighbor graph
MVU	Preserves pairwise distances, starts from LLE, not explicit
FastMVU	Preserves distances, neighbor graph
Landmark MVU	Starts from MVU
CCA	Neighbor angle preserving, starts from LLE or Laplacian eigenmap, not explicit
LLC	Linear, mixture of linear models, global alignment, not explicit
Manifold charting	Mixture of linear factor analyzers
CFA	Mixture of linear factor analyzers, not explicit
Other supervised methods	
NCA, MCML, LMNN	Linear, supervised, Mahalanobis metric

Label	Mapping	Overlap	Function
DSOLM	Direct class Separability Optimization Linear Mapping	DS	Linear
TOLM	Thornton Opt. Linear Map.	Thornton	
CDOLM	Class Distance Opt. Linear Map.	CD	
QKDA	DA with Quadratic Kernel map.	J-index	Quadratic
DSOQM	Direct class Separability Opt. Quadratic Map.	DS	
TOQM	Thornton Opt. Quadratic Map.	Thornton	
CDOQM	Class Distance Opt. Quadratic Map.	CD	
PKDA(D)	DA with Polynomial Kernel map.	J-index	Polynomial ($D = 2,\ldots,7$)
DSOPM(D)	Direct class Separability Opt. Pol. Map.	DS	
TOPM(D)	Class Distance Opt. Pol. Map.	Thornton	
CDOPM(D)	Class Distance Opt. Pol. Map.	CD	

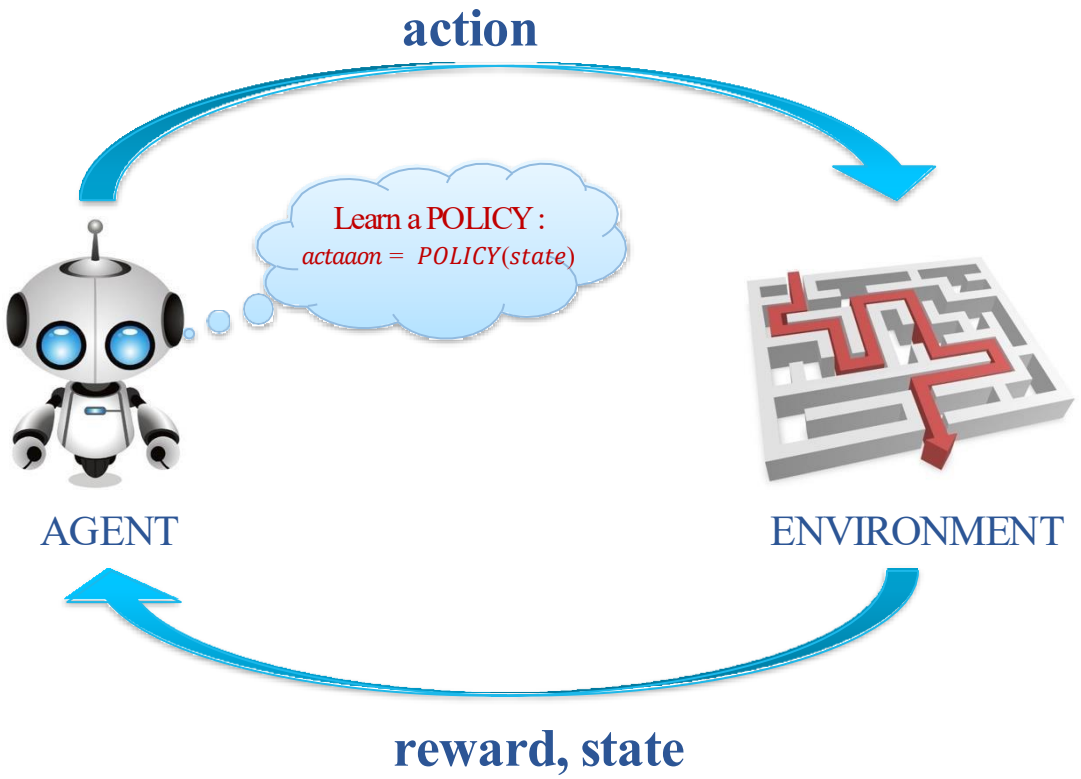
Alawadi, Sadi, Manuel Fernández-Delgado, David Mera, and Senén Barro. "Polynomial Kernel Discriminant Analysis for 2D visualization of classification problems." Neural Computing and Applications: 1-17.

Dimensionality Reduction



reinforcement learning, Semi-Supervised learning, and Interactive / Active Learning

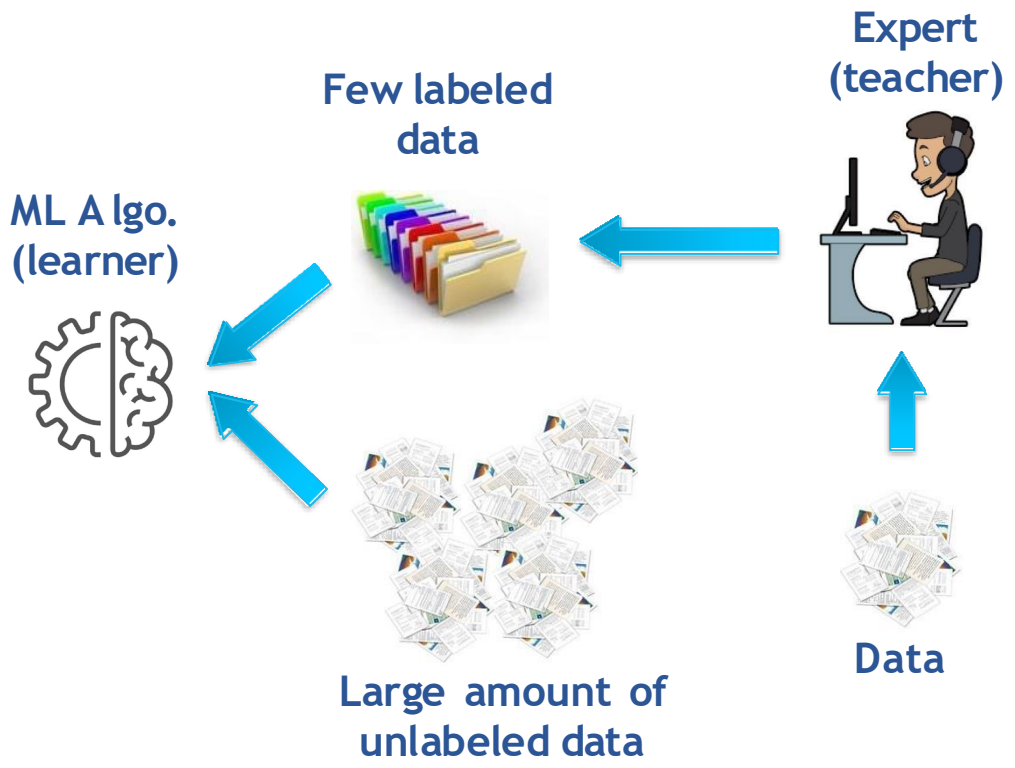
Reinforcement Learning



- How an agent can learn to perform a task by interacting with the environment?

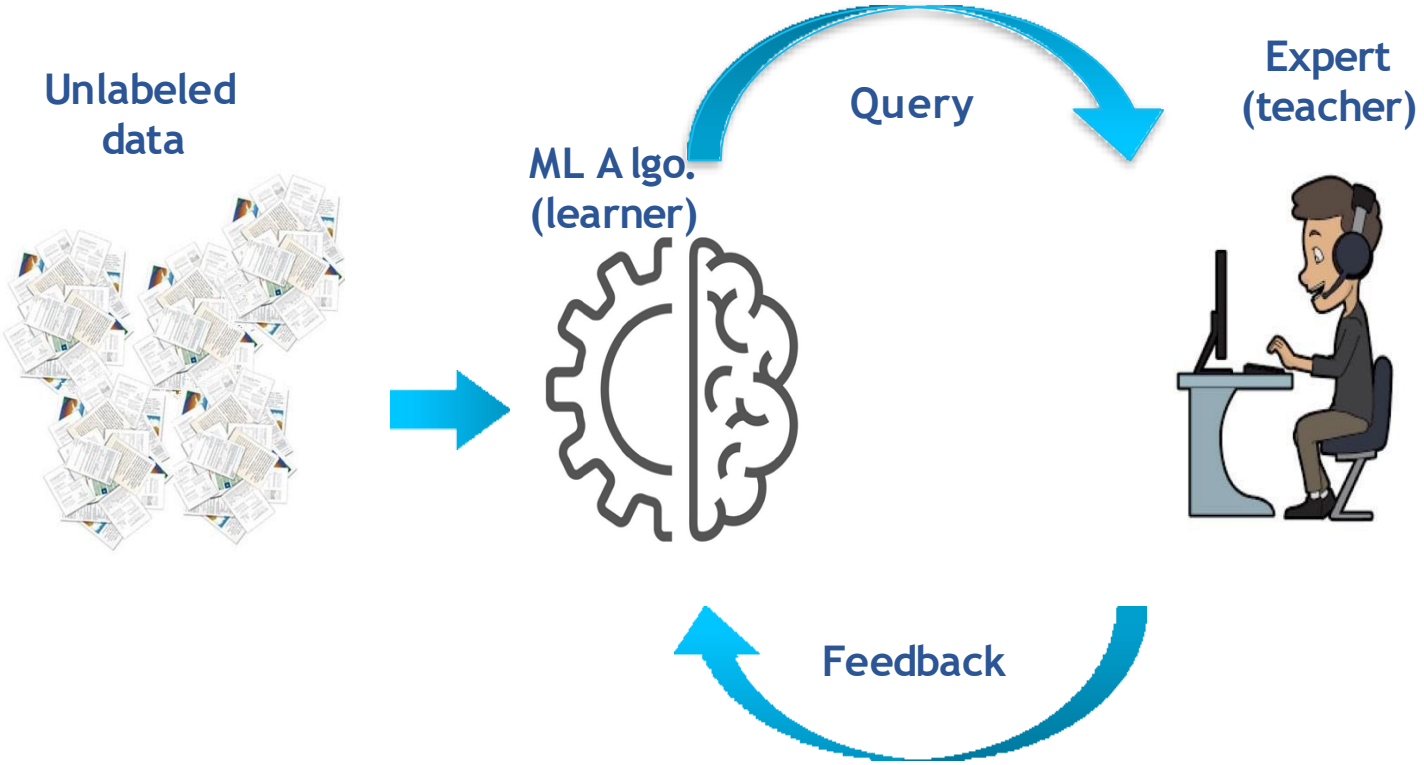
Basic RL methods such as:
- Q-learning.
- Deep-Q-Networks.

Semi-supervised Learning



- Observations
  - Labelling is costly and time-consuming (requires a lot of expert effort).
  - Unlabelled data can be collected cheaply.
- Semi-supervised methods learn using both labelled and unlabelled data.

Interactive/ Active Learning

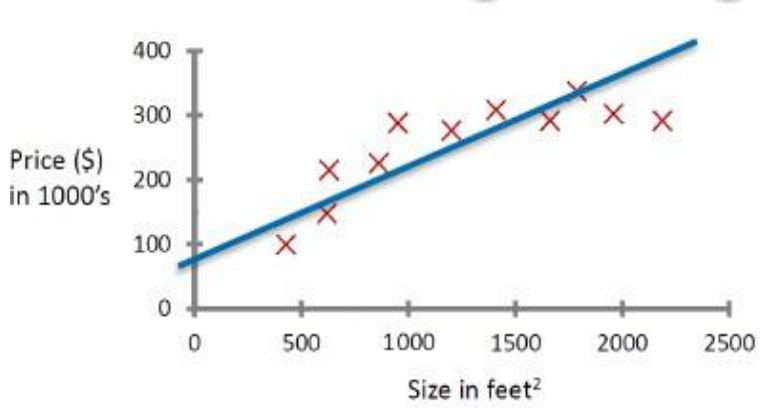


- Interaction between the algorithm and the expert during the learning process.
- **Active learning for classification**
  – The algorithm selects the most informative data to label (by the expert).
- **Interactive anomaly detection**
  – Learning to detect anomalies that are more relevant to the expert.

Overfitting, Generalization, and Regularization

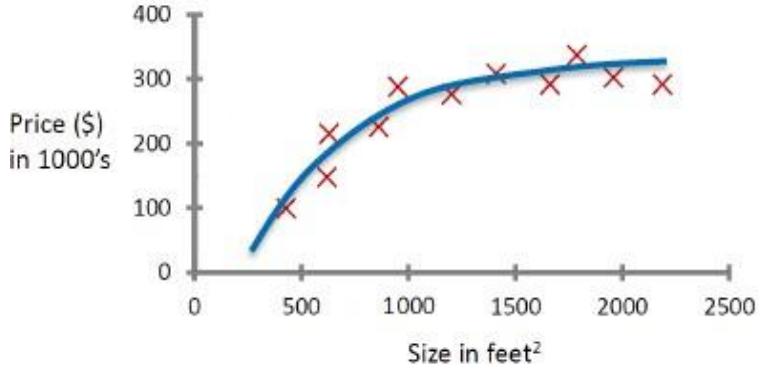
Overfitting Problem

Example: Linear regression (housing prices)

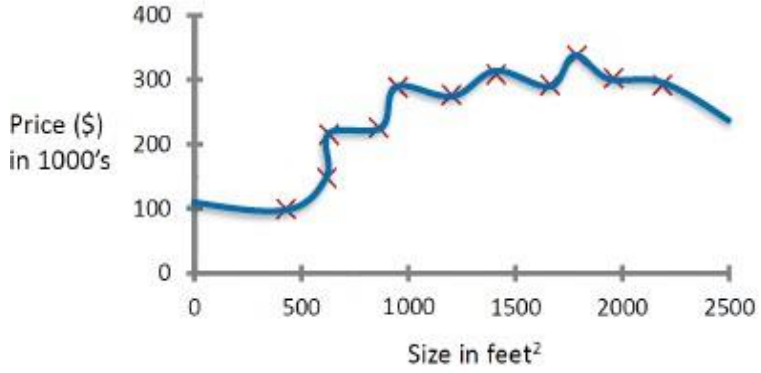


$$h_\theta(x) = \theta_0 + \theta_1 x$$

Which of these models is a good model **for this data**?



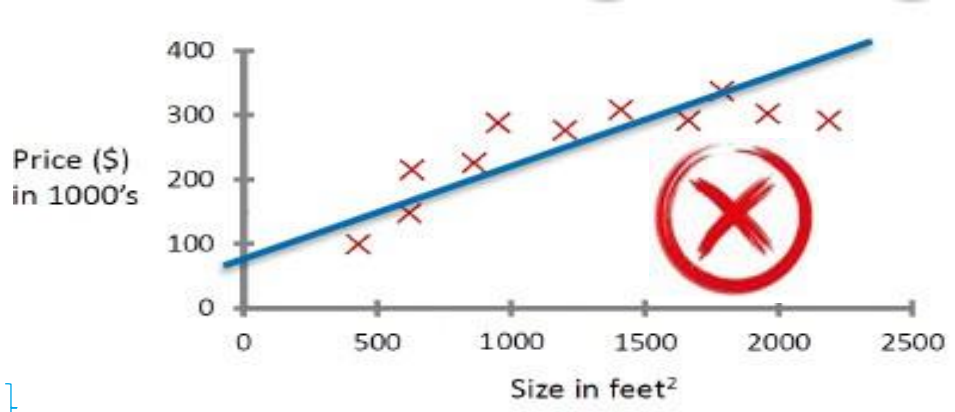
$$h_\theta(x) = \theta_0 + \theta_1 x + \theta_2 x^2$$



$$h_\theta(x) = \theta_0 + \theta_1 x + \theta_2 x^2 + \theta_3 x^3 + \theta_4 x^4 + \theta_5 x^5 + \theta_6 x^6 + \theta_7 x^7 + \theta_8 x^8 + \theta_9 x^9 + \theta_{10} x^{10}$$

Overfitting Problem - regression

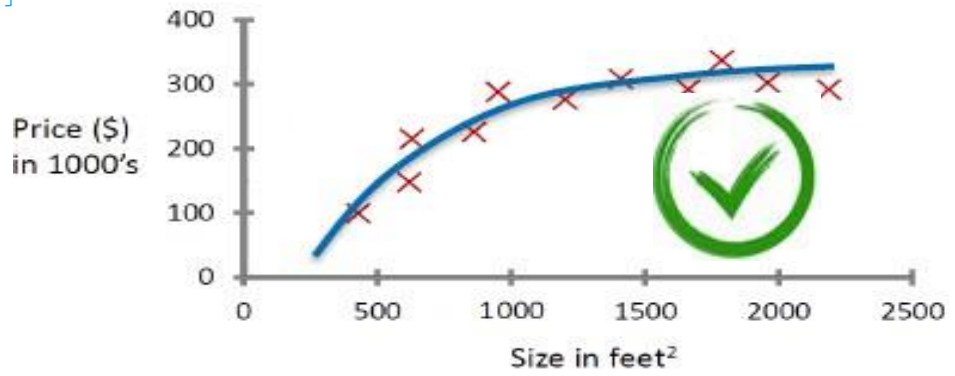
Underfitting
- The model has a high **bias**
- The model makes a strong assumption that the housing prices will vary linearly with their size but ends up not fitting the training data well (poor fit).



Simple model

$$h_\theta(x) = \theta_0 + \theta_1 x$$

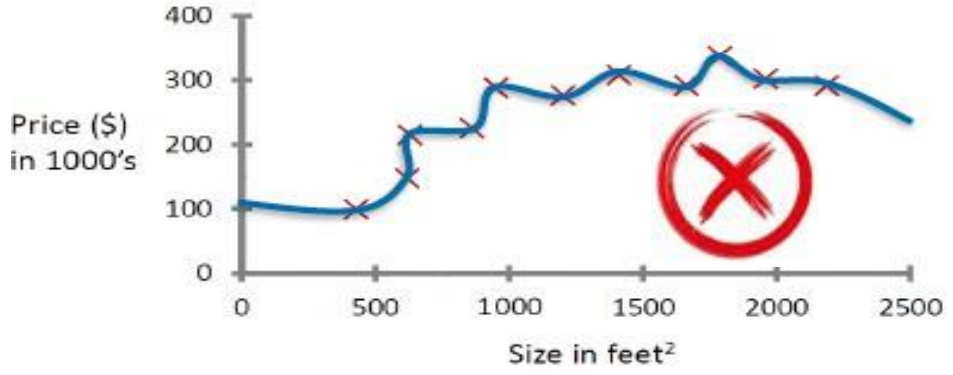
- **Just right**
- **It fits the data pretty well**



More complex model

$$h_\theta(x) = \theta_0 + \theta_1 x + \theta_2 x^2$$

Overfitting
- The model has a high **variance**
  - The space of possible hypothesis functions of this order is too large (too variable), and we do not have enough data to construct a good hypothesis of this type.
- Seems to fit the training data perfectly, but will have very poor performance on new data. It has 0 error on the training data but does not generalize well.

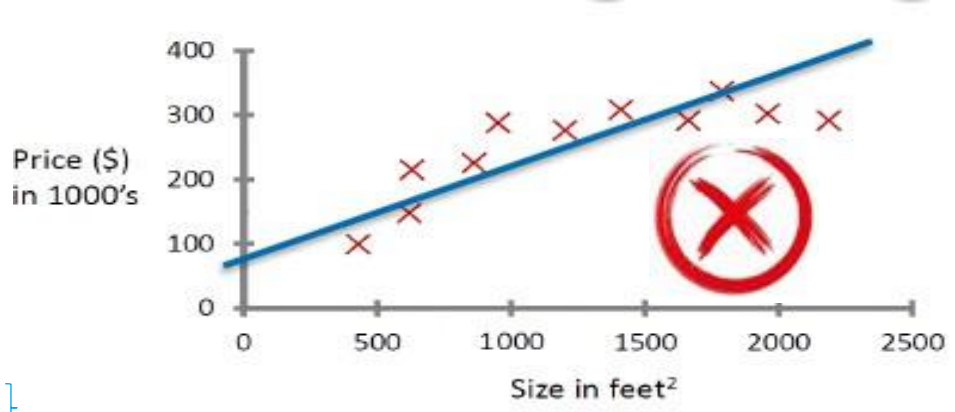


Much more complex model

$$h_\theta(x) = \theta_0 + \theta_1 x + \theta_2 x^2 + \theta_3 x^3 + \theta_4 x^4 + \theta_5 x^5 + \theta_6 x^6 + \theta_7 x^7 + \theta_8 x^8 + \theta_9 x^9 + \theta_{10} x^{10}$$

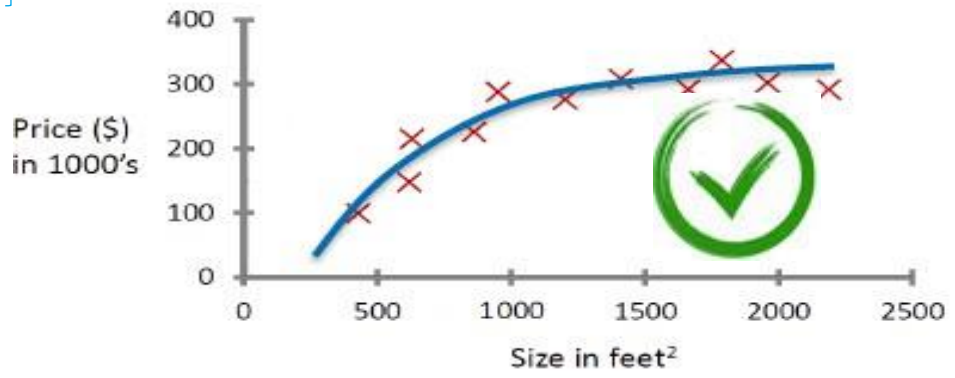
Overfitting Problem - regression

Overfitting: If we have too many features, the learned model may fit the training set very well ($E(\theta) = \frac{1}{n}\sum_{i=1}^{n}\left[h_\theta\left(x^{(i)}\right) - y^{(i)}\right]^2 \approx 0$), but fails to generalize to new examples (predict prices on new examples).



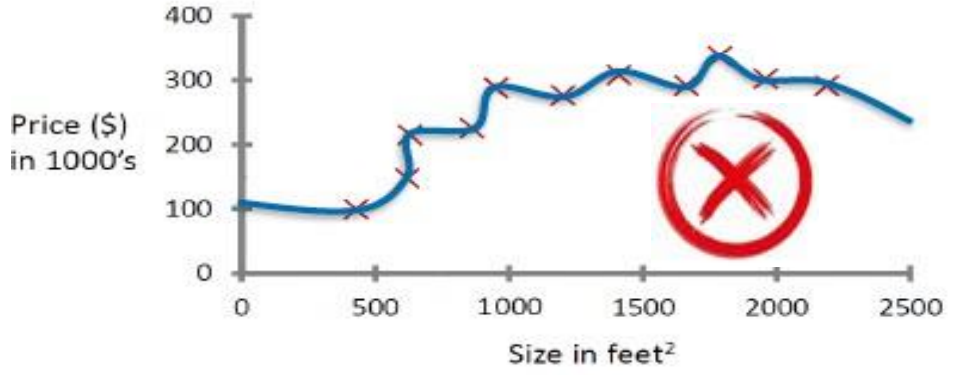
Simple model

$$h_\theta(x) = \theta_0 + \theta_1 x$$



More complex model

$$h_\theta(x) = \theta_0 + \theta_1 x + \theta_2 x^2$$



Much more complex model

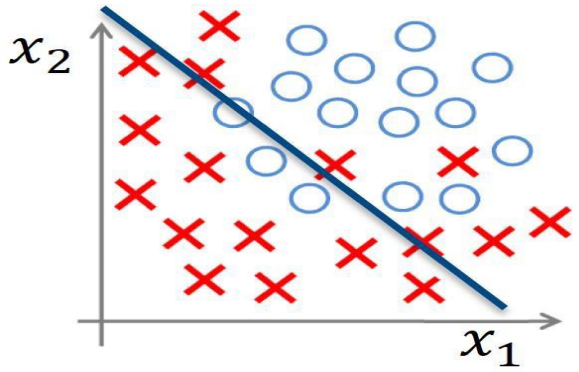
$$h_\theta(x) = \theta_0 + \theta_1 x + \theta_2 x^2 + \theta_3 x^3 + \theta_4 x^4 + \theta_5 x^5 + \theta_6 x^6 + \theta_7 x^7 + \theta_8 x^8 + \theta_9 x^9 + \theta_{10} x^{10}$$

Overfitting Problem - Classification

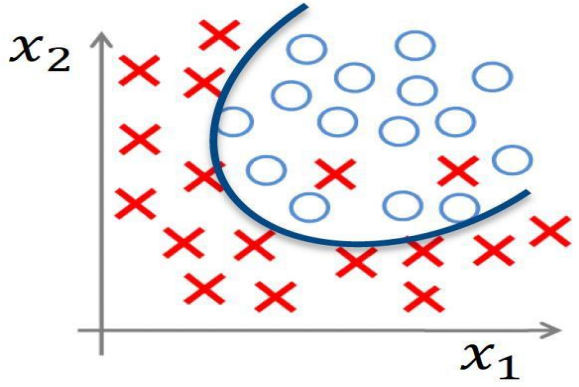
Example: Classification (with Logistic Regression)

NOTE: g here is the sigmoid function.

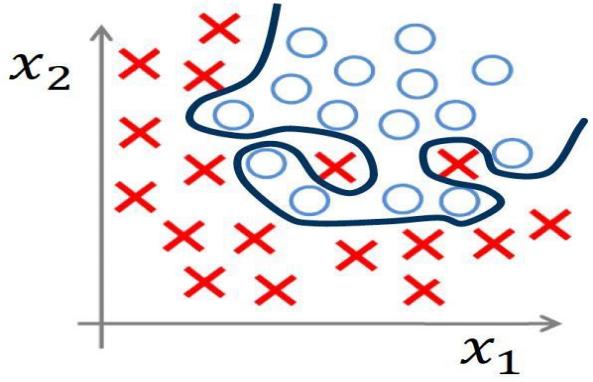
Which of these models is a good model **for this data**?



$$h_\theta(x) = g(\theta_0 + \theta_1 x_1 + \theta_2 x_2)$$



$$h_\theta(x) = g(\theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_1^2 + \theta_4 x_2^2 + \theta_5 x_1 x_2)$$

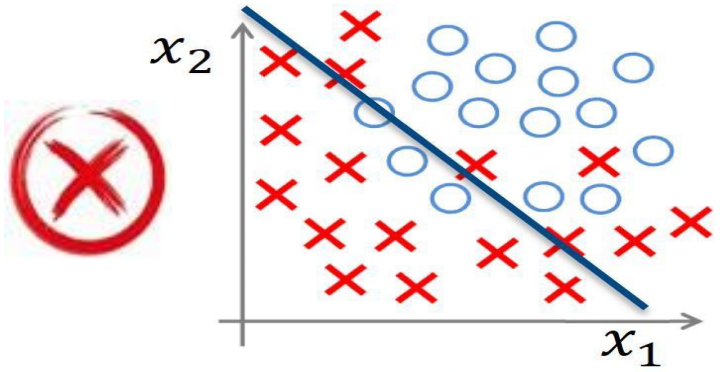


$$h_\theta(x) = g(\theta_0 + \theta_1 x_1 + \theta_2 x_1^2 + \theta_3 x_1^2 x_2 + \theta_4 x_1^2 x_2^2 + \theta_5 x_1^2 x_2^3 + \theta_6 x_1^3 x_2 + \cdots)$$

Overfitting Problem - Classification

Example: Classification (with Logistic Regression)

Underfitting
(high bias)

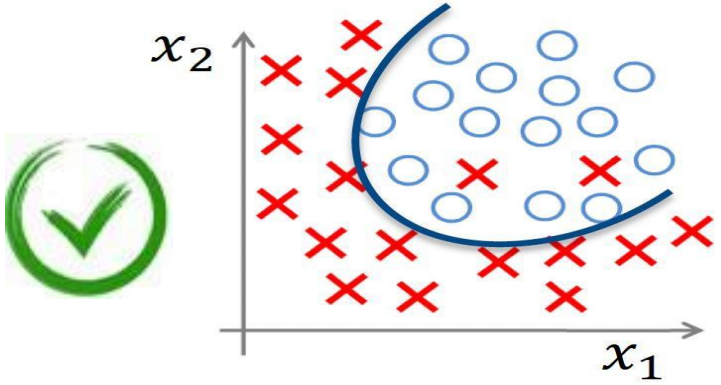


Simple model

$$h_\theta(x) = g(\theta_0 + \theta_1 x_1 + \theta_2 x_2)$$

NOTE: g here is the sigmoid function.

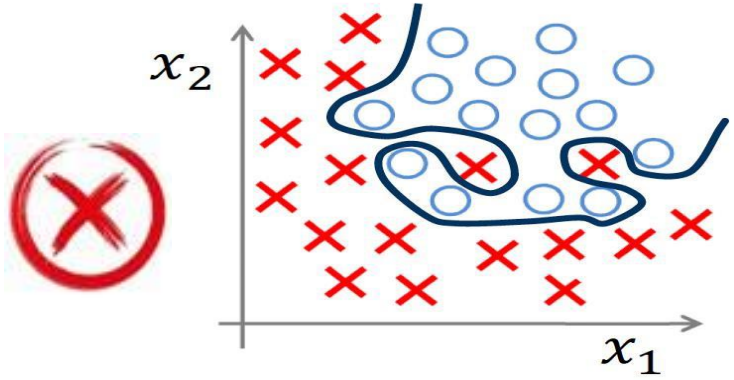
Ok



More complex model

$$h_\theta(x) = g(\theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_1^2 + \theta_4 x_2^2 + \theta_5 x_1 x_2)$$

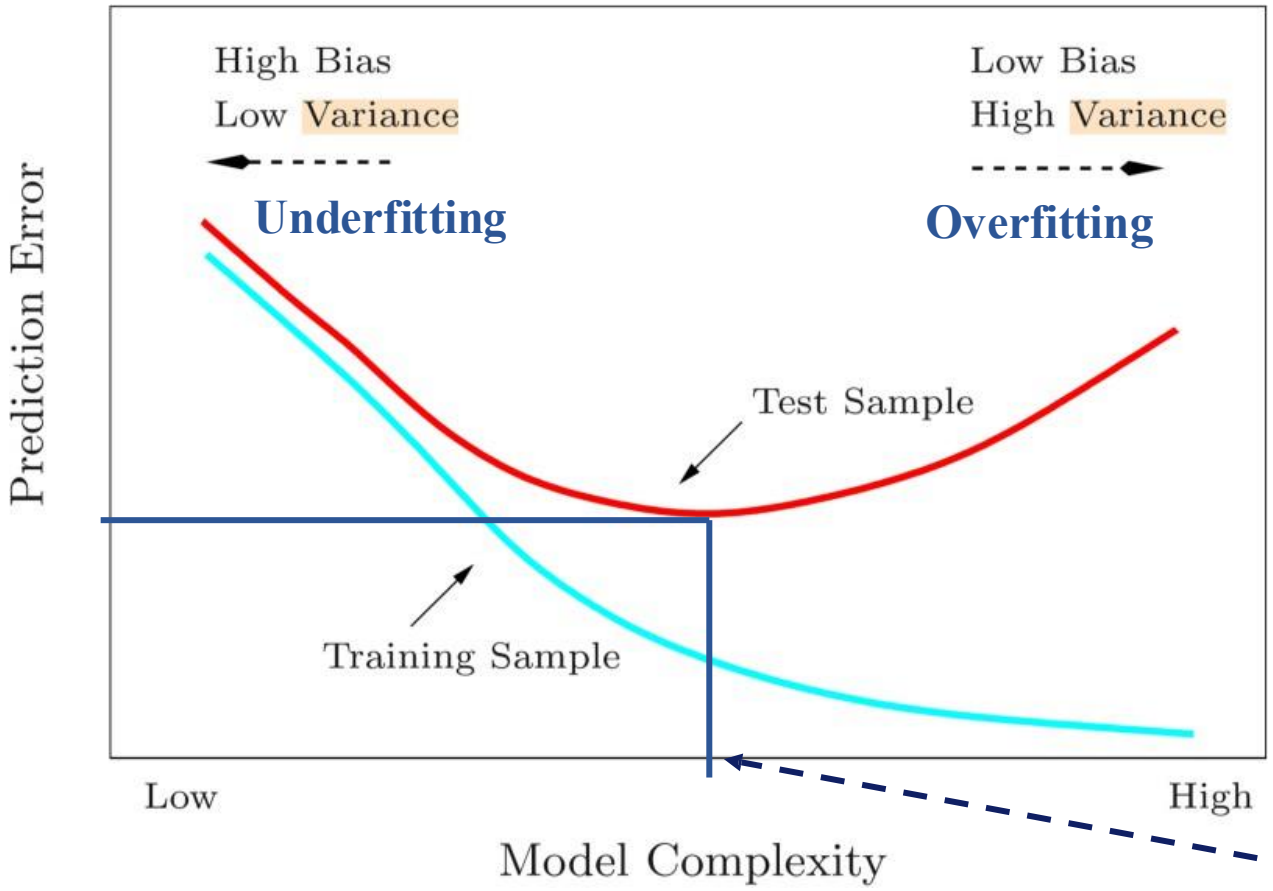
Overfitting
(high bias)



Much more complex model

$$h_\theta(x) = g(\theta_0 + \theta_1 x_1 + \theta_2 x_1^2 + \theta_3 x_1^2 x_2 + \theta_4 x_1^2 x_2^2 + \theta_5 x_1^2 x_2^3 + \theta_6 x_1^3 x_2 + \cdots)$$

Overfitting Problem



What makes it more likely to overfit?

- Not enough training examples (small training dataset)
- Too many features
- Using a non-convenient type of models/hypothesis functions (e.g. too much complex for our problem/data).

Generalization Error

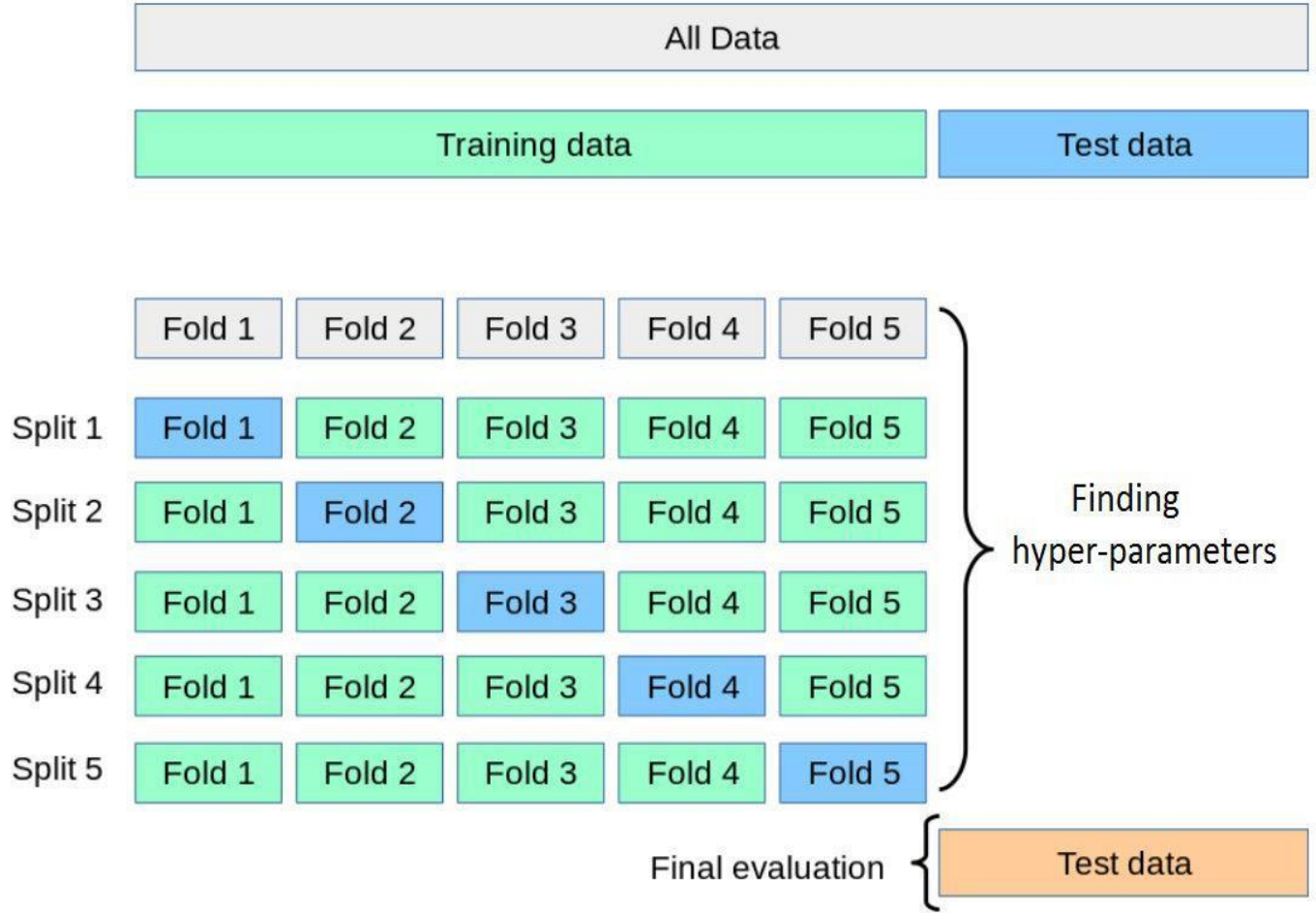
- The **training error** is the error (or accuracy) of the model $h\theta$ on the training examples.
  - ✓ It is **not** an estimation for the error that the model $h\theta$ will have when deployed and applied on new data.
- The **generalization error** is the error of the model $h\theta$ on new (unseen) data.
  - ✓ The generalization error is typically higher than the training error.
- How do we estimate the generalization error of some models?
  - ✓ Using **k-fold Cross Validation** (k-CV)

Generalization Error

Example of k=5-fold cross-validation:

- Split the training dataset into 5 parts (folds).
- Each time, train a model on the training parts (green) and apply it on the remaining part (blue) to estimate the error.

- So, you will finally get 5 estimates of the error. The **generalization error** is estimated as the average of these 5 errors. In the case of classification, you can also compute the generalization accuracy the same way (accuracy% = 100 – error%)



Leave-one-out cross-validation:

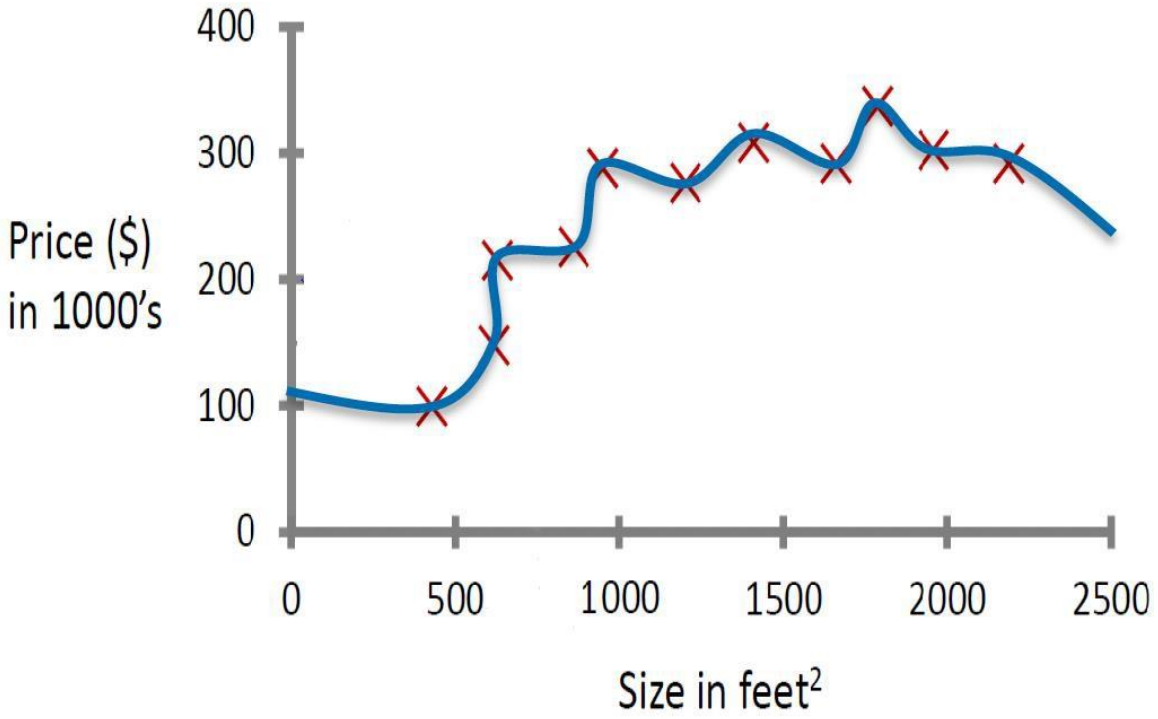
- It is similar to the k-fold cross-validation, but the training dataset is split into n parts, where n is the number of training samples. i.e., we leave one sample each time.
- It is usually used when you have a small training dataset (i.e. when n is small).

Addressing Overfitting

It becomes hard to visualize the data and the model when we have more than 2 or 3 features.

- So, how can we avoid overfitting in general?

$x_1 =$ size of house
$x_2 =$ no. of bedrooms
$x_3 =$ no. of floors
$x_4 =$ age of house
$x_5 =$ average income in neighborhood
$x_6 =$ kitchen size
$\vdots$
x_{100}



Addressing Overfitting

- **Hyperparameters tuning**
  - You can try various models of different complexity (e.g. with various hyperparameters values), compute the generalization error for each of them (as explained previously), and keep the best hyperparameters.
- **Reducing the number of features**
  - We are more likely to overfit when the number of features is high (relative to the size of the dataset).
- **Using ensemble methods**
  - Such as Random Forest, which addresses overfitting in Decision Tree.
- **Using regularization**
  - Keep all features, but reduce the magnitude (value) of parameters θ_j
  - Works well when we have a lot of features, and each feature contributes a bit to predicting the output.

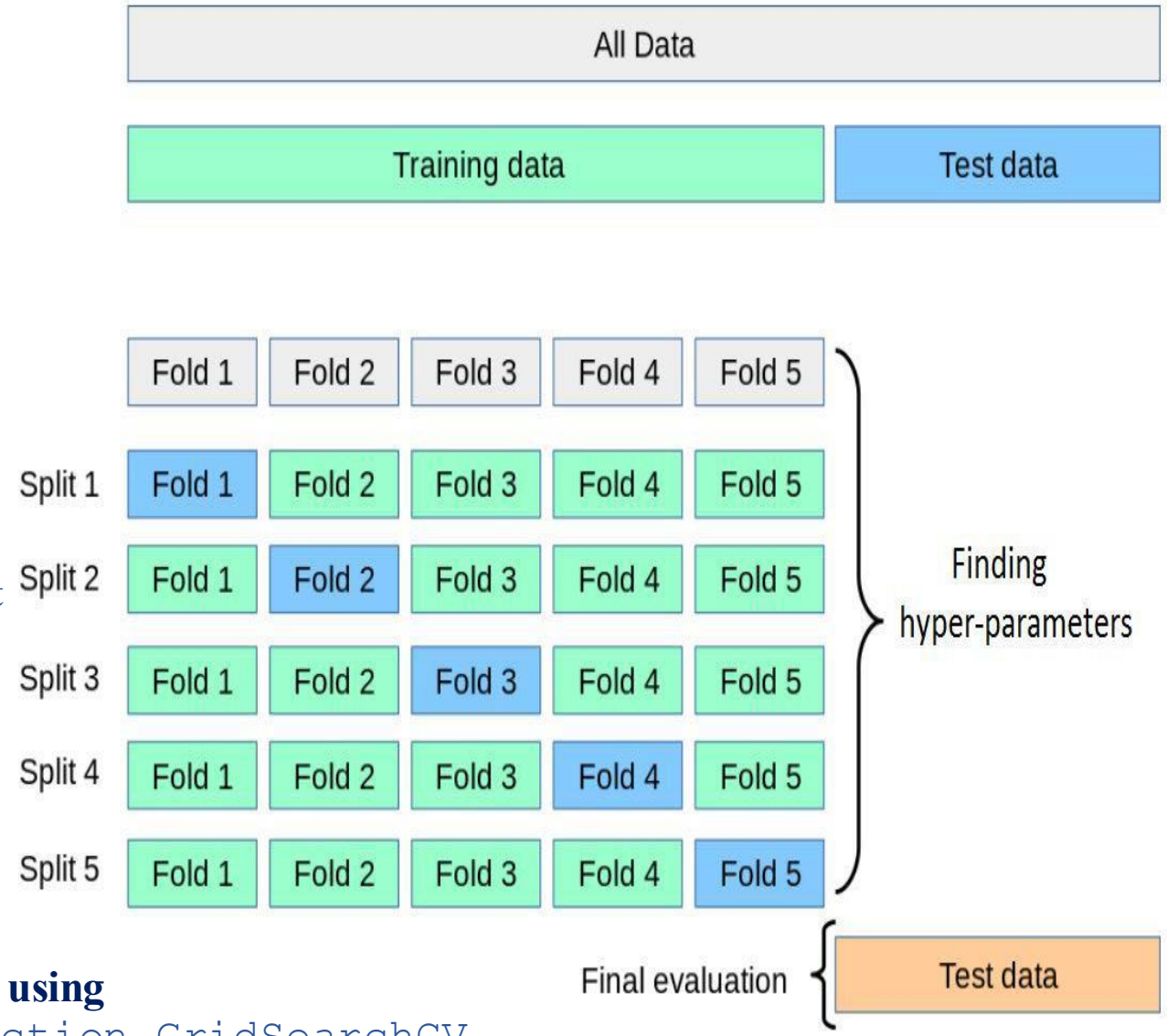
Hyper-parameters Tuning

ML Algos have hyperparameters that to more or less complex models.

- e.g. the *k* in the *k*NN algorithm is a hyperparameter.

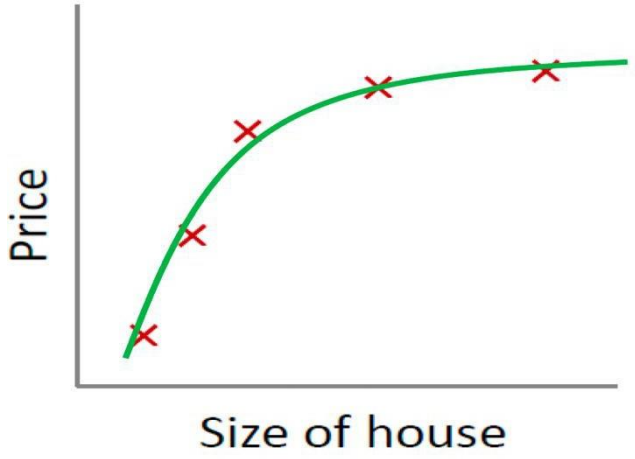
- To find the values of hyperparameters that lead to a good model (not too simple, not too complex), we need to:

1. Pick some values for our hyperparameters.
2. Compute the generalization error using 10-fold cross-validation.
3. Repeat steps 1 and 2 (with various values for the hyperparameters).
4. Keep the hyperparameters that gave you the smallest generalization error, and train a model on the whole training set using those hyperparameters.
5. You can now test your trained model on the test data (see figure) to see how it will perform when deployed in real-world.

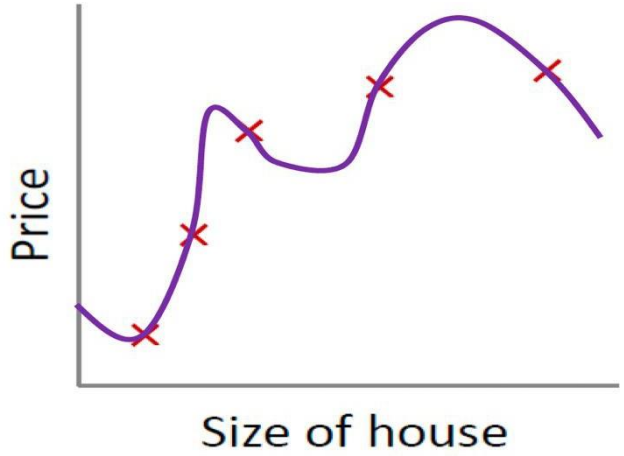


Easy to do in scikit-learn using
`sklearn.model_selection.GridSearchCV`

Features Selection

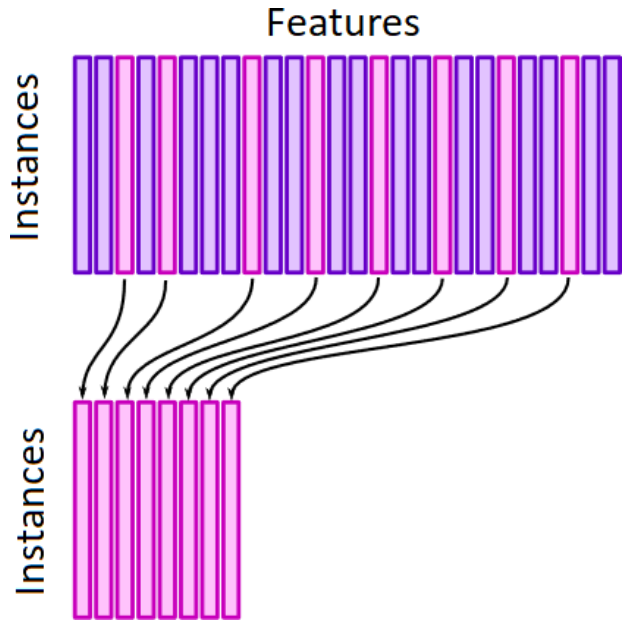


$$h_\theta(x) = \theta_0 + \theta_1 x_1 + \theta_2 x_1^2$$



$$h_\theta(x) = \theta_0 + \theta_1 x_1 + \theta_2 x_1^2 + \boldsymbol{\theta_3 x_1^3 + \theta_4 x_1^4}$$

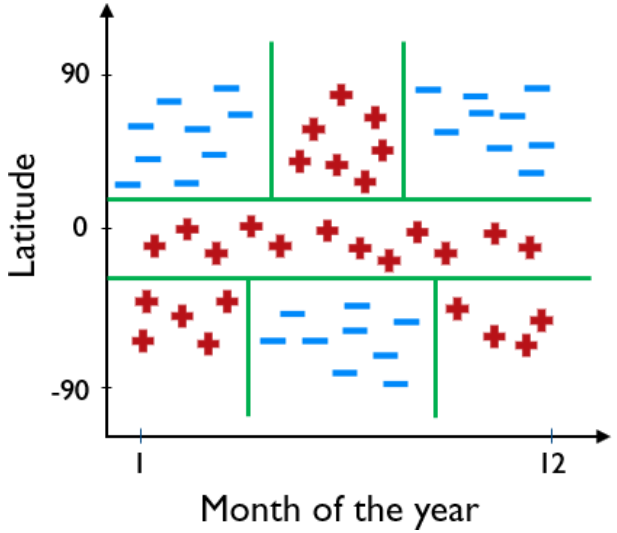
Overfits the data and does not generalize well ☹

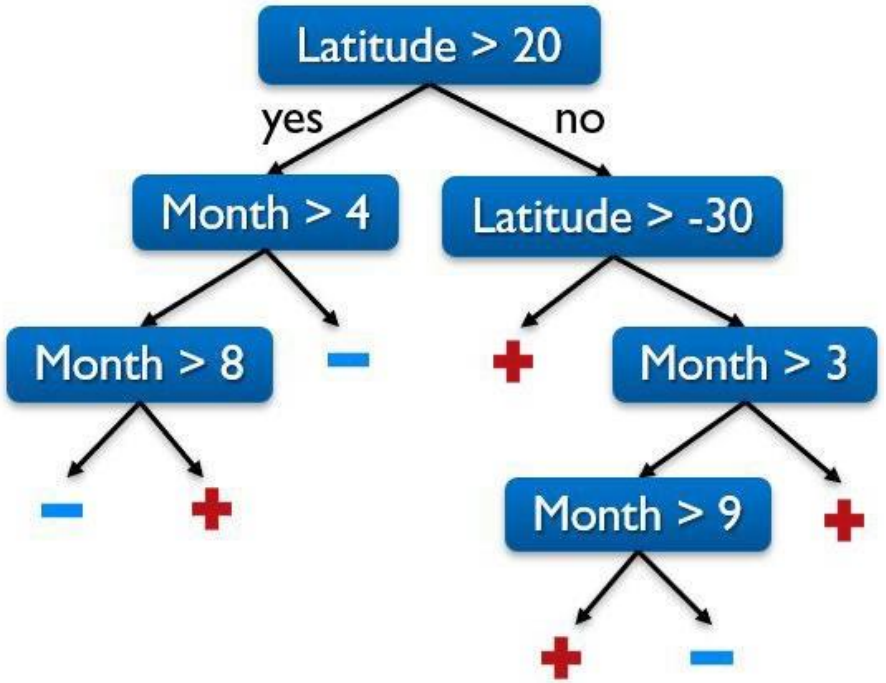


- Datasets with thousands or millions of features are quite usual today.
- A very large d (relative to n), can lead to overfitting, as there is less likelihood of all possible features being encountered in the training process.
  - ✓ If possible, increase n (i.e., collect more data).
  - ✓ Or decrease d (i.e., select the most important features), to construct simpler and faster models.

Features Selection
Tree-based and Features Importance

- Some classification or regression models, such as tree-based models (decision trees, random forest …), can compute features' importance.
  - How much each feature decreases the weighted impurity in a tree (or ensemble of trees).

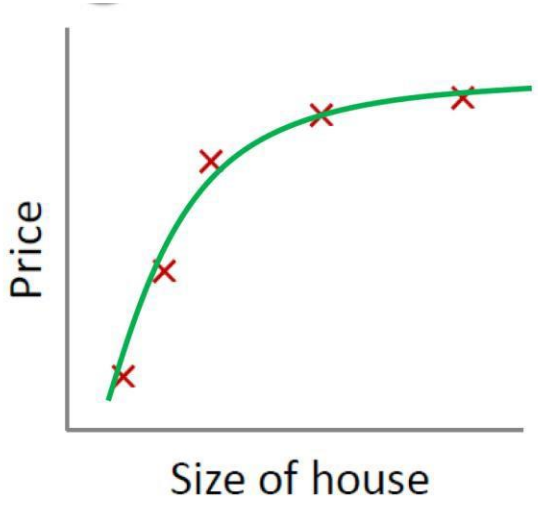




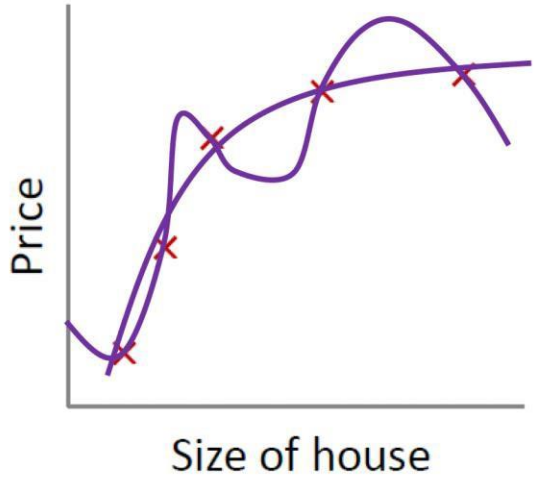
Example with scikit-learn:

```
>>> from sklearn.ensemble import DecisionTreeClassifier
    ...
>>> clf =DecisionTreeClassifier(n_estimators=50)
>>> clf = clf.fit(X, y)
>>> clf.feature_importances_
array([ 0.04...,  0.05...,  0.4...,  0.4...])
```

Regularization



$$h_\theta(x) = \theta_0 + \theta_1 x_1 + \theta_2 x_1^2$$



$$h_\theta(x) = \theta_0 + \theta_1 x_1 + \theta_2 x_1^2 + \theta_3 x_1^3 + \theta_4 x_1^4$$

$\theta_3 x_1^3 \approx 0$, $\theta_4 x_1^4 \approx 0$

We added more features, e.g. x_1^3 and x_1^4

Overfits the data poorly and does not generalize well ☹

Suppose we define the cost function as follows (to make θ_3, θ_4 very small)

$$\min_\theta \frac{1}{2n} \sum_{i=1}^{n} \left[h_\theta\left(x^{(i)}\right) - y^{(i)}\right]^2 + \mathbf{1000}\,\boldsymbol{\theta_3^2} + \mathbf{1000}\,\boldsymbol{\theta_4^2}$$

Then, the only way to make this new cost function small is if θ_3 and θ_4 are small

Regularization

- Small values for parameters $\theta_0, \theta_1, \ldots, \theta_p$
  - Imply a simpler model
  - Less prone to overfitting
- So we just modify our cost function as follows

λ = Regularization parameter (it's a hyper-parameter)

$$E(\theta) = \frac{1}{2n}\left[\sum_{i=1}^{n}[h_\theta(x^{(i)}) - y^{(i)}]^2 + \lambda \sum_{j=1}^{p}\theta_j^2\right]$$

Objective 1:
- Fit the training dataset well

Objective 2:
- Keep the parameter small

λ controls the trade-off between two objectives:

Regularization

What is the relation between λ and overfitting or underfitting?
i.e. what happens if λ is too small or too large.

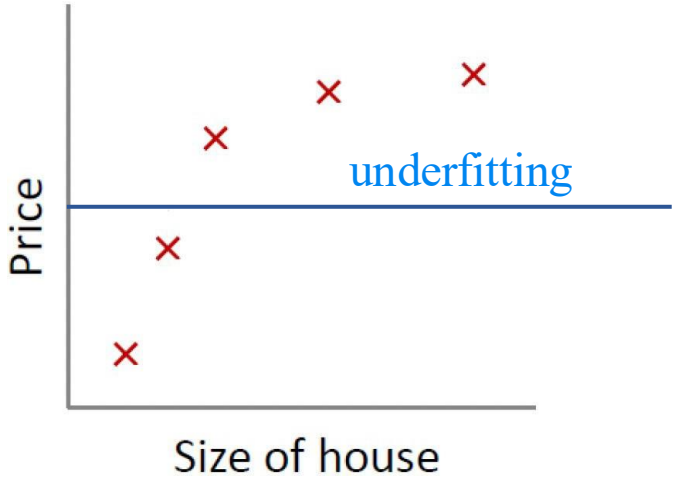
$$E(\theta) = \frac{1}{2n}\left[\sum_{i=1}^{n}[h_\theta(x^{(i)}) - y^{(i)}]^2 + \lambda\sum_{j=1}^{p}\theta_j^2\right]$$

What happens if λ is **set to zero**?

- This becomes our original cost function. **Overfitting** can happen.

What happens if λ is set to an **extremely large value**?

- The algorithm might result in **underfitting**.
- Example for Linear Regression:



So, it's important to try several values for λ and estimate the generalization error each time (i.e. hyper-parameter tuning).

Suppose:

$$h_\theta(x) = \theta_0 + \theta_1 x_1 + \theta_2 x_1^2 + \theta_3 x_1^3 + \theta_4 x_1^4$$

We will end up penalizing θ_1, θ_2, θ_3, θ_4 (their value will be close to 0)



Thanks ☺

And, once again:

Welcome!